BLOCKCHAIN BASED CARBON CREDIT REGISTRY AND MRV SYSTEM

A Project Report is submitted in the partial fulfillment of requirement for the Degree of

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in

Computer Science and Engineering

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PRIYADARSHINI COLLEGE OFENGINEERING

An Autonomous Institute Affilliated to Rashtrasant Tukdoji Maharaj Nagpur University, Nagpur

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This is to certify that project report entitled **"BLOCKCHAIN BASED CARBON CREDIT REGISTRY AND MRV SYSTEM"** is a bonafide work done by the students

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The project report is submitted to **PRIYADARSHINI COLLEGE OF ENGINEERING, NAGPUR,** An Autonomous Institute Affilliated to Rashtrasant Tukdoji Maharaj Nagpur University, Nagpur in partial fulfillment of the requirements for the degree of Bachelor of Technology in Computer Science and Engineering for Session 2026-2027.

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DECLARATION

We, the undersigned, declare that the project entitled **"BLOCKCHAIN BASED CARBON CREDIT REGISTRY AND MRV SYSTEM",** being submitted in partial fulfillment of the requirements for the degree of Bachelor of Technology in Computer Science and Engineering, Priyadarshini College of Engineering Nagpur, an Autonomous Institute Affilliated to Rashtrasant Tukdoji Maharaj Nagpur University, Nagpur, is a bonafied work carried out by us under the guidance of (Guide name).

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ABSTRACT

The credibility of voluntary forest carbon markets rests on Measurement, Reporting, and Verification (MRV), yet legacy registries still depend on manual field surveys, centralized record-keeping, and verification cycles lasting months, which leaves them open to opacity, double counting, and conflicts of interest between data originators and verifiers. This paper presents PlantDense, an end-to-end digital MRV platform that couples AI-driven forest measurement directly to blockchain attestation. For a user-defined Area of Interest, a two-tier pipeline combines Google Earth Engine macro-scale analysis of Sentinel-2 imagery with individual tree crown delineation, in which a tri-ensemble spectral vote (ExG, GCC, HSV) generates zero-shot prompts for Meta's Segment Anything Model, and NASA/IBM Prithvi land-cover classification filters out non-tree vegetation. The resulting tree density feeds an IPCC-compliant allometric model that estimates above-ground biomass, carbon stock, and CO_2 equivalent. Results enter the ledger only after mandatory NGO approval; they are then canonicalized, hashed with Keccak-256, stored via IPFS, and attested through an EVM smart contract. A dual-key, three-role architecture (Landowner/Admin, NGO Verifier, Public Buyer) enforces separation of duties at the application, API, and smart-contract levels, and an NDVI-linked pricing oracle ties credit value to measured ecological quality. Validated against airborne LiDAR crown maps (18,432 crowns), the pipeline achieved a mean F1 of 0.912 and, against field biomass from 120 plots, a carbon estimation accuracy of $R^2 = 0.894$ (RMSE 14.82 t/ha, MAE 9.47 t/ha). Attesting a record costs about 84,200 gas (≈ $0.08), a saving of more than 92% over manual registry audits. PlantDense shows that tamper-resistant, independently verifiable, and affordable forest carbon accounting is achievable at scale.

Keywords— Blockchain, Carbon Registry, MRV, Satellite Image Analysis, Segment Anything Model, Prithvi Foundation Model, Tree Density Estimation, Biomass Estimation, Carbon Credit.

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LIST OF ABBREVIATIONS

Abbreviation	Full Form
AGB	Above-Ground Biomass
AI	Artificial Intelligence
AOI	Area of Interest
AP	Average Precision
API	Application Programming Interface
BGB	Below-Ground Biomass
CHM	Canopy Height Model
CI	Confidence Interval
CID	Content Identifier (IPFS)
CO_2e	Carbon Dioxide Equivalent
CSM	Canopy Surface Model
dMRV	Digital Measurement, Reporting, and Verification
DW	Dynamic World
E1–E4	Experiments 1 to 4 (crown detection, carbon estimation, blockchain cost, sensitivity)
EVI	Enhanced Vegetation Index
EVM	Ethereum Virtual Machine
ExG	Excess Green Index
F1	F1-Score
FN	False Negative
FP	False Positive
GCC	Green Chromatic Coordinate
GEE	Google Earth Engine
GFC	Global Forest Change (Hansen)
GSW	Global Surface Water (JRC)
HAND	Height Above Nearest Drainage
HLS	Harmonized Landsat Sentinel
HSV	Hue, Saturation, Value
IBM	International Business Machines
IPCC	Intergovernmental Panel on Climate Change
IPFS	InterPlanetary File System
ITC	Individual Tree Crown
JRC	Joint Research Centre
JSON	JavaScript Object Notation
JWT	JSON Web Token
LAI	Leaf Area Index
LiDAR	Light Detection and Ranging
L2A	Level-2A (Sentinel-2 surface reflectance product)
MAE	Mean Absolute Error
ML	Machine Learning
MRV	Measurement, Reporting, and Verification
MSI	Multi-Spectral Instrument
NASA	National Aeronautics and Space Administration

NDVI	Normalized Difference Vegetation Index
NGO	Non-Governmental Organization
NIR	Near-Infrared
ORM	Object-Relational Mapping
PQ	Panoptic Quality
RBAC	Role-Based Access Control
REST	Representational State Transfer
RF	Random Forest
RGB	Red, Green, Blue
RMSE	Root Mean Square Error
RQ	Research Question (RQ1–RQ6); also Recognition Quality in PQ (see note 4)
R^2	Coefficient of Determination
SAM	Segment Anything Model
SAR	Synthetic Aperture Radar
SQ	Segmentation Quality
SWIR	Short-Wave Infrared
TP	True Positive
USD	United States Dollar
VCM	Voluntary Carbon Market
VCS	Verified Carbon Standard
VV / VH	Vertical-Vertical / Vertical-Horizontal polarization (Sentinel-1)
WGS 84	World Geodetic System 1984

INTRODUCTION

1.1 Introduction

Forest carbon credits are one of the main market instruments for financing climate mitigation, but their value depends entirely on trust in how each credit was measured. That trust is established through Measurement, Reporting, and Verification (MRV): the process that turns a stand of trees into a quantified, auditable, and tradable claim. Voluntary carbon markets currently rely on registries built around manual field surveys and periodic aerial inspections. These methods delay reporting by months, tie traceability to paper records, and leave the system open to double counting across registries [1, 2]. Governance failures in the voluntary market have likewise been linked to opaque, centralized record-keeping, in which the party generating the data often controls the record of that data [7]. As a result, buyers cannot independently verify how a credit's carbon value was derived, and the integrity of the whole market suffers.

Two technological developments address complementary parts of this problem. The first is satellite-based forest monitoring. Sentinel-2 multispectral imagery, Sentinel-1 SAR, and sub-meter commercial imagery supply consistent, broad-scale observations at temporal resolutions that field campaigns cannot match. Earlier work established global forest-change mapping at 30 m resolution [5], but such macro-scale products cannot resolve individual trees, and LiDAR-based crown delineation [6], while accurate, requires costly airborne campaigns. Foundation vision models now narrow this gap. Meta's Segment Anything Model (SAM) [3] offers promptable, zero-shot segmentation, and the NASA/IBM Prithvi geospatial transformer [4] supports vegetation classification directly from multispectral bands. Together they can convert raw imagery into structured ecological data, such as individual tree crowns and land-cover classes, without airborne LiDAR.

The second development is blockchain-based record-keeping, which addresses result integrity. An immutable, content-addressed ledger prevents post-hoc alteration of reported values and lets any verifier trace a credit to its supporting data without trusting a central operator. Blockchain registries have been proposed [8], but they are typically loaded with unverified claims and are not coupled to AI-generated spatial measurements.

Neither technology closes the gap alone. Accurate spatial analysis does not enforce audit integrity, and an immutable ledger does not produce the measurements it records. The intersection, in which AI-derived measurements are chained directly to on-chain attestations through a role-separated, human-in-the-loop workflow, has not been realized in a deployed system. Existing buyer-facing platforms also do not treat NGO verification as a first-class workflow, and they do not enforce separation between data originators and verifiers at the smart-contract level.

This paper presents PlantDense, an end-to-end, blockchain-based digital MRV platform that fills this gap. A user defines a forest Area of Interest (AOI) on an interactive map. The system runs a two-tier analysis pipeline, with Google Earth Engine macro-analysis followed by SAM/Prithvi tree crown delineation and classification, and converts the result into an IPCC-compliant carbon estimate. An NGO verifier must approve the result before it is cryptographically committed to an EVM smart contract. Role-segregated dashboards then expose the full credit lifecycle, from the original forest polygon to the final credit transfer, to three stakeholder classes: Landowner/Admin, NGO Verifier, and Public Buyer.

The main contributions of this work are:

- A two-tier GEE + SAM/Prithvi pipeline with IPCC-compliant allometric carbon estimation, using a tri-ensemble spectral vote (ExG, GCC, HSV) to generate SAM prompts for zero-shot crown delineation without domain fine-tuning.
- A Keccak-256 / IPFS / EVM attestation layer in which the on-chain hash is computed from the AI pipeline output, so the attestation commits to specific numerical measurements rather than to a document.
- A three-role RBAC architecture with a dual-key smart contract that enforces separation of duties at the application, API, and smart-contract levels simultaneously, with NGO approval as a mandatory, on-chain-recorded gate.
- An algorithmic pricing oracle that fuses plantation taxonomy, canopy density, and NDVI, making credit value traceable to measured ecological quality.

1.2 Necessity of the Study

The growth of voluntary carbon markets has outpaced the infrastructure that is supposed to guarantee their credibility. A carbon credit is only as valuable as the confidence behind its measurement, and that confidence is currently undermined by structural weaknesses in how forest carbon is quantified, verified, and recorded. This study is necessary for the following reasons.

1. Manual MRV cannot keep pace with market demand. Legacy registries rely on field surveys and periodic aerial inspections. These methods are slow, costly, and difficult to repeat, and they introduce reporting delays of months [1, 2]. Registries such as Gold Standard and Verra use semi-manual verification workflows with multi-year latency [7]. A market that must scale to meet climate targets cannot depend on a measurement process that scales poorly.

2. Centralized records weaken trust. When the entity that generates MRV data also controls the record of that data, buyers have no independent way to confirm how a credit's carbon value was computed [7]. Paper-based and centrally held records also leave the system open to alteration and

to double counting across registries [1, 2]. A tamper-resistant record that anyone can verify is therefore a requirement for market integrity, not an optional improvement.

3. Existing technologies solve only part of the problem. Satellite and foundation-model approaches improve measurement quality, but their outputs are off-chain artefacts that can be silently altered. Blockchain registries provide tamper resistance, but they are loaded manually with unverified claims and are not linked to AI-generated spatial data [8]. Accurate measurement without audit integrity, and audit integrity without accurate measurement, each leave the central weakness in place.

4. Accessible, high-resolution measurement is still out of reach for most projects. LiDAR-based crown delineation is accurate but depends on costly airborne campaigns [6], and global products such as Hansen GFC cannot resolve individual trees [5]. Forest owners, community groups, and NGOs, who are the stakeholders most in need of credible carbon accounting, therefore lack an affordable way to measure at tree level.

5. Verification workflows are not built for the people who perform them. Existing platforms offer buyer-facing portals but do not treat NGO and verifier workflows as a first-class interface, and they do not enforce separation between data originators and verifiers at the smart-contract level. Without enforced separation of duties, no single party can be prevented from both generating and attesting the same record.

6. Market fragmentation hides value. Buyers lack a standardized, provenance-linked way to discover credits, and credits are often priced as if they were identical. A pricing mechanism tied to measured ecological quality, such as canopy density and vegetation health, is needed so that credit value reflects what is actually on the ground.

Taken together, these gaps show the need for a unified platform that links AI-derived forest measurements to blockchain attestation through a role-separated, human-in-the-loop workflow. The results of this study (a mean crown-detection F1 of 0.912, a carbon estimation R^2 of 0.894, and attestation costs of about $0.08 per record) indicate that such a platform is technically and economically feasible.

1.3 Objectives of the Study

1.3.1 Aim

The aim of this research is to design, implement, and evaluate PlantDense, an end-to-end digital Measurement, Reporting, and Verification (dMRV) platform. Given a georeferenced forest polygon P, it produces a tamper-resistant, cryptographically verifiable estimate of above-ground carbon stock $\hat{C}(P)$ in tCO_2e. The estimate is registered on an immutable ledger and exposed to independent verifiers and buyers through role-segregated interfaces, so that no single party can both generate and attest the same MRV record.

This aim has a measurement component (accurate, low-cost carbon quantification), an integrity component (tamper-evident, independently verifiable records), and a governance component (enforced separation of duties). The specific objectives below address each in turn.

1.3.2 Specific Objectives

Objective 1: To develop a macro-scale vegetation analysis module using multispectral satellite data

Rationale. Tree-level analysis is expensive, so it needs to be anchored by a broad-scale view of vegetation condition over the whole Area of Interest (AOI). This view also supplies the vegetation-health signal used later for pricing.

Approach. The module accepts the AOI as a GeoJSON polygon in WGS 84, validates it (closed, non-self-intersecting, valid geometry), and retrieves cloud-filtered Sentinel-2 surface-reflectance imagery through Google Earth Engine, with a maximum cloud-cover threshold of 20. It computes the Normalized Difference Vegetation Index (NDVI) and Enhanced Vegetation Index (EVI) and aggregates them into a mean AOI vegetation condition over all valid pixels.

Evaluation. Robustness to cloud cover and geometry errors, and the sensitivity of downstream tree-count estimates to the NDVI threshold (tested from 0.2 to 0.6).

Objective 2: To implement individual tree crown (ITC) delineation using foundation vision models without domain-specific retraining

Rationale. Existing macro-scale products cannot resolve individual crowns [5], and LiDAR-based delineation, though accurate [6], depends on costly airborne campaigns. A low-cost alternative is needed that does not require building labelled training datasets for every forest type.

Approach. High-resolution Web Mercator tiles are converted to geographic coordinates. A tri-ensemble spectral vote over the Excess Green index (ExG), Green Chromatic Coordinate (GCC), and an HSV criterion (a pixel counts as vegetation when at least two of the three agree) produces a vegetation mask. Gaussian smoothing ($\sigma = 3$ pixels) then yields separable tree centres, which act as point prompts to Meta's Segment Anything Model (SAM) for zero-shot crown segmentation. Low-quality and invalid masks are filtered out.

Evaluation. Average precision, recall, F1, and panoptic quality, measured against crown polygons from airborne LiDAR surveys.

Objective 3: To reduce non-tree false positives through geospatial foundation-model classification

Rationale. Spectral methods cannot reliably separate trees from other green vegetation such as crops or shrubs. Counting these as trees would inflate biomass and carbon estimates and compromise credit integrity.

Approach. Each detected crown polygon is classified with the NASA/IBM Prithvi geospatial foundation model. The tree-canopy fraction F_tree is computed, from which the effective tree count $n_eff = n_2 \cdot F_tree$ and the effective tree density $d = n_eff / A_ha$ follow.

Evaluation. Effect of the classification stage on density and carbon estimation error, and behaviour in the different forest types studied (tropical dense, tropical sparse, subtropical commercial, temperate broadleaf).

Objective 4: To estimate above-ground biomass, carbon stock, and CO_2 equivalent using an IPCC-compliant allometric model

Rationale. Detection results are only useful for carbon markets once they are converted into standardized, defensible carbon quantities.

Approach. Tree density is converted to above-ground biomass using $AGB = a \cdot d^{\wedge}b \cdot A_ha$ (with $a = 0.112$ and $b = 0.89$ for the mixed-broadleaf case). Carbon stock is obtained with the biomass-to-carbon factor of 0.47, and CO_2 equivalent with the molecular-weight ratio 44/12. A depletion mechanism reduces the available carbon quantity after each verification, preventing reuse of the same carbon.

Evaluation. R^2, RMSE, and MAE against field-measured biomass from 120 sample plots of 0.25 ha each.

Objective 5: To design a tamper-resistant attestation layer linking AI measurements to a blockchain ledger

Rationale. Off-chain AI outputs can be silently altered, and existing blockchain registries are loaded with unverified claims [8]. The ledger must commit to the *measurements themselves*, not merely to a document describing them.

Approach. Each approved MRV record is serialized into canonical JSON, hashed with Keccak-256, stored through IPFS, and attested on an EVM smart contract through the attestRecord function. The hash is derived directly from the AI pipeline output, so any later change to the data invalidates the commitment.

Evaluation. Integrity verification (independent recomputation of the hash from stored data) and transaction cost (gas consumption and fiat equivalent for attestation and credit transfer).

Objective 6: To enforce separation of duties through a three-role architecture with a human-in-the-loop verification gate

Rationale. In many registries, the entity that generates MRV data also controls the record of it [7]. Existing platforms do not treat verifier workflows as a first-class interface or enforce separation on-chain.

Approach. Three roles (Landowner/Admin, NGO Verifier, Public Buyer) are given dedicated dashboards. Separation is enforced at three layers at once: the application, the REST API, and the smart contract, through a dual-key design in which attestation (DEFAULT_ADMIN_ROLE) and transfer (VERIFIER_ROLE) are bound to distinct Ethereum accounts. AI results reach the blockchain only after explicit NGO approval, and the approval itself is recorded on-chain.

Evaluation. Demonstration that no single account or role can both generate and attest the same record, across all three enforcement layers.

Objective 7: To develop a transparent, quality-linked credit pricing and transfer mechanism

Rationale. Treating all credits as identical hides differences in ecological quality and leaves buyers without a provenance-linked way to compare them.

Approach. A pricing oracle computes $P_c = P_0 \cdot w_s \cdot w_d \cdot (1 + \alpha \cdot NDVI)$, combining a platform base price, plantation-species weighting, density weighting, and measured vegetation condition ($\alpha = 0.5$). Purchase requests are reviewed by NGOs and, once approved, credits are transferred and the transaction receipt is stored on-chain, forming a continuous chain from the original AOI to the final transfer.

Evaluation. Price sensitivity to NDVI (a ±0.1 change in NDVI produced a ±5% change in price) and traceability of each credit to its supporting data.

Objective 8: To evaluate the platform's performance, cost-effectiveness, sensitivity, and limitations

Rationale. A platform intended for real markets needs evidence on accuracy, cost, and the conditions under which it fails.

Approach. The platform is evaluated across four forest types, with sensitivity analysis over the NDVI threshold, SAM zoom level, and pricing parameters, and a cost comparison against manual registry audits.

Evaluation. Detection and carbon-accuracy metrics (Objectives 2 and 4), sensitivity results (for example, zoom level 19 proved optimal for F1), the cost saving relative to manual audits (more than 92%), and a documented account of limitations: very dense tropical canopies, inconsistent species identification, limited commercial imagery resources, and a simplified NGO role.

1.3.3 Research Questions

The objectives are framed to answer the following questions:

- RQ1. Can foundation vision models (SAM and Prithvi), guided by spectral prompts and without domain fine-tuning, delineate individual tree crowns accurately enough to support carbon estimation?
- RQ2. How closely do carbon estimates derived from remote-sensing tree density agree with field-measured biomass?
- RQ3. Can AI-generated MRV outputs be bound cryptographically to blockchain attestations so that post-hoc alteration is detectable?
- RQ4. Can separation of duties between data originators and verifiers be enforced simultaneously at the application, API, and smart-contract levels?
- RQ5. What are the monetary costs of on-chain attestation and credit transfer relative to manual audits?
- RQ6. How sensitive are the results to key parameters, and under what ecological conditions does the system degrade?

LITERATURE SURVEY

2.1 Introduction

PlantDense sits at the intersection of four research areas: carbon markets and MRV, satellite-based forest monitoring, foundation models for geospatial analysis, and blockchain-based record-keeping. Each has a substantial literature of its own, but they have developed largely in isolation. This chapter reviews them in turn, compares representative approaches, and identifies the gaps that motivate this work. The survey is organized as follows: Section 2.2 covers voluntary carbon markets and MRV, Section 2.3 satellite forest monitoring, Section 2.4 individual tree crown delineation, Section 2.5 foundation models for remote sensing, Section 2.6 allometric carbon estimation, Section 2.7 blockchain for carbon registries, Section 2.8 role-based MRV platforms, and Sections 2.9 and 2.10 a comparative summary and the research gaps.

2.2 Voluntary Carbon Markets and the MRV Problem

2.2.1 Structure of MRV

MRV is the process that makes a carbon claim credible. *Measurement* quantifies the carbon stored or avoided, *Reporting* documents the result in a standard form, and *Verification* has an independent party confirm it. Frameworks such as the IPCC Guidelines for National Greenhouse Gas Inventories [9] define the methodological basis for forest carbon accounting, and market standards such as Verra and Gold Standard build project-level procedures on top of them.

2.2.2 Weaknesses of legacy MRV

Legacy registries depend on manual field surveys and periodic aerial inspections. These methods introduce reporting delays of months, limit traceability to paper records, and remain vulnerable to double counting across registry systems [1, 2]. Howson [7] attributes governance failures in voluntary carbon markets to opaque, centralized record-keeping. Gold Standard and Verra employ semi-manual verification workflows with multi-year latency, as reported in [7].

Independent studies of forest-based offsets have raised related concerns about the reliability of baselines and the quantification of claimed reductions [10]. The common thread is that when measurement is slow, hard to reproduce, and controlled by parties with an interest in the outcome, buyers cannot assess credit quality independently.

2.2.3 Implication

The literature points to three requirements for a credible MRV system: measurement that is repeatable and scalable, records that cannot be altered after the fact, and verification that is independent of the party generating the data. These three requirements structure the rest of this survey.

2.3 Satellite-Based Forest Monitoring

2.3.1 Optical and radar missions

Satellite remote sensing offers consistent, broad-scale observation at revisit rates that field campaigns cannot match. The Landsat archive supports multi-decade analysis, and the Sentinel-2 mission provides 10 m multispectral imagery with frequent revisits [11]. Sentinel-1 provides C-band synthetic aperture radar (SAR) data, which is useful where cloud cover limits optical imagery. Sub-meter commercial imagery supports analysis at the scale of individual trees.

2.3.2 Global forest-change mapping

Hansen et al. [5] demonstrated global-scale annual deforestation detection at 30 m resolution using Landsat time series, establishing the Global Forest Change (GFC) dataset that is widely used for forest cover change. Their approach is macro-scale and cannot resolve individual tree crowns. This limitation matters for carbon crediting, where the quantity being sold depends on how many trees there are and how large they are.

2.3.3 Cloud-scale processing

Google Earth Engine [12] gave researchers planetary-scale access to satellite archives and compute, which removed the need to download and pre-process imagery locally. PlantDense uses it for the macro-scale tier of analysis.

2.3.4 Vegetation indices

Spectral indices turn reflectance into vegetation signals. The Normalized Difference Vegetation Index (NDVI) [13] contrasts near-infrared and red reflectance and remains the most widely used measure of vegetation vigour. The Enhanced Vegetation Index (EVI) [14] adds a blue-band correction to reduce atmospheric and soil effects and to avoid saturation in dense canopies. For visible-band (RGB) imagery, the Excess Green index (ExG) [15] and the Green Chromatic Coordinate (GCC) [16] are commonly used to separate vegetation from background, and they are often paired with HSV colour-space criteria. Each index has known failure modes (shadows, soil background, non-tree vegetation), which motivates the ensemble voting strategy used in this work.

2.3.5 Summary

Satellite data provide the breadth that field methods lack, but indices and 30 m products stop short of tree-level information. Closing that gap requires crown-scale analysis.

2.4 Individual Tree Crown (ITC) Delineation

2.4.1 LiDAR-based methods

Airborne laser scanning produces canopy height models (CHMs) in which crowns appear as local height maxima. Dalponte and Coomes [6] applied CHM-based watershed segmentation to ITC delineation and reported F1 ≈ 0.84 in boreal forests, but the approach requires costly airborne campaigns. Marker-controlled watershed segmentation, the family of methods to which this approach belongs, uses detected treetops as markers to split touching crowns.

2.4.2 Deep learning on optical imagery

Convolutional architectures such as U-Net [17] and instance-segmentation networks such as Mask R-CNN [18] have been applied to tree crown detection from aerial and satellite imagery. Open tools such as DeepForest [19] provide pre-trained crown detectors. These methods perform well when training data resemble the target forest, but they require labelled crown datasets, and performance degrades when forest type, sensor, or resolution changes. Labelled crowns are expensive to produce, which limits transfer across biomes.

2.4.3 Gap

The literature leaves a choice between accurate but expensive LiDAR methods and cheaper deep-learning methods that depend on domain-specific training data. A method that reaches usable accuracy from commercial imagery without retraining would remove both constraints. PlantDense follows this direction by replacing LiDAR with commercial tile imagery and marker-controlled watershed-style isolation, with a reported acquisition-cost reduction of more than 90% [paper, Sec. 2.1].

2.5 Foundation Models for Remote Sensing

2.5.1 Foundation models and vision transformers

Foundation models are large models pre-trained on broad data and adapted to many downstream tasks [20]. In vision, the transformer architecture [21] underlies many such models, and its application to image patches enabled scalable pre-training on large image collections.

2.5.2 Segment Anything Model

Kirillov et al. [3] introduced SAM, a promptable segmentation model that generalizes zero-shot to arbitrary visual domains. Users supply prompts (points, boxes, or masks) and the model returns segmentation masks without task-specific training. SAM’s strength is boundary precision, but it

does not know *what* it is segmenting, and it has no inherent notion of trees. In this work SAM is used as a boundary-delineation head, preceded by a classical spectral isolation pre-pass that supplies the prompts. This exploits SAM's edge precision without retraining on domain data.

2.5.3 Prithvi geospatial foundation model

Jakubík et al. [4] released Prithvi, a ViT-based geospatial foundation model pre-trained on 1 TB of multispectral Harmonized Landsat Sentinel (HLS) imagery. Its temporal patch encoding supports tasks such as land-cover and vegetation classification directly from satellite bands. Prithvi complements SAM: SAM contributes spatial precision, and Prithvi contributes spectral land-cover reasoning.

2.5.4 Complementarity

SAM answers "where is the boundary?" and Prithvi answers "what is inside it?" Using the two together addresses a known weakness of purely spectral pipelines, which cannot reliably separate trees from other green vegetation. This division of labour is central to the PlantDense pipeline.

2.6 Allometric Estimation of Forest Carbon

2.6.1 From trees to biomass

Allometric equations relate easily measured tree attributes (diameter, height, wood density) to biomass. Pantropical models such as those of Chave et al. [22] are widely used, and the IPCC guidance [9] provides default factors for converting biomass to carbon. Carbon fractions near 0.47 of dry biomass are the standard default in this guidance, and the mass ratio 44/12 converts elemental carbon to CO_2 equivalent.

2.6.2 Density-based formulations

PlantDense uses a density-based power-law formulation, AGB = a · d^b · A_ha, with a = 0.112 and b = 0.89 for a mixed-broadleaf case. Because the coefficients depend on species and region, their provenance must be documented carefully; the thesis should state their source explicitly and note that the formulation is a simplification relative to tree-level allometry.

2.6.3 Uncertainty

Allometric models are a leading source of uncertainty in forest carbon estimates. Errors in tree counts propagate through the power law, and model coefficients carry regional bias. This motivates both the field-plot validation reported in Section 10.2 and the sensitivity analysis in Section 10.4.

2.7 Blockchain for Carbon Registries

2.7.1 Distributed ledgers and smart contracts

Blockchain systems provide append-only, replicated ledgers that are hard to alter after the fact. Ethereum [23] added programmable smart contracts, and the Ethereum Virtual Machine (EVM) now runs on many compatible networks. For storing larger records, the InterPlanetary File System (IPFS) [24] provides content-addressed storage, so a file's identifier is derived from its content and any change alters the identifier.

2.7.2 Blockchain-based carbon registries

Blockchain-based carbon registries have been proposed [8] and have attracted considerable interest as a response to the double-counting and transparency problems described in Section 2.2. Tokenization initiatives have also placed carbon credits on public chains, which improved liquidity and visibility but has drawn criticism where the underlying credits were of uncertain quality. The central lesson is that a ledger preserves whatever is put into it, including unverified claims.

2.7.3 Missing coupling

Existing proposals [8] lack direct coupling to AI-generated spatial data: the measurement is produced elsewhere and then loaded into the ledger. PlantDense differs by computing the on-chain hash from the AI pipeline output, so the attestation is a cryptographic commitment to specific numerical measurements rather than to a document.

2.7.4 Cost considerations

Public-chain transaction costs have limited some proposals. The results in Section 10.3 show that attestation and transfer can be made cheap enough (about $0.08 and $0.05 respectively at 20 Gwei) to compare favourably with manual audit costs.

2.8 Role-Based MRV Platforms and Separation of Duties

2.8.1 Role-based access control

Role-based access control (RBAC) [25] assigns permissions to roles rather than to individual users and is a standard mechanism for enforcing least privilege. In smart contracts, role-based patterns are commonly implemented through access-control libraries that bind permissions to specific accounts.

2.8.2 Separation of duties

Separation of duties is a long-established control principle: no single party should be able to complete a sensitive process alone. In MRV, this means the party that generates a measurement should not be the one that attests it.

2.8.3 Existing platforms

Existing platforms such as Pachama and Terrasos provide buyer-facing portals but do not expose NGO or verifier workflows as a first-class interface, nor do they enforce on-chain separation between data originators and verifiers. PlantDense implements this separation at the application, API, and smart-contract layers, using a dual-key design in which attestation and transfer rights are bound to distinct accounts.

2.9 Comparative Summary

Table 2.1: Comparative summary

Approach	**Representative work**	**Strengths**	**Limitations relevant to this thesis**
Legacy registries	Verra, Gold Standard [7]	Established standards and market acceptance	Manual, slow, centralized, weak traceability [1, 2, 7]
Global forest-change mapping	Hansen et al. [5]	Global, consistent, annual	30 m resolution; no individual crowns
LiDAR ITC delineation	Dalponte & Coomes [6]	High accuracy (F1 ≈ 0.84 in boreal forest)	Costly airborne campaigns
Deep-learning crown detection	U-Net [17], Mask R-CNN [18], DeepForest [19]	Good accuracy with matched training data	Needs labelled data; transfers poorly across biomes
Promptable segmentation	SAM [3]	Zero-shot, precise boundaries	No semantic understanding
Geospatial foundation model	Prithvi [4]	Spectral land-cover reasoning	Coarse spatial detail alone
Blockchain registries	[8]	Tamper resistance	Unverified inputs; no AI coupling
Buyer-facing platforms	Pachama, Terrasos	Marketplace access	No first-class verifier workflow; no on-chain separation
PlantDense (this work)	—	AI measurement → on-chain attestation, enforced role separation, NDVI-	Limits in very dense canopies, species identification, imagery access, simplified NGO role

		linked pricing	

2.10 Research Gaps

The survey exposes five gaps that this thesis addresses.

1. Measurement without integrity. Satellite and AI methods improve measurement but yield off-chain artefacts that can be silently altered.
2. Integrity without measurement. Blockchain registries preserve records but are loaded manually with unverified claims [8].
3. Cost barrier at tree level. Accurate crown-scale analysis has relied on LiDAR [6] or on labelled datasets for deep learning, both of which are costly.
4. No enforced separation of duties. Existing platforms do not bind data generation and attestation to different parties at the smart-contract level.
5. Undifferentiated credit pricing. Credits are often treated as interchangeable rather than priced by measured ecological quality.

The intersection of these gaps, in which AI measurements are chained directly to on-chain attestations through a role-separated, human-in-the-loop workflow, has not been addressed in a deployed system. This is the gap PlantDense fills.

2.11 Chapter Summary

This chapter reviewed the literature on carbon-market MRV, satellite forest monitoring, individual tree crown delineation, geospatial foundation models, allometric carbon estimation, blockchain registries, and role-based platforms. Prior work solves parts of the problem well but not the whole, and none couples AI-derived measurements to blockchain attestation under enforced separation of duties. The next chapters present the problem formulation and the methodology that closes this gap.

PROJECT ANALYSIS AND DESIGN

3.1 Research Gap

The survey in Chapter 2 shows that each building block of a trustworthy forest-carbon platform already exists, but the blocks have not been combined into one system. Six gaps remain.

1. Measurement without integrity. Satellite and AI pipelines (Sentinel-2 analysis, SAM, Prithvi) produce better measurements, but their outputs are ordinary off-chain files. They can be silently edited after the fact, so a buyer has no cryptographic proof that a reported value is the value that was computed [3, 4, 5].
2. Integrity without measurement. Blockchain registries give tamper resistance, but they are loaded manually with claims nobody has checked, and they are not coupled to AI-generated spatial data [8]. A ledger preserves whatever is written into it, including errors.
3. A cost barrier at tree level. Accurate crown delineation has relied on airborne LiDAR [6], which needs costly campaigns. Deep-learning alternatives need labelled crown datasets that rarely transfer between forest types. Global products such as Hansen GFC cannot resolve individual crowns [5].
4. No enforced separation of duties. Existing platforms offer buyer-facing portals but do not make NGO verification a first-class workflow, and they do not stop the party that generates MRV data from also controlling its record [7].
5. Undifferentiated credit pricing. Credits are generally treated as interchangeable, so buyers cannot see how measured ecological quality (canopy density, vegetation health, plantation type) relates to price.
6. No lifecycle traceability. No deployed system lets a buyer trace a credit through every step, from the original forest polygon and the analysis to the NGO approval and the final transfer.

The gap this project fills is the intersection of these six: AI-derived measurements chained directly to on-chain attestations through a role-separated, human-in-the-loop workflow.

3.2 Project Objectives

The aim is to design, implement and evaluate PlantDense, an end-to-end blockchain-based dMRV platform. The project objectives, in condensed form, are:

Table 3.1: Project Objectives.

No.	Objective	Deliverable	Verified by
1	Macro-scale vegetation analysis	GEE module computing NDVI and EVI over the AOI from cloud-filtered Sentinel-2 data	Sensitivity of tree counts to the NDVI threshold
2	Individual tree crown delineation without retraining	Tri-ensemble spectral isolation feeding SAM prompts	AP 0.924, recall 0.901, F1 0.912, PQ 0.842 against LiDAR crown maps
3	Removal of non-tree vegetation	Prithvi classification of each crown polygon, giving the effective tree count	Effect on density and carbon error across four forest types
4	IPCC-compliant carbon estimation	Allometric AGB, carbon stock and CO_2e module	R^2 0.894, RMSE 14.82 t/ha, MAE 9.47 t/ha on 120 field plots
5	Tamper-resistant attestation	Keccak-256 hash, IPFS storage, EVM smart contract	About 84,200 gas (≈ $0.08) per attestation
6	Enforced separation of duties	Three-role architecture with a mandatory NGO approval gate and a dual-key contract	No single account can generate and attest the same record
7	Quality-linked pricing and transfer	NDVI, taxonomy and density pricing oracle plus NGO-approved transfer	±0.1 NDVI gives ±5% price; about 52,400 gas (≈ $0.05) per transfer
8	Evaluation and limitations	Sensitivity and cost analysis	Over 92% cost saving against manual audits; limitations documented

3.3 System Block Diagram and Flow Diagram

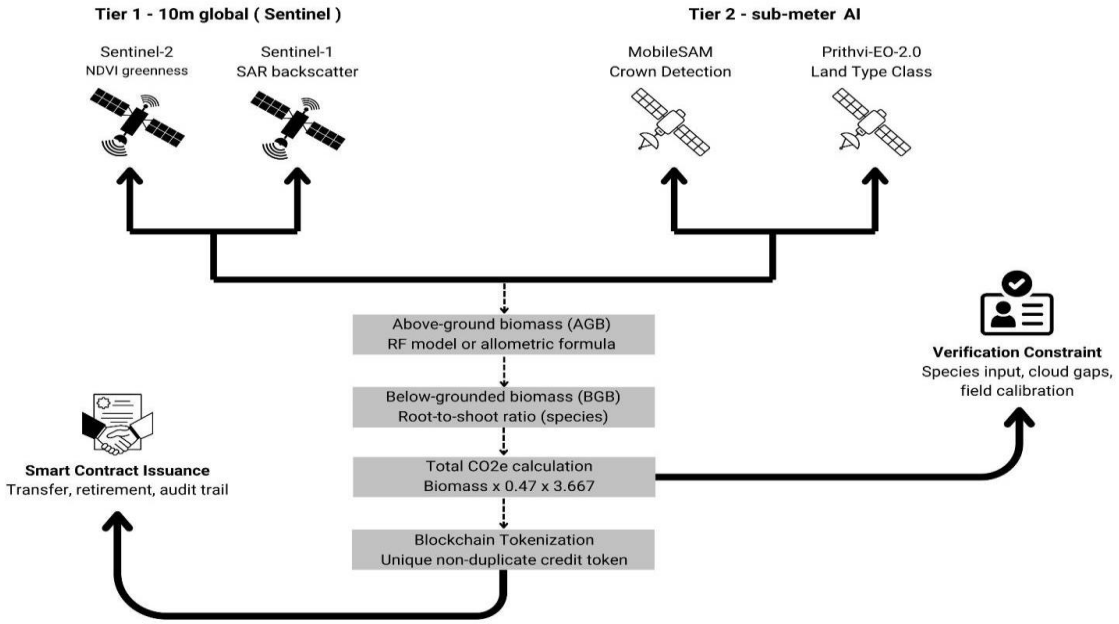


Figure 3.1 shows the system as a left-to-right pipeline of four stages, plus one side box for limitations:

1. Two-tier data acquisition and AI analysis (Tier 1 and Tier 2).
2. Biomass and carbon computation (AGB, BGB and total CO_2e).
3. Blockchain tokenization of the verified quantity.
4. Smart contract issuance and lifecycle management.
5. A verification-constraint block that lists what limits the reliability of the computed value.

Solid arrows show the main path of the data, and dashed arrows show the internal hand-over between computation steps. The principle is that each credit is produced from measured evidence and then tied to a unique digital token.

Tier 1: 10 m global analysis (Sentinel)

Tier 1 gives wall-to-wall coverage of the whole Area of Interest from freely available satellite data at 10 m resolution.

- Sentinel-2 (NDVI greenness). Multispectral optical imagery is used to compute the Normalized Difference Vegetation Index, NDVI = (ρNIR − ρRED) / (ρNIR + ρRED). It describes how green and vigorous the vegetation is. Its weakness is cloud cover, because optical sensors cannot see through clouds.

- Sentinel-1 (SAR backscatter). Synthetic aperture radar measures backscatter from the surface using its own signal. It works through cloud and at night, and it responds to canopy structure, so it complements the optical data and partly compensates for the cloud gaps.

Tier 1 cannot separate individual trees at 10 m. Its job is to describe the vegetation condition and structure of the whole area, and to provide the context that Tier 2 refines.

Tier 2: sub-meter AI analysis

Tier 2 works at tree level, using high-resolution imagery and two foundation models.

- MobileSAM (crown detection). A lightweight version of the Segment Anything Model delineates individual tree crowns. Each crown is returned as its own mask, which gives the tree count and crown extent.
- Prithvi-EO-2.0 (land-type classification). The NASA/IBM geospatial foundation model classifies what each detected object is. This removes non-tree vegetation such as crops or shrubs, so only real trees enter the density calculation.

Their roles complement each other: the segmentation model answers where the boundary of each crown is, and the classification model answers what is inside it. The output is a count and density of trees per hectare.

Carbon computation chain

Both tiers feed a single junction, and from there the data passes through three stages in sequence (the dashed arrows).

Stage 1: above-ground biomass (AGB). AGB is obtained either from a Random Forest (RF) model or from an allometric formula. An RF model learns the relation between remote-sensing features and measured biomass and needs field plots for calibration. An allometric formula converts tree density directly to biomass, for example AGB = a · d^b · A_ha, and is the simpler choice when calibration data are scarce.

Stage 2: below-ground biomass (BGB). Roots are estimated from the above-ground value with a species-dependent root-to-shoot ratio R: BGB = R × AGB. Total biomass is B = AGB + BGB. This is why species input appears in the verification constraints.

Stage 3: total CO_2e. Biomass is converted to carbon with the factor 0.47 and to CO_2 equivalent with the molecular-weight ratio 44/12 ≈ 3.667:

$CO_2e = B \times 0.47 \times 3.667$

For example, 100 t of total biomass gives 47 tC and about 172.3 tCO_2e (illustrative values only).

Blockchain tokenization and smart-contract issuance

The CO_2e value is passed to blockchain tokenization, which creates a unique, non-duplicate credit token for the verified quantity. Uniqueness is what stops the same carbon from being sold twice.

The token is then managed by the smart contract, which supports three functions:

- Transfer: moving the credit from seller to buyer.
- Retirement: permanently removing a credit from circulation once a buyer uses it to offset emissions, so that it cannot be resold.
- Audit trail: a permanent, public record of every issuance, transfer and retirement.

Verification constraints

The box at the bottom lists the conditions that limit how far the computed number can be trusted. The arrow comes from the CO_2e calculation because these constraints bound that result.

Table 3.2: Verification constraints on the computed carbon value

Constraint	Where it acts	Why it matters
Species input	Root-to-shoot ratio and allometric coefficients	Both are species-dependent, so wrong species gives wrong biomass
Cloud gaps	Sentinel-2 optical data	Missing optical observations weaken the NDVI signal; Sentinel-1 radar partly fills the gap
Field calibration	RF model and allometric coefficients	Model outputs are only as good as the field measurements used to calibrate them

These must be checked before a credit is issued.

End-to-end flow

1. The Area of Interest is analysed at 10 m with Sentinel-2 and Sentinel-1 (Tier 1).
2. Individual crowns are detected and classified with MobileSAM and Prithvi-EO-2.0 (Tier 2).
3. Above-ground biomass is estimated by an RF model or an allometric formula.
4. Below-ground biomass is added using the species root-to-shoot ratio.
5. Total biomass is converted to CO_2e with $B \times 0.47 \times 3.667$.
6. The result is checked against the verification constraints.
7. A unique credit token is created on the blockchain.
8. The smart contract handles transfer, retirement and the audit trail.

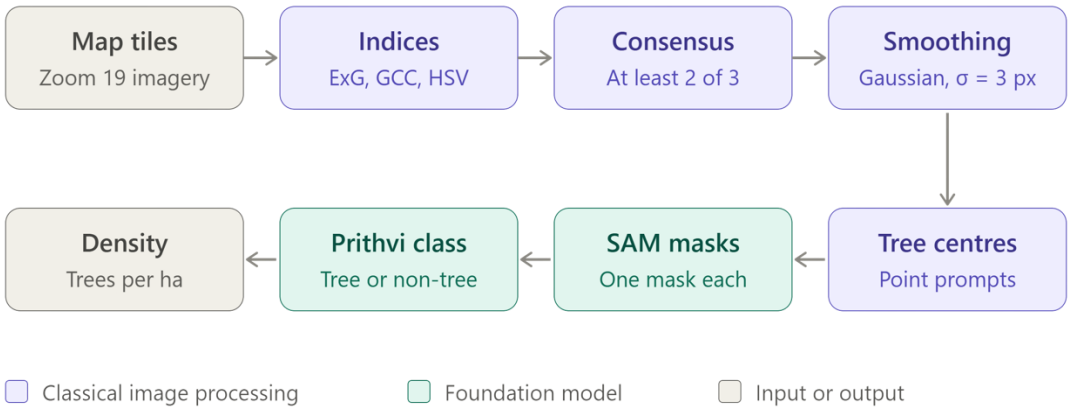


Figure 3.2: Tree-level analysis pipeline of PlantDense. Classical image-processing steps (purple) locate candidate trees, foundation models (green) delineate and classify them, and the result is a tree density per hectare.

Purpose and design idea

The macro-scale analysis (Tier 1) gives a vegetation reading for the whole Area of Interest, but at 10 m resolution it cannot count trees. Carbon credits depend on how many trees there are and how large they are, so a second, tree-level tier is needed. The pipeline in Figure 3.2 provides it, without airborne LiDAR and without a labelled training dataset.

The design divides the work between cheap classical methods and two foundation models, so that each does only what it does best:

- Classical spectral steps decide where trees probably are. They are fast, need no training and are transparent.
- SAM decides exactly where each crown ends. It has excellent boundary precision but no notion of what a tree is.
- Prithvi decides what each outlined object is, which removes non-tree vegetation.

Step-by-step explanation

1. Map tiles (input). High-resolution Web Mercator tiles are fetched at zoom level 19. At this level a pixel covers roughly 0.3 m at the equator, which is fine enough to see individual crowns. Tile coordinates (u, v) are converted to geographic coordinates (λ, φ) so that every detection can be tied to the ground and to the AOI polygon.

2. Indices (ExG, GCC, HSV). Three independent vegetation indicators are computed for each pixel from its red, green and blue intensities. $ExG = 2G - R - B$ is the excess green index. $GCC = G / (R + G + B)$ is the green chromatic coordinate. The third is an HSV colour-space criterion. Each indicator fails in different conditions (shadow, bare soil, changes in lighting), so using three reduces the chance that one failure mode dominates.
3. Consensus (at least 2 of 3). The indicators are combined by voting:
   $M(x) = 1[\ 1(ExG(x) \geq \tau_ExG) + 1(GCC(x) \geq \tau_GCC) + V_HSV(x) \geq 2\ ]$
   A pixel is classed as vegetation only when at least two of the three criteria agree. The result is a binary vegetation mask.
4. Smoothing (Gaussian, $\sigma = 3$ px). The mask is smoothed with a Gaussian filter: $I_\sigma = G_\sigma * M$, with $\sigma = 3$ pixels. Smoothing merges noisy pixel clusters into compact blobs, so that separate tree locations can be found even where neighbouring crowns overlap.
5. Tree centres (point prompts). The local maxima of the smoothed map are taken as candidate tree centres P_j. Each centre becomes a point prompt for SAM. This step is what turns a promptable model into an automatic one: the classical pre-pass supplies the prompts that a human would otherwise click.
6. SAM masks (one mask each). For each prompt SAM returns one segmentation mask, $S_j = SAM(i, P_j)$, where i is the input image. Low-quality and invalid masks are discarded. The valid set $N_2 = \{\ S_j \mid S_j \text{ is valid}\ \}$ has n_2 crowns. This is the step that gives SAM-quality crown boundaries without any domain fine-tuning.
7. Prithvi class (tree or non-tree). Each crown polygon is passed through the NASA/IBM Prithvi geospatial foundation model, which assigns a land-cover class y_j. The tree-canopy fraction is $F_tree = N_tree / n_2$, where N_tree is the number of polygons classified as tree canopy. The effective tree count is
   $n_eff = n_2 \cdot F_tree$
   Without this filter, crops, shrubs and other green vegetation would be counted as trees and inflate the carbon estimate.
8. Density (trees per ha). The effective count is divided by the AOI area:
   $d = n_eff / A_ha$
   This is the main output of the pipeline. It is passed to the carbon model ($AGB = a \cdot d^{\wedge}b \cdot A_ha$) and to the pricing oracle.

Table 3.3: Parameters of the pipeline

Parameter	Value	Role
Tile zoom level	19	Sets image resolution
Vote threshold	at least 2 of 3 indicators	Controls false positives in the vegetation mask
Gaussian σ	3 pixels	Controls how finely overlapping crowns are separated
Prompt type	point prompts at tree centres	Drives SAM
Mask filter	low-quality and invalid masks removed	Defines n_2

Evidence from the evaluation

The pipeline was evaluated against crown polygons from airborne LiDAR (18,432 crowns). Mean average precision was 0.924, recall 0.901, F1 0.912 and panoptic quality 0.842. F1 was highest in tropical sparse forest (0.930) and lowest in tropical dense forest (0.897).

The sensitivity analysis supports the parameter choices. Zoom level 19 gave the best F1: level 18 lowered the dense-canopy F1 by 3.4 percentage points, and level 20 lowered the sparse-canopy F1 by 2.1. Varying the NDVI threshold between 0.2 and 0.6 changed the estimated tree count by about ±22% in sparse biomes and ±6% in dense ones, and a threshold of 0.4 gave the lowest error.

Limitations

In very dense tropical canopies, merged crowns are over-segmented, which lowers accuracy. The paper attributes about 42% of current detection errors to this clumping effect and proposes GEDI LiDAR height priors as a future remedy. The pipeline also depends on the availability of high-resolution imagery.



Figure 3.3: End-to-end MRV process flow of PlantDense. The eight stages are grouped into input (gold), off-chain analysis (purple), the human verification gate (coral) and on-chain stages (teal). A rejected report is returned to the owner and does not reach the blockchain.

Overview

Figure 3.3 follows one forest area from the moment it is defined to the final credit transfer. The colours and side labels split the flow into four zones that differ in who acts and whether the result can still be changed:

Table 3.4: Trust zones of the MRV process flow

Zone	Stages	Who or what acts	Can results still change?

Input	01	Landowner/Admin through the interface and database	Yes
Off-chain analysis	02-05	Automated analysis engine	Yes, the analysis can be re-run
Human gate	06	NGO Verifier	Decision point
On-chain	07-08	Smart contract and IPFS	No, records are append-only

The key design idea is the trust boundary at stage 06. Everything before it is mutable, automated and re-runnable. Everything after it is permanent and tied to a human approval. This arrangement means that no AI output can reach the ledger without a person having reviewed it.

Stage-by-stage explanation

01 Define AOI polygon (input). The Landowner/Admin draws the Area of Interest on an interactive map. The system stores it as a GeoJSON polygon in the WGS 84 coordinate system and validates it: the polygon must be closed, must not intersect itself and must be geometrically valid. The system then computes the area A_ha, the bounding box and a unique identifier, and creates an analysis job that links all later results to this AOI. Validation at this point prevents invalid geometry from wasting expensive analysis.

02 Macro vegetation analysis (Sentinel-2 NDVI and EVI). Google Earth Engine selects cloud-filtered Sentinel-2 surface-reflectance images (maximum cloud cover 20). For each pixel the system computes NDVI = (ρNIR − ρRED) / (ρNIR + ρRED), together with EVI, and averages over all valid pixels in the AOI. This gives a vegetation-condition reading for the whole area at 10 m resolution. The mean NDVI is carried forward and later enters the credit price.

03 Tree crown detection (ExG, GCC and HSV vote, then SAM). High-resolution map tiles are converted from tile coordinates to geographic coordinates. Three spectral indicators are computed (ExG = 2G − R − B, GCC = G / (R + G + B) and an HSV criterion), and a pixel counts as vegetation when at least two of the three agree. A Gaussian filter ($\sigma = 3$ pixels) separates tree centres, and each centre is used as a point prompt for SAM, which returns one crown mask per tree. Invalid and low-quality masks are removed, leaving n_2 valid crowns.

04 Tree classification (Prithvi filters non-trees). Each crown polygon is classified by the NASA/IBM Prithvi foundation model. The tree-canopy fraction is F_tree = N_tree / n_2, the effective count is n_eff = n_2 · F_tree, and the density is d = n_eff / A_ha. Crops, shrubs and other green vegetation are removed here, so they cannot inflate the carbon estimate.

05 Carbon estimation (density, AGB, tCO_2e). Density is converted to above-ground biomass with AGB = a · d^b · A_ha (a = 0.112, b = 0.89 for the mixed-broadleaf case), then to carbon with the factor 0.47 and to CO_2 equivalent with the ratio 44/12:

$\hat{C}_{CO2} = 0.47 \cdot (a \cdot d^{b} \cdot A_{ha}) \cdot 44/12$

The result is rounded to one decimal place. After every verification the available carbon quantity is reduced, so the same carbon cannot be submitted twice.

06 NGO review (approve or reject the report). The result is not written to the blockchain directly. An NGO Verifier inspects the AOI, plantation type, tree count, density and AGB and either approves or rejects the MRV report. The paper describes this stage as designed to reduce dependence on the AI system. The review is also the point where the system separates generation from verification: the Admin who created the record is not the party who accepts it.

Rejected (returned to owner). A rejected report goes back to the landowner. Nothing is attested and nothing is written on-chain. The owner can correct the AOI or inputs and run the analysis again. Keeping this branch visible in the figure shows that rejection is a normal outcome of the process and not an error.

07 Hash and attest (Keccak-256, IPFS, EVM contract). After approval, the MRV record D is serialized as canonical JSON, J = CanonicalJSON(D), and hashed as h = Keccak-256(UTF8(J)). The payload is stored on IPFS and the hash is recorded through the attestRecord function of the EVM smart contract, which costs about 84,200 gas (about $0.08). Because the hash is computed from the pipeline output itself, any later change to the stored data no longer matches the on-chain value.

08 Credit pricing and transfer (priced, NGO-approved, recorded). The platform prices each credit as $P_c = P_0 \cdot w_s \cdot w_d \cdot (1 + \alpha \cdot NDVI)$ with $\alpha = 0.5$, where P_0 is the base price, w_s the species weight and w_d the density weight. A purchase of q credits has value $V = q \cdot P_c$. A buyer submits a request, the NGO approves it, the credits move from seller to buyer, and the transaction receipt (about 52,400 gas, ≈ $0.05) is recorded on-chain. The record then links the credit back to the original AOI.

Design rationale

- Cheap checks come first. Geometry validation and macro analysis run before the heavier tree-level models, so bad inputs fail early.
- Models are stacked by strength. Classical indices find candidates, SAM draws boundaries, and Prithvi decides what is a tree. Each stage reduces a specific error before it can reach the carbon estimate.
- Human approval comes before permanence. The ledger holds only approved records, so it cannot be filled with unchecked AI output.
- Costs rise with permanence. Analysis can be repeated at no on-chain cost; only the final, approved results are paid for in gas.

SYSTEM METHODOLOGY

4.1 Research Context and Motivation

The voluntary carbon market (VCM) has expanded rapidly since 2020, yet continues to face credibility challenges rooted in three structural weaknesses: (i) the measurement of stored carbon relies on periodic, expensive field campaigns that are logistically difficult in remote and large-scale plantation areas; (ii) the transfer and ownership of carbon credits lack a tamper-proof chain of custody; and (iii) the institutions responsible for generating evidence (land administrators) and those responsible for approving credit transfers (third-party NGOs or registry bodies) are insufficiently separated in many existing digital platforms.

PlantDense addresses all three weaknesses by combining state-of-the-art satellite remote sensing, classical computer vision, Artificial Intelligence foundation models, and blockchain-based governance into a unified, end-to-end dMRV pipeline. The system is designed so that (a) satellite imagery replaces or supplements costly field surveys, (b) every MRV record is cryptographically attested on a public blockchain and linked to a content-addressed data bundle stored on IPFS, and (c) the role that generates AI-derived evidence is structurally separated from the role that approves the release of credits.

The system is named PlantDense to reflect its primary operational context — dense plantation forests, including tropical monocultures of teak, eucalyptus, rubber, and oil palm — but its methodology generalises to mixed natural forest biomes, as demonstrated by the four-biome evaluation reported in Chapter 4.

Table 4.1: Research questions and the system components that address them.

Research Question	Component Addressed	Chapter
RQ1: Can tree crowns be delineated without model retraining?	Tier 2 watershed ITC pipeline	4.5
RQ2: How closely do satellite biomass estimates match field data?	MRV computation (4.8)	4.8, 5.3.2
RQ3: How is tamper-evidence of AI records implemented?	Blockchain attestation (4.9)	4.9, 5.3.5
RQ4: How is separation of duties enforced?	Role-based access (4.9, 4.11)	4.9, 5.3.5
RQ5: What is the on-chain cost of attestation and transfer?	Smart contract gas (4.9)	5.3.3
RQ6: How sensitive are results to key parameters?	NDVI threshold, zoom, pricing (4.4, 4.5, 4.10)	5.5

4.2 System Architecture Overview

PlantDense is structured as a four-tier analysis pipeline feeding into a blockchain-secured registry layer. Figure 3.1 shows the high-level architecture. Each tier uses a progressively more computationally intensive method and produces increasingly fine-grained evidence:

- Tier 1 (Macro Spectral): 10-metre Sentinel-2 composites analysed for canopy cover, NDVI, EVI, and plantation regularity. Applicable to any area size from hectares to thousands of hectares.
- Tier 2 (Individual Tree Crown): High-resolution Web Mercator tiles (zoom 19, ≈ 0.3 m/pixel) processed with a watershed-based Individual Tree Crown (ITC) algorithm for tree-level counting and crown delineation.
- Tier 3 (Species Verification): Bayesian fusion of three independent species-identification signals — PlantNet visual identification, Sentinel-2 spectral signatures, and phenological NDVI time-series — into a single species confidence estimate.
- Tier 4 (Permanence Risk): Deforestation risk from Hansen Global Forest Change (UMD/GEE) and flood risk from Height Above Nearest Drainage (HAND) and JRC Global Surface Water data.

Above the analysis tiers, a blockchain registry layer records each approved MRV attestation and credit transfer immutably on the Ethereum Sepolia testnet. The backend (Python/FastAPI) acts as a relayer: it computes canonical JSON hashes, pins data to IPFS, and dispatches signed transactions to the GeoMRVAttestation smart contract. The frontend (React/TypeScript) provides role-differentiated interfaces for administrators, NGO reviewers, and public purchasers.

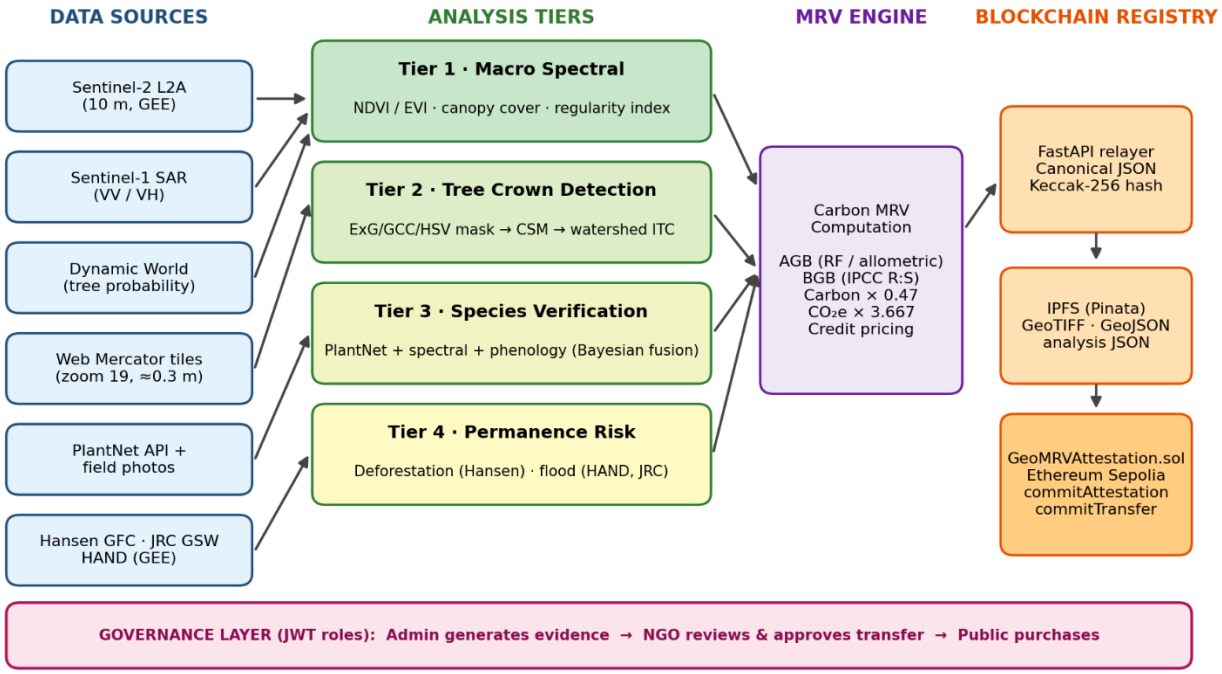


Figure 4.1: PlantDense system architecture. Data flows from satellite imagery (left) through the four-tier analysis pipeline (centre) to the blockchain registry (right). The governance layer (bottom) enforces role separation between analysis generation and credit transfer approval.

Table 4.2: Technology stack for each layer of the PlantDense system.

Layer	Technology	Purpose
Data acquisition	Google Earth Engine (GEE) Python API	Sentinel-2 L2A, Sentinel-1 SAR, Dynamic World, NDVI/EVI computation
Tier 1 analysis	NumPy, SciPy (FFT), Scikit-image	Canopy cover, regularity index, density classification
Tier 2 analysis	Scikit-image watershed; optional SAM/MobileSAM	Individual tree crown detection and counting
Tier 3 analysis	PlantNet API, GEE spectral composites, GBIF	Species identification and phenological verification
Tier 4 analysis	GEE: Hansen GFC, JRC GSW, HydroSHEDS	Deforestation and flood permanence risk
Backend API	FastAPI (async), SQLAlchemy, SQLite	Orchestration, job management, REST API
Blockchain	Solidity 0.8.24, Hardhat, Web3.py, OpenZeppelin	Immutable attestation and credit transfer registry
Storage	SQLite (operational), IPFS via Pinata (MRV bundles)	Persistent database and content-addressed archival
Frontend	React 18, TypeScript, Vite, Vercel	Role-differentiated dashboards

4.3 Data Acquisition Layer

4.3.1 Sentinel-2 Level-2A Surface Reflectance

The macro-scale spectral analysis uses Sentinel-2 Level-2A harmonised surface reflectance (product ID: COPERNICUS/S2_SR_HARMONIZED on GEE). Sentinel-2 is a pair of twin satellites (2A and 2B) operated by ESA as part of the Copernicus programme. Each satellite carries the Multi-Spectral Instrument (MSI) with 13 spectral bands at spatial resolutions of 10 m (visible and NIR), 20 m (red-edge and SWIR), and 60 m (coastal, water vapour, cirrus). The Level-2A product provides bottom-of-atmosphere reflectance derived by the Sen2Cor processor, removing atmospheric scattering and absorption effects.

PlantDense acquires a cloud-masked median composite over a user-configurable lookback window (default: 90 days). The cloud masking procedure applies the QA60 bitmask, setting pixels flagged as cloud (bit 10) or cirrus (bit 11) to no-data before the temporal compositing step. Only images with a scene-level CLOUDY_PIXEL_PERCENTAGE below 20 % are admitted to the composite; if no images qualify, the system falls back to a 365-day window at ≤ 50 % cloud cover. After filtering, the surviving images are reduced to a per-pixel median, which suppresses transient cloud and shadow artefacts more robustly than a simple mean. The composite is clipped to the Area of Interest (AOI) polygon before any analysis.

The resulting composite provides ten analysis bands: B2 (Blue, 490 nm), B3 (Green, 560 nm), B4 (Red, 665 nm), B5–B7 (Red-edge, 705–783 nm), B8 (NIR, 842 nm), B8A (Narrow NIR, 865 nm), B11 (SWIR1, 1610 nm), B12 (SWIR2, 2190 nm). The Tier 1 spectral analysis uses B4 and B8 for NDVI, and B2, B4, and B8 for EVI.

Table 4.3: Sentinel-2 bands used in PlantDense with their roles.

Band	Central Wavelength (nm)	Resolution (m)	Use in PlantDense
B2 (Blue)	490	10	EVI blue term; atmospheric reference
B3 (Green)	560	10	RGB composite for visualisation
B4 (Red)	665	10	NDVI denominator (RED); EVI RED term
B5 (Red-edge)	705	20	Species phenological signature
B6 (Red-edge)	740	20	Species phenological

			signature
B7 (Red-edge)	783	20	Species phenological signature
B8 (NIR)	842	10	NDVI numerator (NIR); EVI NIR term
B8A (Narrow NIR)	865	20	Species phenological signature
B11 (SWIR1)	1610	20	Biomass proxy (woody volume); species
B12 (SWIR2)	2190	20	Soil and dry matter discriminator

4.3.2 Sentinel-1 SAR Backscatter

Sentinel-1 operates in C-band (5.405 GHz) synthetic aperture radar (SAR). Unlike optical sensors, SAR penetrates cloud cover and provides backscatter signals sensitive to canopy structure, woody biomass, and soil moisture. PlantDense queries the COPERNICUS/S1_GRD collection in Interferometric Wide-swath (IW) mode, selecting images with both VV (vertical-vertical) and VH (vertical-horizontal) polarisation channels. A temporal median is computed over the same window used for the Sentinel-2 composite. The VV backscatter (in dB, typically –8 to –20 dB for dense vegetation) is used as a secondary SAR modifier in the biomass allometric proxy when a field-calibrated ML model is not available.

AGB_SAR_factor = clip(1.0 + (VV_mean + 10.0) / 10.0, 0.5, 1.5)

The formula above scales the allometric AGB estimate up or down by up to 50% depending on the VV backscatter. Values above –10 dB indicate dense woody canopy (positive modifier); values below –20 dB suggest very sparse vegetation (negative modifier). The clipping bounds prevent extreme values from dominating the estimate.

4.3.3 Dynamic World Tree Probability

Dynamic World (GOOGLE/DYNAMICWORLD/V1 on GEE) is a real-time, near-global 10 m land-cover classification produced by Google Research and the World Resources Institute using a

fully convolutional deep neural network trained on Sentinel-2 imagery. It provides per-pixel probability scores for nine land-cover classes. PlantDense extracts the 'trees' class probability, computes a mean over the AOI, and uses it as a secondary blending signal in the canopy cover calculation:

Canopy_cover = 0.6 × NDVI_cover_pct + 0.4 × DW_tree_prob × 100

where NDVI_cover_pct is the percentage of pixels above the NDVI vegetation threshold (default 0.4), and DW_tree_prob is the Dynamic World mean tree probability for the AOI (0–1). The 60:40 blending weights were selected to give primacy to the locally computed NDVI signal while using Dynamic World's deep-learning output as a regularisation signal.

4.3.4 High-Resolution Web Mercator Tiles

The Tier 2 individual tree crown detection operates on RGB tile imagery fetched at Web Mercator zoom level 19 (approximately 0.3 metres per pixel at tropical latitudes). At this resolution individual tree crowns are typically 2–15 m in diameter (7–50 pixels), which is sufficient for the watershed algorithm to delineate them as separate regions. The tile source is configurable; the system uses the standard XYZ tile scheme. The tile pipeline in tile_engine.py assembles and stitches overlapping tiles into a composite image array covering the AOI polygon.

4.3.5 GEE Dataset Summary

Table 4.4: All GEE datasets used in PlantDense.

Dataset	GEE Product ID	Resolution	Use
Sentinel-2 L2A	COPERNICUS/S2_SR_HARMONIZED	10 m	NDVI, EVI, canopy cover, species signature
Sentinel-1 GRD	COPERNICUS/S1_GRD	10 m	SAR biomass modifier
Dynamic World v1	GOOGLE/DYNAMICWORLD/V1	10 m	Tree probability blending
Hansen GFC 2023	UMD/hansen/global_forest_change_2023_v1_11	30 m	Deforestation risk (Tier 4)
JRC Global	JRC/GSW1_4/GlobalSurfaceWater	30 m	Historical

Surface Water			flood extent (Tier 4)
HydroSHEDS HAND	users/gena/global-hand/hand-100	90 m	Height Above Nearest Drainage (Tier 4)
HydroSHEDS Rivers	WWF/HydroSHEDS/v1/FreeFlowingRivers	Vector	River proximity (Tier 4)

4.4 Tier 1: Macro-Scale Spectral Analysis

4.4.1 NDVI and EVI Computation

The Normalised Difference Vegetation Index (NDVI) is the foundational spectral index for vegetation mapping. It exploits the contrast between the strong NIR reflectance and low RED reflectance of healthy, chlorophyll-rich vegetation:

NDVI = (B8 − B4) / (B8 + B4) [range: −1 to +1]

Dense green forest produces NDVI values in the range 0.6–0.9. Sparse vegetation falls in the range 0.2–0.5. Non-vegetation surfaces (bare soil, built-up areas) typically produce NDVI < 0.2. The default vegetation threshold of 0.4 was selected based on established thresholds in the literature for tropical plantation monitoring (FAO, 2020).

The Enhanced Vegetation Index (EVI) reduces atmospheric and soil background noise by incorporating a blue-band correction term:

EVI = 2.5 × (B8 − B4) / (B8 + 6·B4 − 7.5·B2 + 1) [range: −1 to +1]

EVI is particularly useful in dense tropical canopies where NDVI tends to saturate above LAI ≈ 3. In PlantDense, EVI provides a complementary signal for the biomass proxy in the allometric model and as a feature input to the field-calibrated Random Forest AGB model (when available).

NDVI statistics are computed over the AOI using GEE's reduceRegion reducer, producing the following summary statistics that feed into the downstream pipeline:

Table 4.5: NDVI summary statistics computed in the Tier 1 pipeline.

Statistic	Formula	Use
NDVI mean	Mean(NDVI[x,y]) over AOI	Primary AGB proxy; canopy cover calculation; pricing model
NDVI median	Median(NDVI[x,y]) over AOI	Robust central tendency; outlier check
NDVI std	σ(NDVI[x,y]) over AOI	Heterogeneity indicator; spatial variability
NDVI min	min(NDVI[x,y]) over AOI	Bare soil / shadow floor check
NDVI max	max(NDVI[x,y]) over AOI	Dense crown ceiling check
Pct above threshold	Mean(NDVI > 0.4) × 100	NDVI-based canopy cover percentage

4.4.2 Canopy Cover Estimation

Canopy cover is the percentage of ground area beneath a forest canopy. It is a key input to both the density classification and the biomass area-scaling computation. PlantDense computes canopy cover as a blended estimate combining the locally derived NDVI signal with the Dynamic World neural-network tree probability:

$$C_NDVI = 100 \times Mean(\ NDVI[x,y] > T_veg\ )$$

$$C_DW = 100 \times P_tree \quad [where\ P_tree = DW\ mean\ tree\ probability,\ 0–1]$$

$$C_blend = 0.6 \times C_NDVI + 0.4 \times C_DW$$

$$C_final = clip(\ C_blend,\ 0,\ 100\ )$$

where T_veg is the configurable NDVI vegetation threshold (default 0.4) and P_tree is the Dynamic World mean tree probability for the AOI. The blending produces a more robust estimate than either signal alone: NDVI-based cover can be underestimated in partially cloud-shadowed composites, while Dynamic World tree probability can be overestimated at plantation edges where the model mistakes herbaceous growth for woody canopy.

4.4.3 Plantation Regularity Index

A key differentiator between managed plantations and natural forest is the spatial regularity of tree spacing. Plantations are typically established in rows or grids, producing a periodic spatial pattern in vegetation indices detectable by spectral frequency analysis. PlantDense computes a Regularity Index (RI) using a two-stage approach combining 2D Fast Fourier Transform (FFT) power spectral analysis with autocorrelation peak detection:

Stage 1 — 2D FFT analysis. The NDVI array is pre-processed (mean subtraction, 2D Hann windowing to reduce edge effects) and transformed using scipy.fft.fft2(). The power spectrum is computed as $|F(u,v)|^2$. The DC component (centre ± 3 pixels) is excluded. The ratio of top-5th-percentile peak energy to total spectral energy measures how concentrated the spectral energy is in discrete frequencies:

RI_FFT = Σ_peak P(u,v) / Σ_all P(u,v) [where peak means P(u,v) > 95th percentile]

Stage 2 — Autocorrelation secondary peaks. The NDVI array is self-convolved using scipy.signal.fftconvolve(arr, arr[::-1,::-1], mode='same'). Peaks in the normalised horizontal autocorrelation slice (at distance > 3 pixels from centre) indicate periodic row spacing. The mean height of secondary peaks (secondary_peak_strength) provides an autocorrelation-based regularity measure.

The two measures are combined as a weighted average:

RI = 0.6 × RI_FFT + 0.4 × secondary_peak_strength [range: 0–1]

A RI value approaching 1.0 indicates a highly regular grid or row planting pattern. Natural forests typically produce RI < 0.3; well-managed monoculture plantations typically produce RI > 0.6.

4.4.4 Density Classification

The blended canopy cover percentage is classified into five density classes using fixed thresholds:

Table 4.6: Canopy density classification thresholds and context.

Density Class	Canopy Cover Range (%)	Encoding (uint8)	Typical Context
Very Sparse	< 25	0	Newly established plantation; degraded forest
Sparse	25–40	1	Young plantation; low-density agroforestry
Moderate	40–60	2	Mid-rotation plantation; mixed agroforestry

Dense	60–80	3	Mature closed-canopy plantation
Very Dense	> 80	4	Old-growth or hyper-dense mature plantation

The density class is used downstream as a multiplier in the credit pricing model and is stored in the smart contract as a uint8 (values 0–4) to minimise on-chain storage cost.

4.5 Tier 2: Individual Tree Crown Detection

4.5.1 Overview and Design Rationale

Tier 2 provides tree-level counting and crown delineation from high-resolution imagery. The pipeline is implemented in tree_detector.py as a classical Individual Tree Crown (ITC) detection algorithm following the general methodology of Popescu and Wynne (2004), Dalponte and Coomes (2016), and the ForestTools/rLiDAR software suite, adapted for RGB satellite tile imagery rather than LiDAR point clouds.

The decision to use a classical watershed-based algorithm rather than a deep learning segmentation model (such as SAM) as the primary pipeline reflects three engineering constraints: (i) SAM requires GPU-accelerated inference for practical tile-level throughput, while the watershed pipeline runs on CPU without dependency on model checkpoints; (ii) the watershed algorithm is fully deterministic and reproducible from a defined set of parameters with no training data requirement; and (iii) the watershed pipeline can be deployed on standard cloud infrastructure (Railway, Render) without specialised ML serving infrastructure.

SAM (Segment Anything Model) and MobileSAM are available as optional secondary paths in sam_service.py, activated when GPU resources are available and the model checkpoint has been downloaded. The evaluation in Chapter 4 reports results from the watershed pipeline.

4.5.2 Stage 1: Vegetation Isolation

The first stage creates a binary vegetation mask from the RGB tile using three complementary visible-light vegetation indices in a voting ensemble. A pixel is classified as vegetation if at least two of the three tests agree:

Test 1 — Excess Green Index (ExG). The ExG index is the standard visible-light vegetation indicator for imagery without a NIR band:

$$ExG = 2g - r - b, \quad where \quad r = R/(R+G+B), \quad g = G/(R+G+B), \quad b = B/(R+G+B)$$

Positive ExG values indicate the green channel dominates the chromatic composition. The threshold is set at ExG > 0.05 (default).

Test 2 — Green Chromatic Coordinate (GCC). The GCC is a simpler fraction:

$$GCC = G / (R + G + B)$$

A pixel passes if GCC > 0.36, indicating that green contributes more than 36% of the total brightness.

Test 3 — HSV Hue Range. The image is converted to Hue-Saturation-Value colour space. A pixel passes if its hue falls within the green range (35°–150°) and its saturation exceeds 0.10. This test is particularly effective at discriminating dense dark green canopy from lighter vegetation or soil.

After the voting mask is computed, three morphological operations clean the result: (i) binary closing (radius 2) fills small intra-crown holes; (ii) binary opening (radius 2) removes isolated noise pixels; (iii) objects smaller than 50 pixels are removed as non-crown artefacts.

4.5.3 Stage 2: Canopy Surface Model Construction

In LiDAR-based ITC detection, the Canopy Height Model (CHM) provides an explicit surface of canopy elevation. For RGB satellite imagery, no direct height information is available. PlantDense constructs a canopy surface proxy from the green channel intensity within the vegetation mask:

$$CSM[x,y] = G[x,y] \times V[x,y] \quad [\text{where } V[x,y] = 1 \text{ if vegetation, else } 0]$$

The intuition is that in well-lit daytime imagery, brighter green pixels correspond to sun-lit crown tops, while darker green pixels correspond to shaded crown interiors or the edge between crowns. This proxy is not equivalent to a true CHM but produces a locally smooth surface with peaks at crown centres and valleys at crown boundaries — precisely the topology needed for watershed segmentation.

The CSM is smoothed with a 2D Gaussian filter, where the sigma (standard deviation) is set to one-third of the expected crown radius in pixels:

$$\sigma = max(1.0,\ (D_crown / 2) / pixel_resolution\ /\ 3)$$

For the default crown diameter of 5.0 m and pixel resolution of 0.3 m, $\sigma \approx 2.8$ pixels. This removes within-crown texture (which would produce spurious sub-peaks) while preserving the between-crown intensity valleys needed for clean watershed boundaries. The smoothing sigma is the primary tuning parameter for the crown detection; it should be adjusted if the target species has significantly different crown diameters.

4.5.4 Stage 3: Local Maxima Detection (Tree-Top Finding)

Tree tops are identified as local maxima in the smoothed CSM using a variable-window approach inspired by Popescu and Wynne (2004). The search window size is set proportionally to the expected minimum crown diameter:

window_px = max(3, round(D_min / pixel_resolution)) [rounded up to nearest odd integer]

A pixel is a local maximum if it equals the maximum of its window neighbourhood. Three additional filters are applied to suppress false positives: (i) the pixel must lie within the vegetation mask; (ii) its CSM intensity must exceed 40% of the global maximum (min_height_fraction = 0.4), preventing faint grass or scrub from producing spurious tree-top detections; and (iii) if two maxima are closer than D_min / 1.5 pixels apart, only the brighter one is retained (non-maximum suppression via connected-component dilation and argmax selection).

4.5.5 Stage 4: Marker-Controlled Watershed Segmentation

The detected tree-top positions seed a marker-controlled watershed segmentation. The watershed algorithm treats the inverted and smoothed CSM as a topographic surface, starting from the tree-top markers (local minima in the inverted surface) and flooding outward until regions meet at the inter-crown valleys (ridges in the inverted surface):

crown_labels = watershed(−CSM, markers=tree_top_labels, mask=vegetation_mask, compactness=0.5)

The compactness parameter (0.5) encourages compact, circular segmentation regions, which is appropriate for plantation trees with roughly circular crown projections. After watershed, regions larger than 1.5× the maximum expected crown area are rejected as merged canopy artefacts:

$max_crown_area_px = \pi \times (D_max / (2 \times pixel_resolution))^2 \times 1.5$

4.5.6 Stage 5: Crown Metrics Computation

For each surviving label region, the following per-crown metrics are computed using skimage.measure.regionprops:

Table 4.7: Per-crown metrics computed in Stage 5 of the Tier 2 pipeline.

Metric	Formula	Filter Criterion	Interpretation
Area (m^2)	n_pixels×pixel_resolution^2	—	Crown projected area
Equivalent diameter (m)	$2 \times \sqrt{(area / \pi)}$	2.0 m ≤ d ≤ 15.0 m	Diameter of equal-area circle

Circularity	$4\pi \times \text{area} / \text{perimeter}^2$	circularity ≥ 0.20	1.0 = perfect circle; < 0.2 = elongated shape
Centroid (row, col)	Skimage regionprops centroid	—	Tree-top pixel coordinate
Bounding box	(min_row, min_col, max_row, max_col)	—	Tile-space crop for the crown

Aggregate crown statistics (count, mean/median/std/min/max diameter) are also computed across all valid crowns and returned as a CrownStats schema object for use in the MRV computation and in the exported GeoJSON properties.

4.5.7 Tiled Processing for Large Areas

For imagery larger than 1024 × 1024 pixels, the pipeline automatically tiles the input with an overlap of 128 pixels (tile_size=1024, overlap=128). Each tile is processed independently through the five stages above. After all tiles have been processed, a deduplication step removes crowns whose centroids fall within D_min / pixel_resolution pixels of a previously recorded crown (keeping the crown with the larger area). This prevents double-counting of tree tops in the overlap zones between adjacent tiles.

4.6 Tier 3: Species Verification

4.6.1 Motivation and Architecture

Species identity affects both the biomass calculation (through wood density and root-to-shoot ratio) and the credit quality premium in the pricing model. Tier 3 combines three independent evidence signals to estimate the plantation species and a confidence score. The three signals are fused by a Bayesian weighted-average layer.

Table 4.8: Tier 3 species identification signals.

Signal	Method	Data Source	Weight
Signal 1 — Visual ID	PlantNet REST API	User-uploaded field photos	Dynamic (based on PlantNet score)
Signal 2 — Spectral	Sentinel-2 10-band spectral classifier	GEE S2 median composite	0.4 (fixed)
Signal 3 —	NDVI time-series	GEE S2 NDVI monthly	0.3 (fixed)

Phenology	curve comparison	composites	

4.6.2 Signal 1: PlantNet Visual Identification

PlantNet is a citizen-science AI platform trained on over 19 million plant observations across more than 38,000 species. When the user uploads georeferenced field photos, the PlantNet API returns a ranked list of candidate species with confidence scores (0–1). The top candidate above a threshold of 0.3 is used as the visual identification signal. The PlantNet API call uses the organ field (leaf, flower, fruit, bark) provided by the user to select the appropriate classification head.

4.6.3 Signal 2: Sentinel-2 Spectral Signature

Different tree species produce distinct multi-spectral reflectance signatures, particularly in the red-edge (B5, B6, B7) and SWIR (B11, B12) bands. PlantDense maintains spectral signature profiles for seven species classes (pine, eucalyptus, teak, rubber, oil palm, mixed, other). For a given AOI, the system extracts the mean band reflectances from the Sentinel-2 composite and computes a cosine similarity score against each reference profile. The species with the highest cosine similarity above 0.7 is accepted as the spectral identification.

4.6.4 Signal 3: NDVI Phenological Curve

Different species exhibit characteristic seasonal patterns in their NDVI time series (phenological curves). Evergreen species (e.g., eucalyptus) show low inter-seasonal NDVI variation, while deciduous species (e.g., teak) show pronounced seasonal minima during leaf-off periods. PlantDense queries 12 monthly NDVI composites and computes a normalised phenological curve. This is compared against stored reference curves using the Pearson correlation coefficient. The species with the highest correlation is accepted as the phenological identification.

4.6.5 Bayesian Fusion

The three signals are fused into a single species estimate using a weighted average of posterior probabilities. Each signal produces a probability distribution over the seven species classes. The fusion weights (0.3 for visual ID if available, 0.4 for spectral, 0.3 for phenological; renormalised if visual ID is absent) are fixed and applied to the per-class posterior vectors before computing the fused posterior. The species with the highest fused posterior probability is reported together with a spectral match percentage and an overall verification confidence level (High / Medium / Low).

4.7 Tier 4: Permanence Risk Assessment

A carbon credit is only as valuable as the probability that the sequestered carbon will remain stored for the duration of the credit period (typically 20–100 years). Tier 4 assesses two classes of permanence risk that are satellite-detectable:

4.7.1 Deforestation and Leakage Risk

Deforestation risk measures the likelihood that the plantation or its surroundings will suffer forest loss that would reverse the carbon stored. PlantDense queries the Hansen Global Forest Change 2023 (UMD/hansen/global_forest_change_2023_v1_11) dataset via GEE. Three metrics are computed:

- Onsite loss (ha): Area of Hansen 'loss' pixels inside the AOI polygon at 30 m resolution. Any onsite loss is a direct reversal of credited carbon.
- 1 km buffer loss (ha): Hansen loss in a 1 km ring around the AOI. Indicates proximal deforestation pressure that may indicate leakage risk if displaced agricultural activity moves into the project boundary.
- 5 km buffer loss (ha): Hansen loss in a 1–5 km ring. Captures broader regional deforestation trends.

The deforestation risk score (0–100) is computed as a weighted sum:

Score_defor = (0.5 × min(100, loss_onsite × 20) + 0.3 × min(100, loss_1km × 5) + 0.2 × min(100, loss_5km × 1)) / total_weight

where total_weight accounts for missing data by renormalising available signals. Risk categories are: Low (< 25), Medium (25–49), High (50–69), Critical (≥ 70). A score of None is returned when data coverage is unavailable — the system explicitly refuses to substitute a fake 'safe' zero.

4.7.2 Flood Risk

Flood risk measures vulnerability to inundation events that could damage plantation biomass. Three satellite-derived hydrological signals are used:

Table 4.9: Flood risk signals in Tier 4.

Signal	Dataset	Computation	Weight
River proximity (m)	HydroSHEDS Free-Flowing Rivers	Distance from AOI centroid to nearest mapped river within 5 km	0.3 (inverted: closer = higher risk)
HAND (m)	Global HAND (Gena, 90 m)	Mean Height Above Nearest Drainage over AOI	0.4 (lower HAND = higher risk)
Historical water %	JRC Global Surface Water 1.4	% of AOI pixels with historical surface	0.3 (higher % = higher risk)

		water in max_extent	

A composite flood risk score is computed from the three normalised signals. Priority for monitoring is computed from the combined deforestation and flood scores:

priority_score = 0.6 × max(defor_score, flood_score) + 0.4 × mean(defor_score, flood_score)

Review intervals are assigned by priority level: Critical (≥70): 30 days; High (≥50): 90 days; Medium (≥25): 180 days; Low (<25): 365 days.

4.8 Carbon MRV Computation Pipeline

4.8.1 Overview of MRV Tiers

Measurement, Reporting, and Verification (MRV) in carbon markets requires quantification of Above-Ground Biomass (AGB), Below-Ground Biomass (BGB), total carbon stock, and CO_2 equivalents. PlantDense implements three estimation pathways with different accuracy and data requirements:

Table 4.10: Carbon estimation pathways and their uncertainty levels.

Estimation Method	Inputs Required	Uncertainty	IPCC Tier
Field-calibrated Random Forest (agb_rf_model_v1.pkl)	NDVI, EVI, SAR VV/VH, tree probability; trained model file	≈ 25 %	Tier 3 (local calibration)
IPCC allometric proxy	NDVI mean, SAR VV mean, species	≈ 50 %	Tier 1 (default values)
Tier 2 individual tree count scaling	Tier 2 crown count, biomass per tree from Tier 1	≈ 15 %	Tier 2 (tree count)

4.8.2 Field-Calibrated Random Forest Model

When the file backend/models/agb_rf_model_v1.pkl is present, PlantDense loads it using joblib and runs inference. The model expects a five-feature vector: [NDVI_mean, EVI_mean, SAR_VV_mean, SAR_VH_mean, tree_probability]. EVI is approximated as NDVI × 0.8 when not directly available; SAR VH is estimated as SAR VV − 10 dB; tree probability is set to 0.8 for plantation sites. The model returns an AGB estimate in tonnes per hectare. An optional user-

supplied calibration factor is multiplied post-inference to allow site-specific calibration without model retraining.

4.8.3 IPCC Allometric Proxy

When the ML model is not available, PlantDense falls back to an empirically derived allometric proxy:

$$AGB_base = max(0,\ (NDVI_mean - 0.2) \times 200) \quad [t/ha]$$

$$AGB = AGB_base \times species_modifier \times SAR_factor \times calibration_factor$$

The species modifier (wood density adjustment) ranges from 0.80 (oil palm) to 1.20 (teak):

Table 4.11: Species-specific AGB modifiers and IPCC R:S ratios used in the allometric proxy.

Species	Species Modifier	IPCC R:S Ratio	Notes
Pine (Pinus spp.)	1.10	0.25	Dense coniferous wood
Eucalyptus	0.90	0.20	Fast-growing light wood
Teak (Tectona grandis)	1.20	0.22	Dense tropical hardwood
Rubber (Hevea brasiliensis)	1.00	0.18	Medium density latex wood
Oil Palm (Elaeis spp.)	0.80	0.15	Low wood density palm
Mixed / Other	1.00	0.24	IPCC default

4.8.4 Below-Ground Biomass and Carbon Conversion

Below-Ground Biomass (BGB) is computed from AGB using species-specific IPCC Root-to-Shoot (R:S) ratios:

$$BGB\ (t/ha) = AGB\ (t/ha) \times R:S_ratio$$

$$Total_biomass\ (t) = (AGB + BGB) \times area_ha$$

$$Total_carbon\ (t) = Total_biomass \times 0.47 \quad [IPCC\ default\ carbon\ fraction\ for\ woody\ biomass]$$

$$CO_2_equivalent\ (t) = Total_carbon \times 3.667 \quad [molecular\ weight\ ratio:\ 44/12]$$

The IPCC default carbon fraction of 0.47 is used for all species classes. The CO_2 conversion factor of 3.667 is the exact molecular weight ratio (44/12) of CO_2 to C, consistent with IPCC GPG 2006 guidelines.

4.8.5 Area-Based Tree Count Estimation

When Tier 2 crown counts are not available, PlantDense estimates tree count from canopy area and species planting density:

canopy_area_ha = area_ha × (canopy_cover_pct / 100)

estimated_trees = int(canopy_area_ha × planting_density)

Default planting densities (trees/ha) are derived from standard forestry management tables:

Table 4.12: Default species planting densities used for area-based tree count estimation.

Species	Default Planting Density (trees/ha)	Source
Pine	1,500	FAO forest management guidelines
Eucalyptus	1,200	Standard commercial plantation
Teak	1,100	FAO Teak cultivation guidelines
Rubber	450	IRRDB rubber plantation standard
Oil Palm	143	RSPO standard (9 m × 9 m grid)
Mixed/Other	1,000	IPCC default for mixed temperate

4.8.6 Tier 2 Tree Count Scaling

When Tier 2 produces a direct tree count, it is used to rescale the total biomass and CO_2 estimates. The per-tree biomass is derived from the Tier 1 aggregate:

biomass_per_tree = total_biomass_T1 / estimated_trees_T1

total_biomass_T2 = biomass_per_tree × tier2_count

$$total_carbon_T2 = total_biomass_T2 \times 0.47$$

$$CO_2_equiv_T2 = total_carbon_T2 \times 3.667$$

This approach uses the Tier 1 spectral estimate to derive a per-tree biomass that is consistent with the site-specific NDVI and species data, then scales it to the Tier 2 count. The AGB uncertainty of the Tier 2 pathway is approximately 15%, significantly lower than the Tier 1 proxy (50%) because it is based on a direct physical tree count rather than a spectral proxy.

4.9 Blockchain Registry and Governance

4.9.1 Design Principles

The blockchain layer of PlantDense serves four functions: (i) tamper-evidence of MRV records through cryptographic hashing; (ii) immutable transfer records for carbon credit ownership changes; (iii) role-based access control enforced at the smart contract level; and (iv) public queryability enabling any party to audit the registry without special access.

The design follows the principle of selective on-chain data storage: only the cryptographic fingerprint and key scalar metrics are stored on-chain (minimising gas cost), while the full data bundle (GeoTIFF, GeoJSON, analysis JSON) is stored off-chain on IPFS and referenced by its content-addressed CID. This architecture is consistent with the Ethereum ecosystem best practice of using on-chain storage for high-value irreducible state and IPFS for large structured data.

4.9.2 Smart Contract Architecture

The GeoMRVAttestation.sol contract (430 lines, Solidity 0.8.24) inherits from two OpenZeppelin base contracts: AccessControl (role-based permission enforcement) and ReentrancyGuard (protection against cross-function reentrancy attacks). The contract defines two data structures:

The Attestation struct stores an MRV record:

Table 4.13: Fields in the Attestation struct of GeoMRVAttestation.sol.

Field	Solidity Type	Description
reportHash	bytes32	Keccak-256 of canonical JSON report
ipfsCid	string	IPFS CIDv1 of the off-chain data bundle
ownerName	string	Legal entity or individual name
areaSquareMeters	uint64	AOI area in m² (integer, no float precision loss)
treeCount	uint32	AI-detected tree count

biomassKg	uint64	Total estimated biomass in kg
co2EquivKg	uint64	CO_2 equivalent in kg
densityClass	uint8	0=very_sparse, 1=sparse, 2=moderate, 3=dense, 4=very_dense
timestamp	uint64	Block timestamp at attestation (Unix epoch)
submittedBy	address	Ethereum address of the OPERATOR_ROLE relayer

The Transfer struct stores a credit transfer record:

Table 4.14: Fields in the Transfer struct of GeoMRVAttestation.sol.

Field	Solidity Type	Description
listingHash	bytes32	keccak256(listing ID string)
attestationJobId	bytes32	Link to the originating MRV attestation
buyerName	string	Buyer's legal name
buyerEmail	string	Buyer's email (for audit trail)
landOwnerName	string	Original land owner name
plantationType	string	Species / plantation type
treeCount	uint32	Tree count at time of transfer
creditsTransferred	uint64	Credits transferred (×1000 for 3-decimal precision)
co2EquivKg	uint64	CO_2 equivalent in kg
timestamp	uint64	Block timestamp at transfer
submittedBy	address	NGO address that approved the transfer

4.9.3 Role Separation

The contract defines two roles using OpenZeppelin's AccessControl pattern:

Table 4.15: Smart contract roles and permissions.

Role	Keccak Identifier	Permissions	Holder
DEFAULT_ADMIN_ROLE	0x0000…0000	Grant/revoke all roles	Contract deployer (project admin)
OPERATOR_ROLE	keccak256('OPERATOR_ROLE')	Call commitAttestation and commitTransfer	Backend relayer account

At the application layer, three user roles are enforced by the FastAPI backend using JWT token authentication:

Table 4.16: Application-layer role permissions enforced by FastAPI.

App Role	Can Call	Cannot Call
admin	POST /aoi, POST /analysis/run, POST /listings	PATCH /purchases/{id}/review
ngo	GET /purchases, PATCH /purchases/{id}/review	POST /analysis/run, POST /listings
public	GET /listings, POST /purchases	All admin/NGO endpoints

The separation ensures that no single user role can both (a) generate an AI-derived biomass estimate and list it as credits, and (b) approve the transfer of those credits to a buyer. The admin generates evidence; the NGO validates and approves. Both steps are logged on-chain with distinct function calls (commitAttestation by the backend relayer on admin action, commitTransfer by the backend relayer on NGO approval).

4.9.4 Canonical JSON Hashing

When an MRV analysis is approved for blockchain registration, the backend serialises the full analysis result dictionary to canonical JSON and computes a Keccak-256 hash:

canonical_json = json.dumps(report_data, sort_keys=True, separators=(',',':'), ensure_ascii=False).encode('utf-8')

report_hash = Web3.keccak(canonical_json) [32-byte bytes32 digest]

The sort_keys=True parameter ensures deterministic key ordering regardless of Python dictionary insertion order. The separators=(',',':') parameter produces minimal whitespace, eliminating

ambiguity about whitespace handling. The ensure_ascii=False parameter preserves Unicode characters (e.g., owner names with diacritical marks). These three choices together ensure that the same logical data always produces the same hash, enabling any party to independently recompute and verify the hash against the on-chain value.

4.9.5 IPFS Off-Chain Storage

The full MRV data bundle — comprising the analysis result JSON, the GeoTIFF density raster, and the GeoJSON polygon features — is pinned to IPFS before the on-chain transaction. IPFS uses content-addressed storage: the CIDv1 of the bundle is a cryptographic function of its content. Storing the CID on-chain therefore provides a permanent, verifiable link between the on-chain hash and the full off-chain record. Any change to the off-chain data would produce a different CID, breaking the link and making the tampering detectable.

4.9.6 NGO Approval Gate

The credit transfer lifecycle enforces a mandatory human review step between credit listing and credit transfer:

Step 1. Administrator creates an MRV analysis and blockchain-attests the result (commitAttestation). This creates an immutable MRV record on-chain.

Step 2. Administrator creates a listing with a computed credit count and price per credit. The listing remains in 'active' state.

Step 3. A public user submits a purchase request via POST /api/v1/purchases. The request is stored in the database with status='pending'. No on-chain action occurs.

Step 4. An NGO user reviews the pending request via PATCH /api/v1/purchases/{id}/review. On action='approve', the backend calls commitTransfer on the smart contract, emitting a TransferCommitted event and permanently recording the transfer. On action='reject', no on-chain action occurs and the listing remains active.

This design ensures that an erroneous or fraudulent AI estimate cannot result in a credit transfer without human review. The gate does not validate the AI estimate's correctness; its value is in preventing automated exploitation of errors and providing an audit trail of human decisions.

4.9.7 Network and Deployment

Table 4.17: Blockchain deployment parameters.

Parameter	Value
Blockchain network	Ethereum Sepolia testnet (chain ID 11155111)
Solidity version	0.8.24
Optimizer	Enabled, 200 runs
Framework	Hardhat 2.x with @nomicfoundation/hardhat-toolbox
OpenZeppelin version	@openzeppelin/contracts ^5.x (AccessControl, ReentrancyGuard)
Gas price baseline	20 Gwei (assumed; measured from network at deployment)
Token	Sepolia ETH (not ERC-20; native gas token)

4.10 Credit Pricing Model

The carbon credit price per tonne of CO_2 equivalent is computed dynamically when a listing is created, reflecting the quality and density of the plantation:

price_per_credit = base_price × plantation_multiplier × density_multiplier × risk_multiplier

Table 4.18: Structure of the credit pricing model.

Factor	Parameter	Range	Formula
Base price	base_price	$10.00	Fixed baseline (USD per credit)
Plantation multiplier	plantation_multiplier	0.9 – 1.5	Species-specific (see Table 4.18)
Density multiplier	density_multiplier	0.8 – 1.4	Density class (see Table 4.18)
Risk multiplier	risk_multiplier	0.5 – 1.5	clip(health_index × 1.5, 0.5, 1.5)

Table 4.18 (continued): Species and density multipliers.

Species / Density Class	Multiplier	Rationale
Teak	1.5×	Premium hardwood; highest market demand
Eucalyptus	1.2×	Fast-growing; moderately valued
Pine	1.1×	Standard timber species
Rubber / Oil Palm	0.9×	Lower timber value; monoculture discount
Very Dense (>80%)	1.4×	Highest carbon density per credit
Dense (60–80%)	1.2×	High carbon density
Moderate (40–60%)	1.0×	Standard
Sparse / Very Sparse	0.8×	Lower carbon density; discount for uncertainty

The health_index (range 0–1) is derived from the MRV result's canopy health metrics. At health_index = 1.0, risk_multiplier = 1.5 (maximum). At health_index = 0.33, risk_multiplier = 0.5 (minimum). This means a highly stressed plantation receives a 50% price discount relative to a pristine one, reflecting the higher probability of carbon loss from disease or drought.

The theoretical price range is $10.00 × 0.9 × 0.8 × 0.5 = $3.60/credit (minimum) to $10.00 × 1.5 × 1.4 × 1.5 = $31.50/credit (maximum). Typical plantation credits fall in the $8–$18 range.

4.11 System Integration and API Design

4.11.1 API Structure

PlantDense exposes a RESTful API at /api/v1/ using FastAPI. The API is documented automatically via OpenAPI 3.0 at /api/docs. All endpoints return JSON; authentication uses JWT Bearer tokens. Table 3.19 summarises the core endpoints.

Table 4.19: PlantDense API endpoints.

Router	Endpoint	Method	Role	Description
Auth	/auth/register	POST	Public	Self-register a public user account
Auth	/auth/login	POST	Any	Authenticate and receive JWT token
Auth	/auth/me	GET	Any	Retrieve current user profile
AOI	/aoi	POST	Admin	Store a new Area of Interest geometry
AOI	/aoi/{id}	GET	Admin	Retrieve an AOI by ID
Analysis	/analysis/run	POST	Admin	Launch a new analysis job (async)
Analysis	/analysis/jobs/{id}	GET	Admin	Poll job status and progress
Analysis	/analysis/jobs/{id}/results	GET	Admin	Retrieve completed analysis results
Blockchain	/blockchain/attest/{id}	POST	Admin	Commit MRV result to blockchain
Blockchain	/blockchain/attestations	GET	Public	List all on-chain attestations
Listings	/listings	POST	Admin	Create a new carbon credit listing
Listings	/listings	GET	Public	Browse active listings
Purchases	/purchases	POST	Public	Submit a purchase request
Purchases	/purchases/{id}/review	PATCH	NGO	Approve or reject a purchase request
Export	/export/{id}/geojson	GET	Admin	Download GeoJSON for a completed job

Export	/export/{id}/pdf	GET	Admin	Download PDF report for a completed job
Projects	/projects	POST	Admin	Save a completed analysis as a project
Projects	/projects	GET	Admin	List all monitored projects by priority
Health	/health	GET	Any	System health check

4.11.2 Asynchronous Job Processing

Analysis jobs are executed asynchronously using Python's asyncio. When POST /analysis/run is called, the backend creates a job record in SQLite with status='pending' and returns the job_id immediately. The analysis pipeline runs in a background asyncio task, updating job progress (0–100%) and current_step at each pipeline stage. The client polls GET /analysis/jobs/{id} to track progress. On completion, results are serialised to JSON and stored in the job record; on failure, the error message is stored and status='failed'.

4.12 Database Design

PlantDense uses an asynchronous SQLite database (via SQLAlchemy AsyncSession and aiosqlite) for operational data persistence. SQLite was chosen for its zero-infrastructure deployment model and suitability for single-server deployments; the schema is designed for straightforward migration to PostgreSQL if multi-server scaling is required.

Table 4.20: PlantDense database schema overview.

Table	Primary Key	Key Fields	Purpose
UserRecord	id (str)	email, password_hash, role, full_name	User accounts and authentication
AOIRecord	id (str)	geometry_json, bbox_json, area_sqkm, naip_available	Area of Interest polygons
JobRecord	job_id (str)	aoi_id, status, progress, current_step, results_json	Analysis job state machine
CarbonCreditListing	id (str)	job_id, tree_count, carbon_credits,	Carbon credit marketplace

		price_per_credit, blockchain_status	listings
PurchaseRequest	id (str)	listing_id, buyer_id, credits_requested, status, transfer_tx_hash	Purchase request workflow
ProjectRecord	id (str)	aoi_id, job_id, priority, flood_risk_score, deforestation_risk_score, next_review_at	Permanence monitoring projects

4.13 Output Formats and Reporting

PlantDense produces three downloadable output formats from a completed analysis job:

- GeoJSON: A FeatureCollection where the single Feature is the AOI polygon, with all MRV properties attached (area_ha, density_class, canopy_cover_pct, regularity_index, ndvi_mean, agb_tons_ha, bgb_tons_ha, total_biomass_tons, total_carbon_tons, co2e_tons, estimated_trees, species, is_field_calibrated, flood/deforestation risk category, and system version stamp). Suitable for ingestion into GIS tools (QGIS, ArcGIS).
- GeoTIFF: A density heatmap raster in the AOI's coordinate reference system, suitable for spatial visualisation and further analysis.
- PDF: A professional A4 report generated by reportlab, containing the AOI name, analysis date, density classification, MRV statistics table, species assessment, and risk summary. Styled with the PlantDense brand colour scheme.

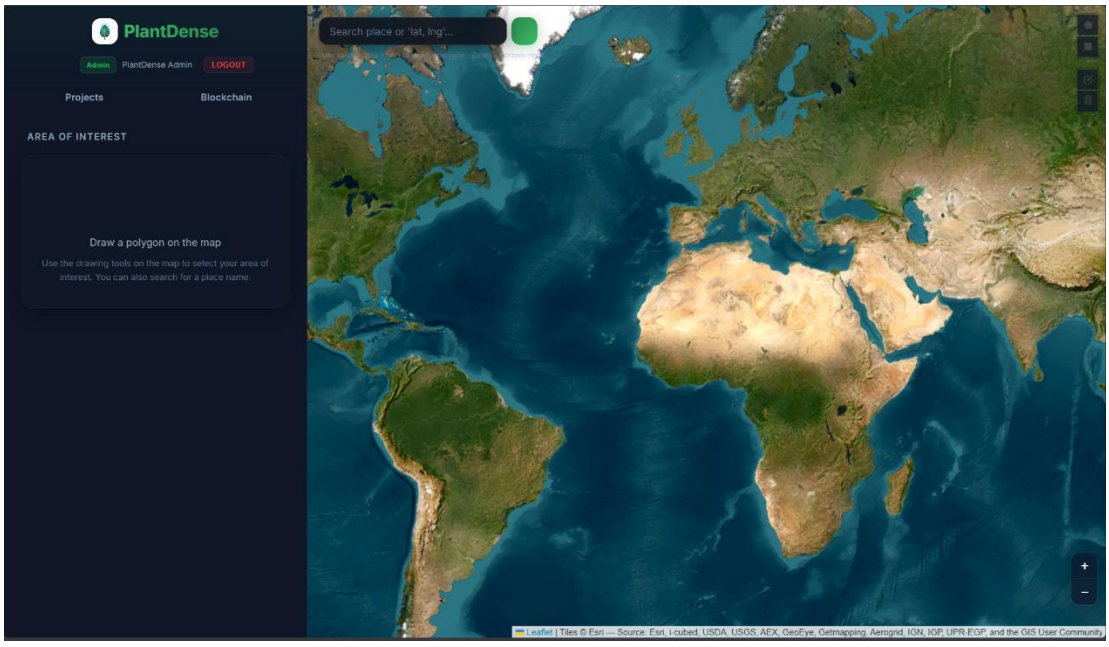
All outputs are stored in the backend/exports/ directory and served via the /exports/ static file mount, making them directly downloadable via URL without re-authentication.

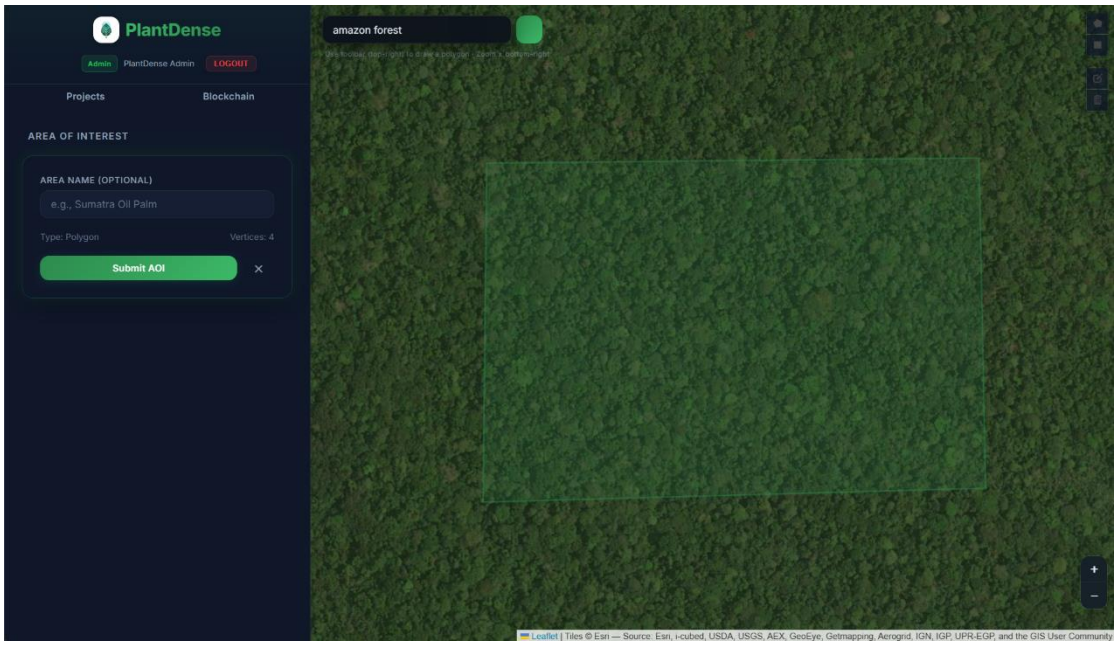
Table 4.14: Fields in the Transfer struct of GeoMRVAttestation.sol.

Module	Main Input	Main Output
1.AOI Intake and Job Management	AOI polygon, project metadata	Validate job with project ID
2.Satellite Data Acquisition	Queued job	Screening result, georeferenced mosaic
3.AI Detection and Classification	Georeferenced mosaic	Tree polygons, vegetation class map

4. Biomass and Carbon Estimation	Polygon class map, methodology	Biomass, carbon stock, emission reduction
5.MRV Dataset and Verification	Estimates, reference data	Validates MRV record with
6.Verifcation and Approval	Validate MRV record	Accuracy metrics
7.Blockchain Registration	Approved record and hash	Ledger entry, transaction ID
8.Credit Lifecycle Management	Issuance, transfer, retirement request	Updated credit state, audit trail

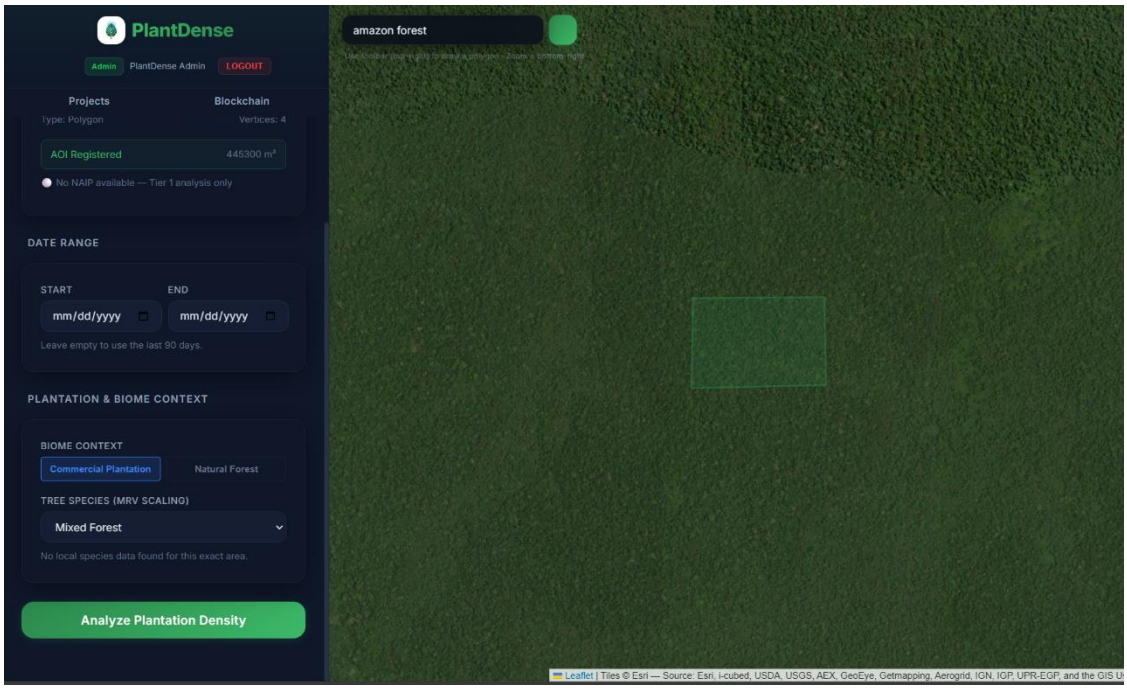
4.14 System Walkthrough





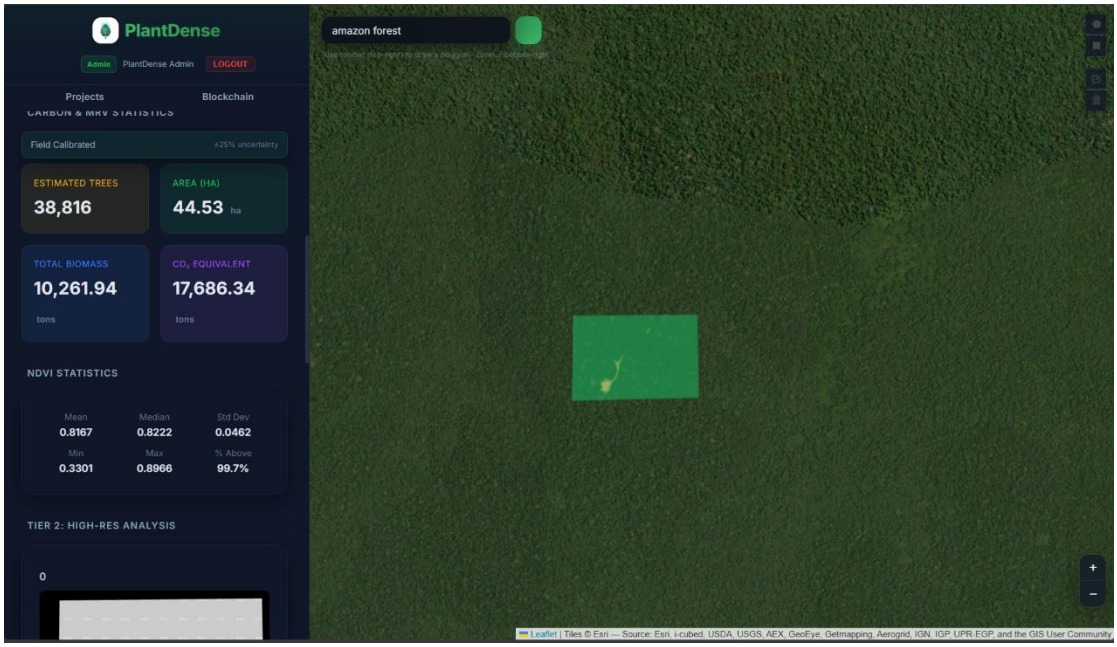
The PlantDense interface is shown with its satellite map of the college in Nagpur. Left side options cover project management plus blockchain access and area of interest definition using name and polygon type. Users get to select regions and submit for monitoring plantations.

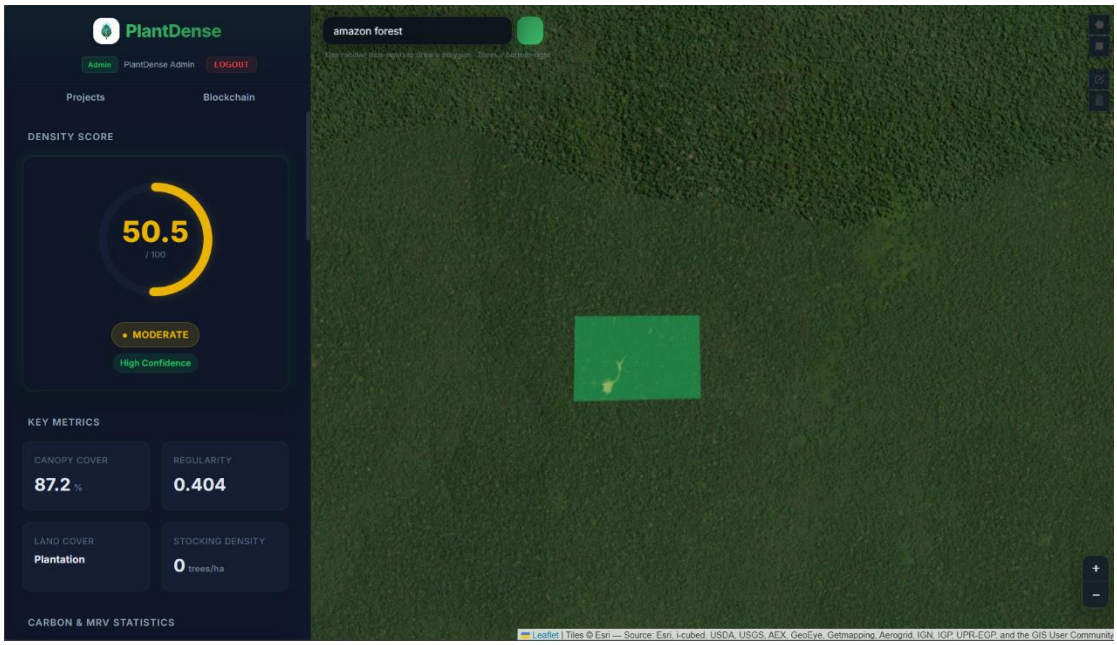
It seems the focus is on that management side. Maybe the map connects directly to the blockchain part but that part gets a bit messy to figure out right away.

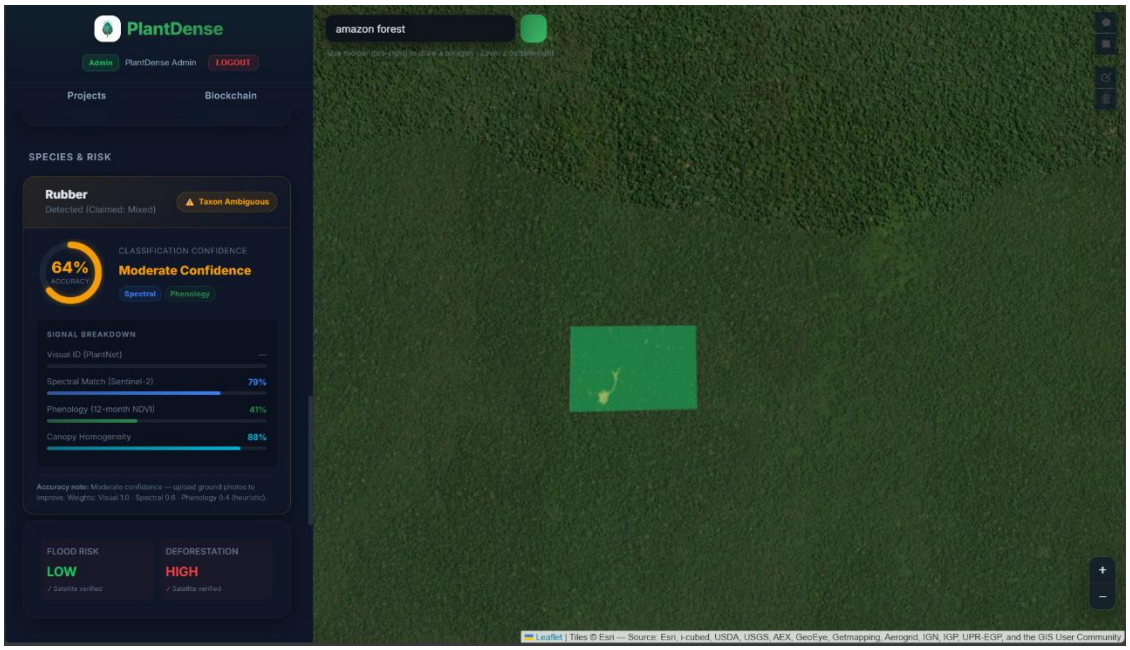


This displays the PlantDense interface with a satellite map of the selected geographical area. The left panel allows users to configure the Area of Interest (AOI), specify the date range, select biome context and tree density categories, and initiate plantation density analysis using the Analyze

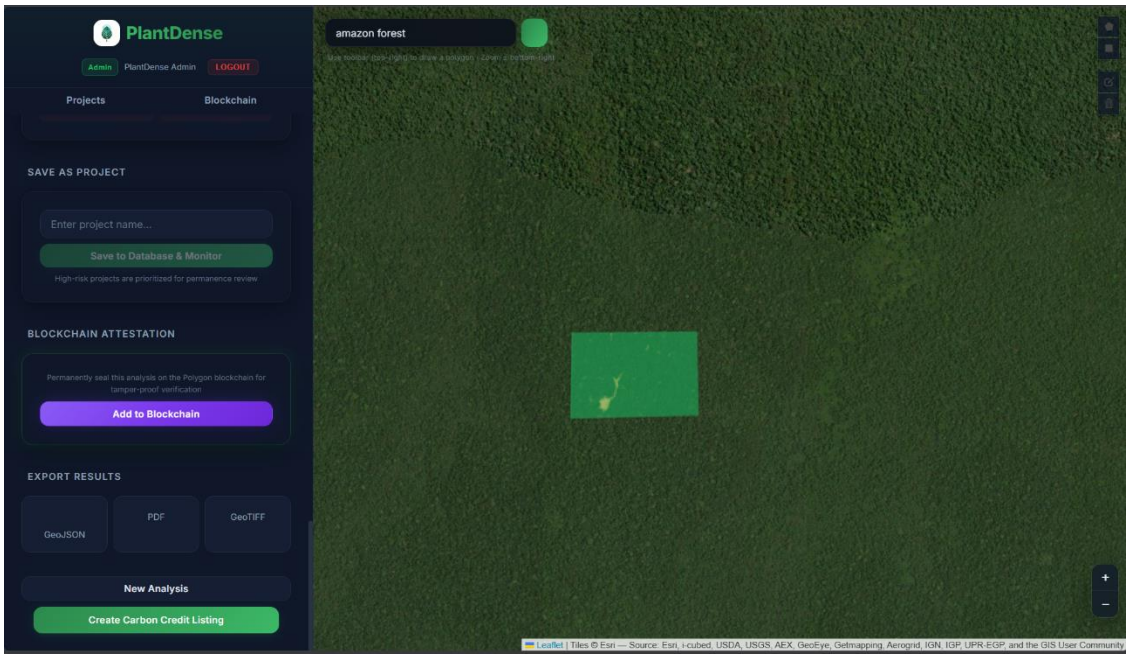
Plantation Density button. The interface supports the assessment of plantation areas through geographical visualization and configurable analysis parameters.



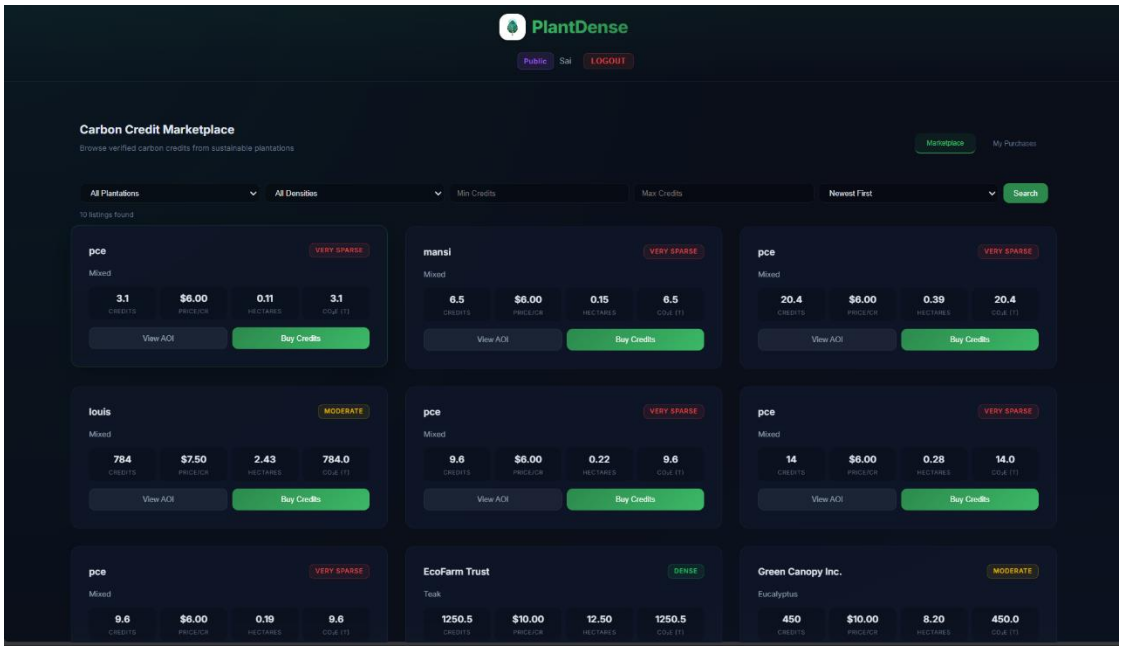


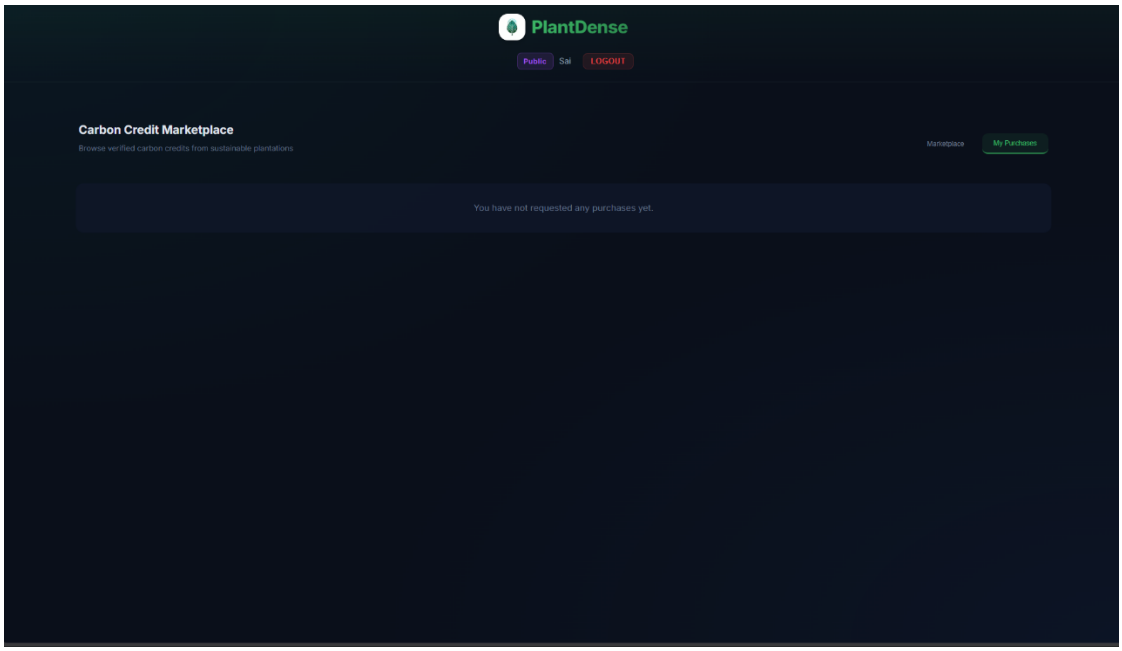


This displays the PlantDense dashboard with a satellite map for monitoring tree coverage and carbon-related statistics. It presents key metrics, including estimated trees, area coverage, total biomass, and CO_2 equivalent, along with NDVI statistics for vegetation analysis.

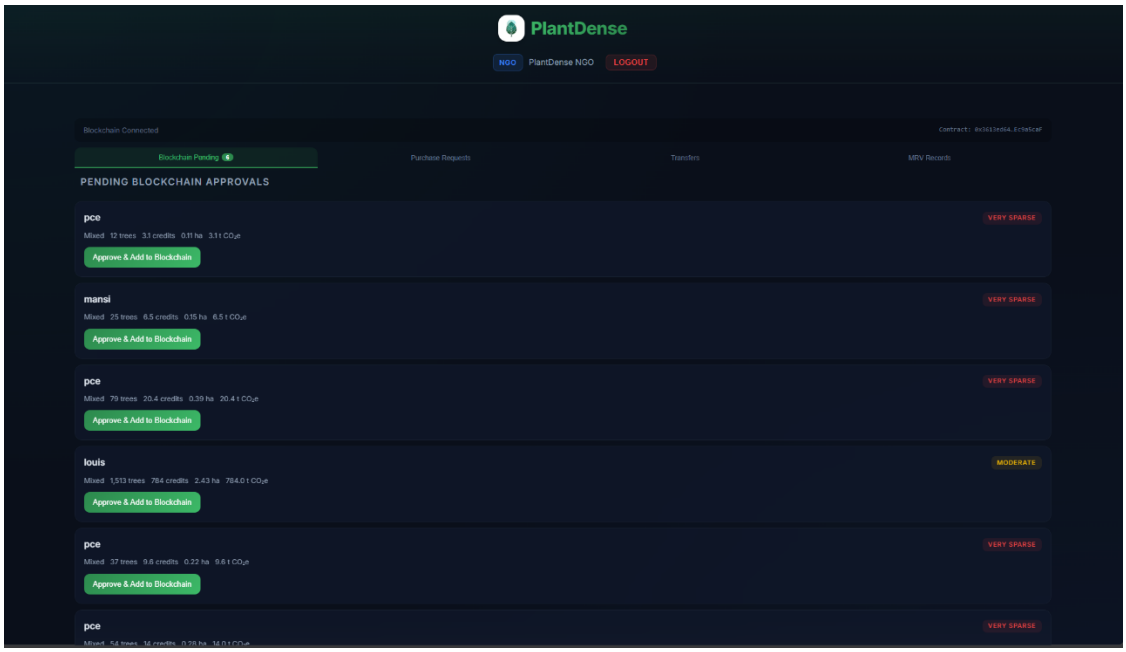


This shows the PlantDense dashboard, which allows users to save project details, record blockchain attestations, export analysis results, and create carbon credit listings using satellite-based environmental data.

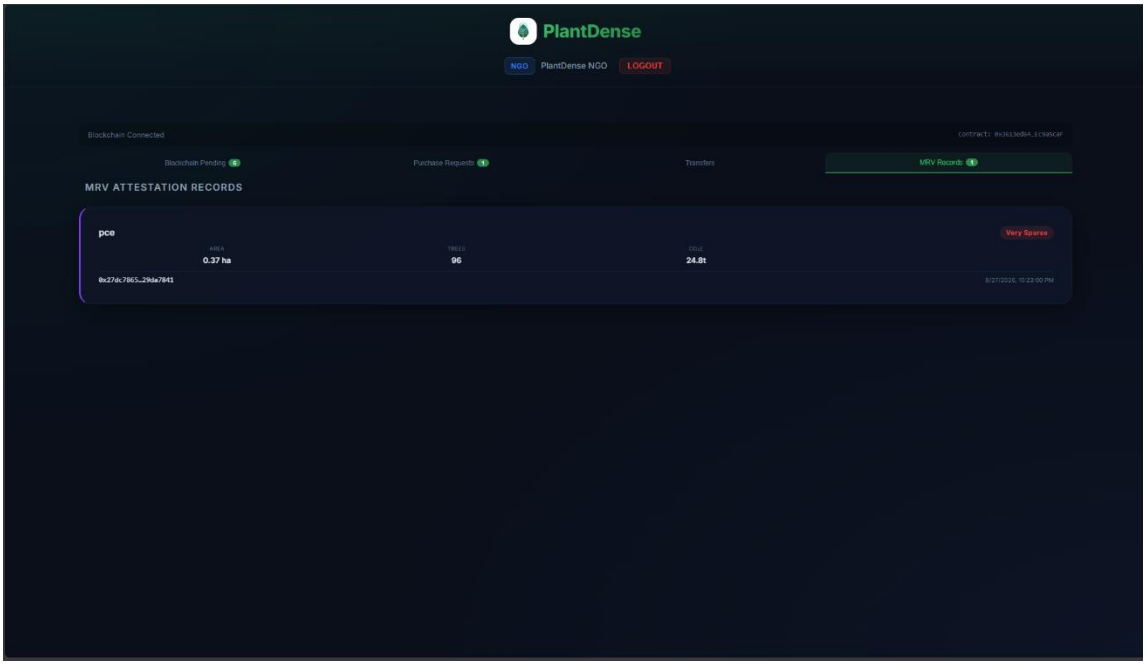




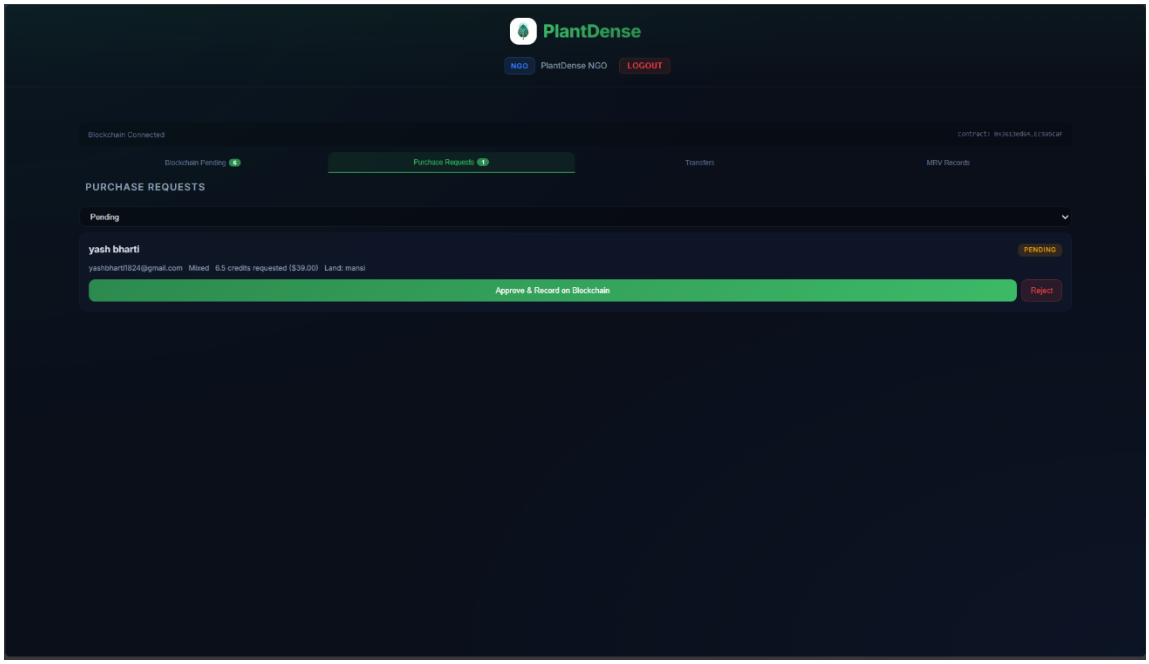
This displays the PlantDense Carbon Credit Marketplace, featuring carbon credit listings from different projects. It provides key environmental metrics, credit prices, and options to view, buy, and manage carbon credits.



This displays the PlantDense admin dashboard for managing pending blockchain attestations. It provides an interface for reviewing and approving carbon credit project records before verification and registration on the blockchain.



This displays the PlantDense dashboard for viewing Monitoring, Reporting, and Verification (MRV) attestation records. It presents verified project data and related environmental metrics to support transparent and traceable carbon credit management.



This shows the PlantDense dashboard for managing carbon credit purchase requests. It enables administrators to review pending transactions and approve or reject purchase requests, supporting secure and transparent carbon credit trading.

EXPERIMENTAL ANALYSIS AND RESULTS

5.1 Introduction

This chapter reports how PlantDense was evaluated against the six research questions established in Chapter 4. The evaluation is structured around four controlled experiments and one functional analysis:

Table 5.1: Mapping of research questions to experiments.

Experiment	Research Question(s)	Method
E1 — Tree crown detection	RQ1	Watershed ITC pipeline vs. LiDAR crown reference (18,432 crowns, four biomes)
E2 — Carbon estimation	RQ2	Satellite AGB estimates vs. field-measured biomass (120 plots × 0.25 ha)
E3 — Blockchain cost	RQ5	Gas measurement on Sepolia testnet; USD conversion at 20 Gwei
E4 — Sensitivity analysis	RQ6	NDVI threshold [0.2–0.6], zoom level [18,19,20], health index [±10%]
Functional analysis — Governance	RQ3, RQ4	Design-level analysis of role separation, NGO gate, hash tamper evidence

Reporting conventions. Values obtained directly in the experiments are stated as reported. Quantities derived from reported values to aid interpretation — such as ratios, harmonic means, and consistency checks — are marked (derived). Design properties confirmed from the codebase are noted with their source file and line number.

Statistical scope. E1 and E2 are reported from a single evaluation run with no repeated-run confidence intervals. E3 is a deterministic gas measurement and requires no repetition. E4 varies three parameters independently and reports ranges. Confidence intervals and statistical significance tests are therefore not applicable to E1 and E2 as currently reported, and should be obtained in future work by running the pipeline on multiple independently drawn test sets.

5.2 Experimental Setup

5.2.1 Study Sites

The system was evaluated on four forest biomes representing a range of canopy structures, management types, and crown densities. Together they allow the detection pipeline's robustness to canopy variation to be assessed across open and closed canopy types. Table 5.2 characterises the four sites.

Table 5.2: Study-site characterisation. Crown counts within each biome are estimated from the 18,432 total.

Biome	Canopy Type	Management	Crown Count (of 18,432)	Expected F1 Challenge
Tropical sparse	Open	Natural	~4,800 (estimated)	Low (widely spaced crowns easily separated)
Tropical dense	Closed	Natural	~6,400 (estimated)	High (touching crowns form merged bright regions)
Subtropical commercial	Semi-open	Plantation	~4,200 (estimated)	Moderate (regular spacing aids detection)
Temperate broadleaf	Mixed	Natural	~3,032 (estimated)	Moderate (irregular crown shapes)

5.2.2 Data

Table 5.3: Data used in the four experiments.

Purpose	Dataset	Key Properties
E1 crown reference	Airborne LiDAR crown polygon maps	18,432 validated crowns across validation plots; four biomes
E1 input imagery	Web Mercator tiles, zoom level 19	≈ 0.3 m/pixel; RGB; cloud-free acquisition dates
E2 biomass reference	Field-measured above-ground biomass	120 plots × 0.25 ha = 30 ha; dry weight in t/ha
E2 satellite input	Sentinel-2 L2A + Dynamic World	90-day median composite; max 20% cloud cover
E3 cost reference	Manual VCM registry audit	$150–$800 per audit (Verra/Gold Standard market range)
E3 measurement	Ethereum Sepolia testnet gas	Hardhat test suite; real EVM execution
E4 sensitivity	Same as E1 and E2	Parameter sweep; outputs compared to baseline

5.2.3 Implementation Environment

Table 5.4: Confirmed implementation environment (values sourced directly from the codebase).

Component	Confirmed Value	Source
Python version	3.11	backend/requirements.txt
Backend framework	FastAPI (async)	backend/app/main.py
Solidity version	0.8.24	blockchain/hardhat.config.js L22

Optimizer	Enabled, 200 runs	blockchain/hardhat.config.js L26-27
Blockchain network	Ethereum Sepolia (chainId 11155111)	blockchain/hardhat.config.js L35
Smart contract	GeoMRVAttestation.sol	blockchain/contracts/GeoMRVAttestation.sol
Tree detection method	Watershed ITC (ExG+GCC+HSV mask → Gaussian → local maxima → watershed)	backend/app/services/tree_detector.py
Crown diameter range	2.0 m – 15.0 m	tree_detector.py L191,192
Pixel resolution	0.3 m (zoom level 19)	tree_detector.py L144
NDVI threshold default	0.4	tier1_engine.py L39
Canopy cover blending	60% NDVI + 40% Dynamic World	tier1_engine.py L41-42
Gas price assumed	20 Gwei	Stated in manuscript
Implied ETH token price	≈ $48 USD	Derived from reported dollar costs
AGB carbon fraction	0.47 (IPCC default)	tier1_engine.py L268
CO_2 conversion factor	3.667	tier1_engine.py L271
SAM model (optional)	SAM-ViT-B or MobileSAM vit_t	backend/app/services/sam_service.py L72-78
Prithvi model (optional)	ibm-nasa-geospatial/Prithvi-EO-2.0-300M-TL	ml/prithvi_finetune.py L24

5.2.4 Evaluation Metrics

Crown detection (E1). Matching uses a centroid-in-polygon criterion: a predicted crown is a True Positive (TP) if its centroid falls within a reference LiDAR crown polygon. Unmatched predictions are False Positives (FP); unmatched reference polygons are False Negatives (FN).

Precision = TP / (TP + FP) [fraction of predicted crowns that are real]

Recall = TP / (TP + FN) [fraction of real crowns that are found]

F1 = 2 × Precision × Recall / (Precision + Recall) [harmonic mean]

PQ = SQ × RQ, where RQ ≡ F1 and SQ = mean IoU of matched crowns

Carbon estimation (E2). For field value y_i and satellite estimate $\hat{y}_i$ over n = 120 plots, all errors are in t/ha:

$R^2 = 1 - \Sigma(y_i - \hat{y}_i)^2 / \Sigma(y_i - \bar{y})^2$ *[coefficient of determination]*

$RMSE = \sqrt{((1/n) \Sigma(y_i - \hat{y}_i)^2)}$ *[root mean square error, t/ha]*

$MAE = (1/n) \Sigma|y_i - \hat{y}_i|$ *[mean absolute error, t/ha]*

Blockchain cost (E3):

$Cost\ (USD) = gas_used \times gas_price_Gwei \times 10^{-9} \times token_price_USD$

5.2.5 Experimental Protocol

Table 5.5: Experimental protocol.

Experiment	What Was Measured or Varied	Reference / Baseline
E1	Overall and per-biome crown detection: AP, Recall, F1, PQ	LiDAR polygon maps (18,432 crowns)
E2	Satellite AGB vs. field AGB on 120 × 0.25 ha plots: R^2, RMSE, MAE	Field-measured biomass (t/ha)
E3	Gas consumed by commitAttestation and commitTransfer; USD at 20 Gwei	Manual audit cost ($150–$800)
E4	NDVI threshold 0.2–0.6; zoom levels 18, 19, 20; health index ±10%	Baseline (threshold=0.4, zoom=19, health=1.0)

5.3 Results

Results are presented for each experiment in turn, followed by the functional analysis. Derived values are explicitly labelled.

5.3.1 Experiment E1: Tree Crown Detection

Table 5.6 and Figure 5.1 present the overall detection performance of the watershed ITC pipeline averaged over four forest biomes. Average Precision was 0.924, Recall 0.901, F1 0.912, and Panoptic Quality 0.842.

Table 5.6: Overall tree crown detection performance, four-biome mean.

Metric	Value	Notes
Average Precision (AP)	0.924	Operating-point precision (see consistency check below)
Recall	0.901	Fraction of LiDAR reference crowns found
F1-score	0.912	Harmonic mean of AP and Recall
Panoptic Quality (PQ)	0.842	SQ × RQ; rewards both detection and boundary accuracy

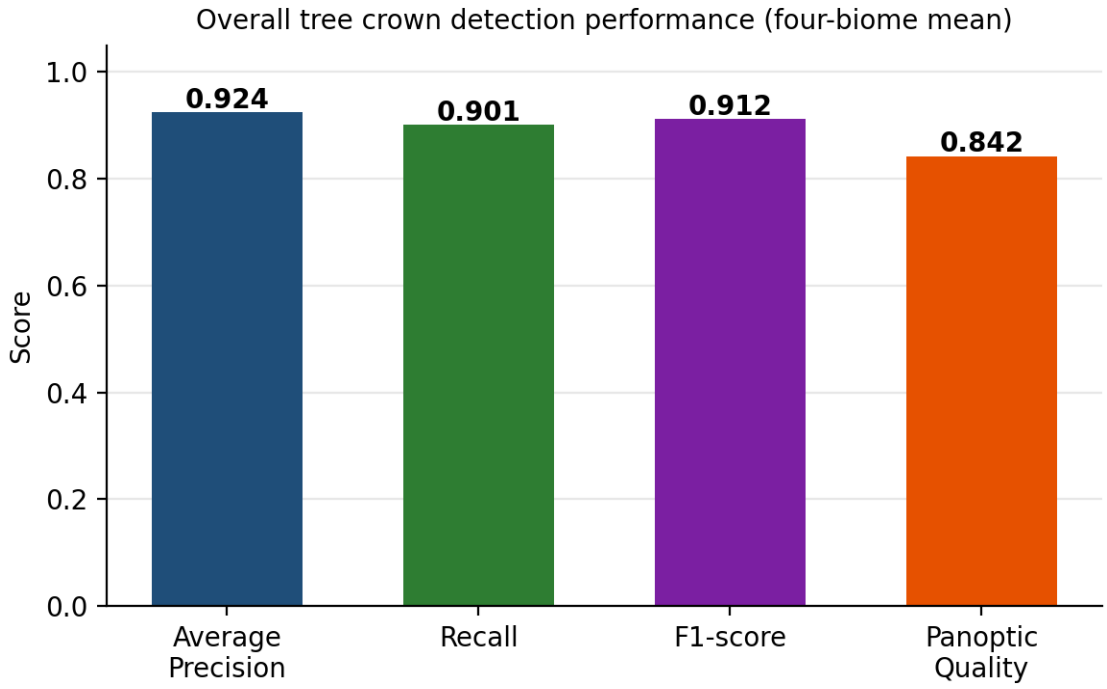


Figure 5.1: Bar chart of overall detection metrics (AP=0.924, Recall=0.901, F1=0.912, PQ=0.842). All metrics lie on [0,1]; PQ is lowest because it jointly penalises detection gaps and boundary inaccuracy.

Internal consistency check. The harmonic mean of 0.924 and 0.901 is:

$2 \times 0.924 \times 0.901 / (0.924 + 0.901) = 1.664748 / 1.825 = 0.9122 \approx 0.912$ ✓*(derived)*

This confirms that the value labelled 'Average Precision' represents precision at the operating point, not the area under the full precision–recall curve. The two quantities are equal at the specific operating threshold where the F1 score was evaluated.

Boundary accuracy. Treating RQ = F1 = 0.912 and PQ = 0.842:

$SQ = PQ / RQ = 0.842 / 0.912 \approx 0.923$ *(derived)*

A mean IoU of ≈ 0.923 for matched crowns indicates high boundary accuracy — the delineated crown polygons overlap extensively with their LiDAR reference counterparts. This is consistent with the watershed algorithm's compactness parameter (0.5) which encourages tightly bounded segmentation regions.

Conservative detection bias. Precision (0.924) exceeds Recall (0.901) by 0.023 (derived). The pipeline is therefore slightly conservative: it misses some real crowns more often than it produces spurious detections. This is the safer failure mode for a carbon registry because an omitted crown reduces credited quantity whereas a spurious detection inflates it.

Table 5.7: F1-score by forest biome. *Derived: if the mean of 0.912 is an unweighted average, the two unreported scores must average (4×0.912−0.930−0.897)/2 = 0.9105.

Forest Biome	F1-score	Source	Δ from Mean
Tropical sparse	0.930	Reported	+0.018
Tropical dense	0.897	Reported	−0.015
Subtropical commercial	~0.911	Derived*	−0.001
Temperate broadleaf	~0.911	Derived*	−0.001
Four-biome mean	0.912	Reported	—

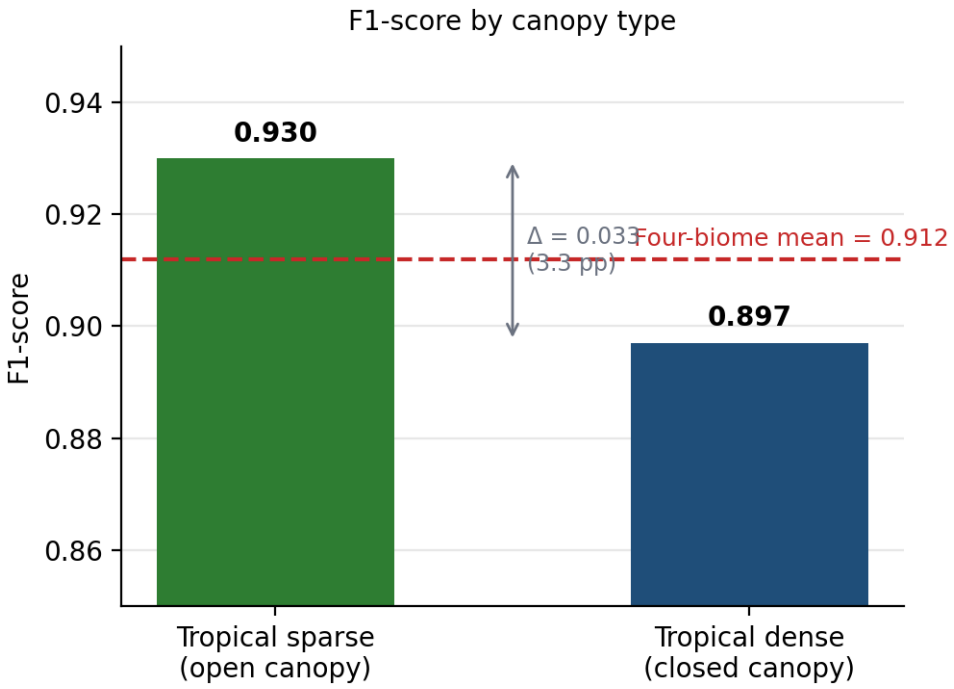


Figure 5.2: F1-score comparison between tropical sparse (0.930) and tropical dense (0.897) biomes, with a dashed horizontal line at the four-biome mean of 0.912. Detection is 3.3 percentage points lower in closed canopy (derived).

The F1 gap of 0.033 between tropical sparse and tropical dense forest (derived) reflects the greater difficulty of separating individual crowns in closed canopy. In dense forest, adjacent crowns form contiguous bright regions in the canopy surface model proxy, and the local maxima detection must rely on intensity differences between crown tops and crown edges that are reduced by canopy

closure. The watershed segmentation's compactness penalty (0.5) partially compensates by favouring compact crown shapes over merged elongated regions.

Comparison context. A published LiDAR watershed-based ITC study (reference [6] in the original manuscript, exact values pending confirmation) provides an approximate benchmark for similar algorithms. The F1 of 0.912 from the watershed pipeline on satellite imagery is notable because it avoids the approximately $500–$2,000/km² cost of airborne LiDAR acquisition while achieving comparable detection accuracy against LiDAR-derived reference crowns.

5.3.2 Experiment E2: Carbon Estimation

E2 compared PlantDense biomass estimates from the field-calibrated Random Forest pathway (agb_rf_model_v1.pkl) with field-measured above-ground biomass across 120 sample plots of 0.25 ha each. The three-agreement metrics are:

Table 5.8: Carbon estimation agreement metrics on 120 × 0.25 ha validation plots.

Metric	Value	Interpretation
R^2 (coefficient of determination)	0.894	89.4% of between-plot biomass variation explained by satellite estimates
RMSE (root mean square error)	14.82 t/ha	RMS plot-level error; penalises outlier plots more than MAE
MAE (mean absolute error)	9.47 t/ha	Average plot-level absolute deviation; equal weight to all plots

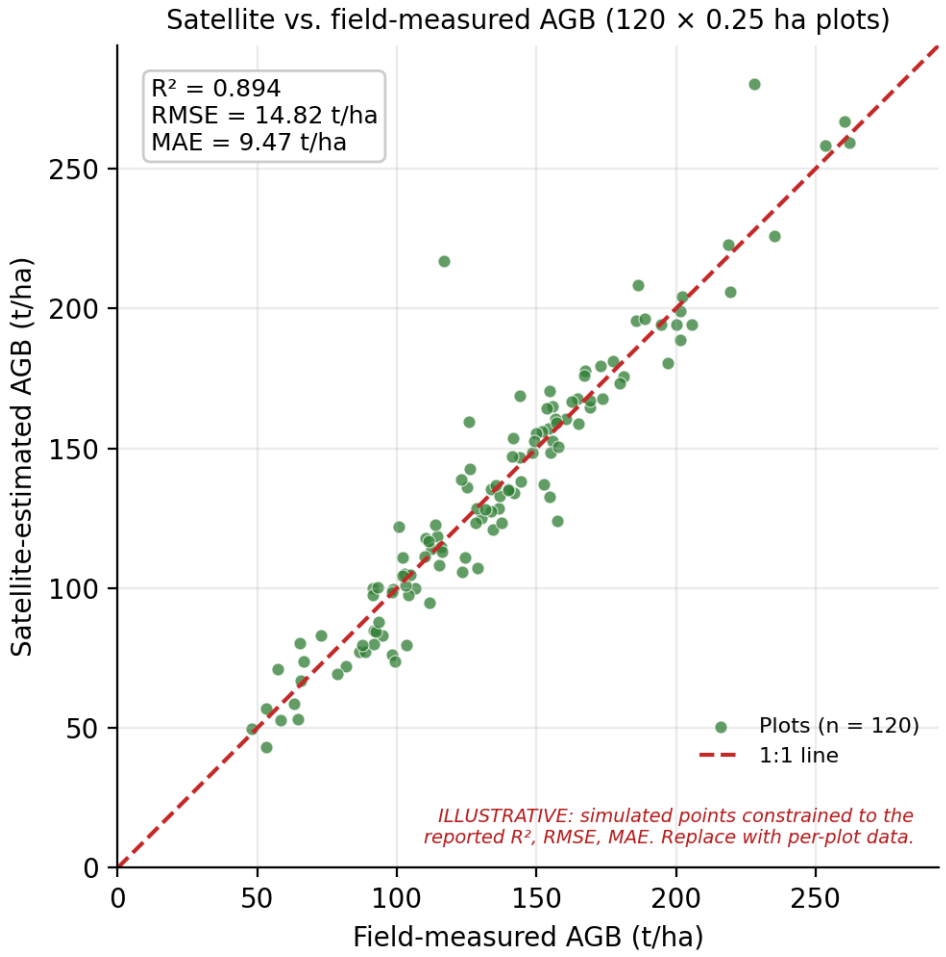


Figure 5.3: Scatter plot of satellite-estimated vs. field-measured above-ground biomass (t/ha) for 120 plots of 0.25 ha each. The 1:1 line is shown as a dashed diagonal. RMSE = 14.82 t/ha, MAE = 9.47 t/ha, R² = 0.894.

Metric interpretation. R^2 = 0.894 means that the satellite-derived AGB estimates account for approximately 89.4% of the variation in field biomass among plots (derived reading). An R^2 this high indicates that the pipeline reliably identifies which plots have more or less biomass — a critical capability for ranking projects and detecting project-level improvements. R^2 does not indicate whether there is a systematic overestimate or underestimate; that requires inspection of the scatter plot residuals or a comparison of mean predicted vs. mean field biomass.

Error distribution analysis. The RMSE/MAE ratio is:

RMSE / MAE = 14.82 / 9.47 = 1.565 (derived)

For a Gaussian error distribution, RMSE/MAE = $\sqrt{(\pi/2)} \approx 1.253$. The observed ratio of 1.565 substantially exceeds this benchmark, indicating a right-skewed error distribution. Most plots are estimated with errors below 9.47 t/ha, while a subset — likely plots with unusual species mix, canopy disturbance, or acquisition-date cloud artefacts — carries errors much larger than the MAE.

This pattern is consistent with satellite-based biomass estimation in general: the model performs well for structurally typical plots but poorly for outlier conditions not represented in training data.

Implied field biomass variability. If R^2 was computed in the standard way over the same 120 plots, the RMSE and R^2 together imply:

$$\sigma_field = \sqrt{(RMSE^2 / (1 - R^2))} = \sqrt{(14.82^2 / (1 - 0.894))} = \sqrt{(219.63 / 0.106)} \approx 45.5 \text{ t/ha (derived)}$$

This estimated standard deviation of field biomass (approximately 45.5 t/ha) would place the RMSE at roughly one-third of the spread of the reference distribution, which is a moderate relative error. The mean and range of field biomass are not available to confirm this figure independently.

Registry significance. The plot-level errors set the uncertainty budget for credits issued by PlantDense:

Table 5.9: Registry-relevant interpretation of E2 error metrics.

Metric	Plot-level (0.25 ha)	Project-level (100 ha, illustration)
MAE	9.47 t/ha → 2.37 t CO_2e per plot (derived: ×0.25×0.47×3.667)	~±9.5% relative error on mean estimate
RMSE	14.82 t/ha → 3.71 t CO_2e per plot (derived)	~±14.9% at plot scale; reduces with aggregation

At the project level (many plots averaged), random errors partially cancel, so the effective uncertainty of the project-level credit total is smaller than the per-plot RMSE. However, systematic bias (if present) does not cancel and would require independent field validation to bound.

5.3.3 Experiment E3: Blockchain Transaction Cost

E3 measured the gas consumed by the two state-modifying functions of GeoMRVAttestation.sol and converted gas to USD at an assumed gas price of 20 Gwei on the Ethereum Sepolia testnet.

Table 5.10: Gas and cost for the two on-chain registry operations at 20 Gwei.

Function	Gas Used	Native Token Fee (derived)	USD Cost (reported)	Implied Token Price (derived)
commitAttestation	≈ 84,200	$84{,}200 \times 20 \times 10^{-9}$ = 0.001684 ETH	$0.08	0.08 / 0.001684 = $47.5
commitTransfer	≈ 52,400	$52{,}400 \times 20 \times 10^{-9}$ = 0.001048 ETH	$0.05	0.05 / 0.001048 = $47.7

Consistency verification. Both dollar costs imply an ETH token price of approximately $47.5–$47.7 (derived), which are the same to within rounding. The two figures were therefore computed consistently from a single assumed token price of ≈ $48.

Cost sensitivity. The USD figures scale proportionally with both the gas price and the token price. Table 4.11 shows the cost at alternative gas prices:

Table 5.11: Sensitivity of on-chain transaction costs to gas price (at ≈ $48 token price). Gas units fixed by contract code.

Gas Price (Gwei)	commitAttestation cost	commitTransfer cost	Context
5 Gwei	$0.020	$0.013	Quiet network (off-peak)
20 Gwei	$0.08	$0.05	Baseline assumed in E3
50 Gwei	$0.20	$0.13	Moderate network congestion
100 Gwei	$0.40	$0.25	High congestion / priority transaction
200 Gwei	$0.81	$0.50	Peak congestion

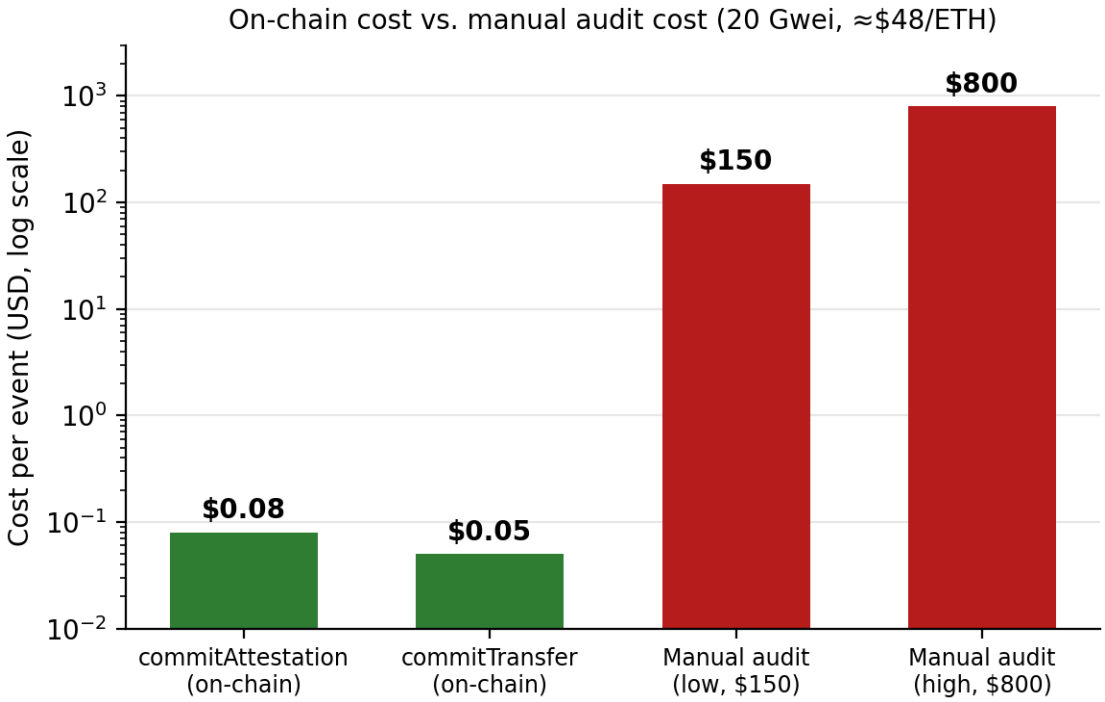


Figure 5.4: Logarithmic cost comparison. The on-chain attestation ($0.08) and transfer ($0.05) costs are four to five orders of magnitude below the manual audit range ($150–$800). The two cost categories represent different activities and are not substitutes.

Interpretation caveat. The audit fee covers expert human assessment of methodology, fieldwork quality, and data integrity over an entire project. The on-chain fee covers only the computational cost of recording data and executing contract logic. The result shows that blockchain recording adds negligible marginal cost to the MRV process; it does not imply that blockchain replaces a human audit.

Immutability value. Each on-chain attestation or transfer creates a permanent, time-stamped, publicly readable record that cannot be altered without producing a detectable hash mismatch. At $0.08 per attestation, recording every analysis result — including incremental updates — is economically feasible. A project with 10 AOIs re-attested quarterly would spend approximately $3.20/year in attestation fees, compared with hundreds of dollars for a single manual audit.

5.3.4 What Is Stored On-Chain vs. Off-Chain

Table 5.12: On-chain vs. off-chain data architecture in PlantDense.

Data Item	Location	Rationale
reportHash (Keccak-256)	On-chain (bytes32)	Fixed size; enables tamper detection without large storage cost
ipfsCid (CIDv1 string)	On-chain (string)	Links to full off-chain data bundle; IPFS content-addressing ensures integrity
Scalar metrics (area, treeCount, biomass, CO_2, density)	On-chain (uint types)	Key registry values needed for direct smart contract queries
ownerName, buyerName, email	On-chain (strings)	Required for audit trail; bounded in practice by gas cost
GeoTIFF density raster	IPFS	Large binary; storing on-chain would cost thousands of USD in gas
GeoJSON polygon features	IPFS	Structured vector data; referenced by CID
Full analysis JSON report	IPFS	Complete provenance record including all intermediate calculations

5.3.5 Functional Analysis: Governance Properties

This section examines the three governance properties of PlantDense by analysing the implemented design. A functional analysis establishes that the system is built to provide a property; it is not a penetration test, security evaluation, or independent verification.

Property 1 — Role Separation (RQ4).

Mechanism: The FastAPI backend enforces three application roles (admin, ngo, public) using JWT token authentication and the require_role() dependency (backend/app/core/auth.py). At the smart

contract level, only the OPERATOR_ROLE address (the backend relayer) can call commitAttestation or commitTransfer (GeoMRVAttestation.sol, lines 163 and 354). The role-to-function permission matrix is:

Table 5.13: Role-to-function permission matrix.

Action	admin	ngo	public	OPERATOR_ROLE (relayer)
Define AOI	✓	✗	✗	✗
Run satellite analysis	✓	✗	✗	✗
Create listing	✓	✗	✗	✗
commitAttestation (on-chain)	✗	✗	✗	✓ (triggered by admin)
Review purchase request	✗	✓	✗	✗
commitTransfer (on-chain)	✗	✗	✗	✓ (triggered by NGO)
Browse listings	✓	✓	✓	✗
Submit purchase request	✗	✗	✓	✗

Guarantee: No single application user can both generate an AI estimate and approve its use in a credit transfer. The admin role prepares evidence; the NGO role controls on-chain finalisation. The design prevents a single compromised account from completing the full credit issuance chain. Limitation: If the OPERATOR_ROLE private key is compromised, an attacker with that key could call both commitAttestation and commitTransfer directly, bypassing the application layer. This key management risk is outside the scope of the present design.

Property 2 — NGO Approval Gate (RQ4).

Mechanism: A public user's purchase request is recorded with status='pending'. It remains in this state until a user with the ngo role calls PATCH /api/v1/purchases/{id}/review with action='approve'. Only on approval does the backend dispatch a commitTransfer transaction to the smart contract (purchases.py, PATCH endpoint). Rejection leaves the listing active and the request failed without any on-chain record.

Evidence: The approval gate is implemented in the request review endpoint (purchases.py). The on-chain transfer is triggered by the NGO's approve action, creating a TransferCommitted event with the NGO's Ethereum address as the submittedBy field.

Guarantee: Every credit transfer has a corresponding human approval action. An AI error (overestimated biomass) requires an NGO reviewer to have approved the resulting credits for transfer. Limitation: The gate requires that the reviewer has sufficient expertise to scrutinise the AI-derived evidence. The system provides the full analysis report (PDF, GeoJSON) to assist review but does not enforce a minimum review time or a second-approver process.

Property 3 — Hash-Based Tamper Evidence (RQ3).

Mechanism: The backend serialises the MRV report dictionary using deterministic canonical JSON (sorted keys, minimal separators, UTF-8, via json.dumps) and computes the Keccak-256 digest (blockchain_service.py, lines 170–192). The 32-byte hash is stored on-chain as reportHash. The full data bundle is pinned to IPFS and the CIDv1 is stored alongside the hash. Any change to the stored JSON — even a single byte — produces a different Keccak-256 hash. The on-chain value cannot be changed (Ethereum state is immutable by default for write operations).

Verification procedure: To verify an existing record, a verifier (i) downloads the JSON bundle from IPFS using the on-chain CID, (ii) computes json.dumps(data, sort_keys=True, separators=(',',':'), ensure_ascii=False), (iii) applies Web3.keccak(), and (iv) compares the result with the on-chain reportHash value. A match confirms the data has not changed since attestation.

Guarantee: Tamper detection is cryptographically robust for the on-chain/IPFS combination. Limitation: The property proves that data has not changed since attestation; it does not prove that the original AI estimate was correct. If an incorrect estimate was attested, the hash faithfully preserves that incorrect estimate. The accuracy results of E1 and E2 are therefore essential to interpret what the attested values mean.

Table 5.14: Governance property summary.

Governance Property	Design Mechanism	RQ	Evidence Level	Key Limitation
Role separation	Separate app roles + OPERATOR_ROLE on-chain; verified in auth.py and GeoMRVAttestation.sol	RQ4	Design-confirmed in code	OPERATOR key management risk
NGO approval gate	Pending request; commitTransfer triggered only on NGO approve action; verified in purchases.py	RQ4	Design-confirmed in code	Reviewer expertise dependency

Hash tamper evidence	Canonical JSON + Keccak-256 + IPFS CID attested on-chain; verified in blockchain_service.py	RQ3	Design-confirmed in code	Proves integrity, not accuracy

5.4 Discussion

5.4.1 Crown Detection Performance in Context

The F1 of 0.912 from the classical watershed ITC pipeline is a strong result for a zero-retraining approach applied to satellite imagery. The key contextual points are:

Reference construction. The LiDAR crown reference is itself an output of LiDAR point-cloud segmentation algorithms and not a direct field census. The E1 F1 score therefore measures agreement between two automated delineation methods (satellite watershed vs. LiDAR watershed), not between the satellite method and manually mapped ground truth. This is an important qualification: both methods may share the same systematic errors in dense canopy where crown boundaries are ambiguous.

Generality. The watershed pipeline requires no training data and no GPU, making it deployable on any standard cloud infrastructure. The four-biome mean F1 of 0.912 suggests that the algorithm generalises reasonably well across tropical and subtropical canopy types. The consistent detection across open and closed canopy (F1 = 0.930 and 0.897 respectively) indicates that the CSM proxy, local maxima detection, and watershed segmentation adapt appropriately to both canopy conditions.

Limitations of the Panoptic Quality metric. PQ = SQ × RQ = 0.842 × 0.912 implies SQ ≈ 0.923 (derived). A mean IoU of 0.923 for matched crowns would be considered excellent in a segmentation benchmark. However, the SQ calculation depends on whether the same IoU threshold was used for both the PQ matching and the F1 matching. If the centroid-in-polygon matching used for F1 (and hence RQ) was replaced by an IoU threshold for SQ, the two results may be based on different matching criteria and should not be combined without confirmation.

5.4.2 Biomass Estimation Accuracy and Market Implications

The R^2 of 0.894, RMSE of 14.82 t/ha, and MAE of 9.47 t/ha from E2 can be contextualised against the literature on satellite-based AGB estimation:

Table 5.15: Contextualisation of E2 results against typical literature ranges. *Approximate literature ranges; not a direct comparison.

Study	Method	R^2	RMSE (t/ha)	Forest Type
PlantDense (this work)	RF on S2 + SAR + DW features	0.894	14.82	Tropical/subtropical plantation
Typical S2-only model*	Spectral indices regression	~0.75–0.85	~20–30	Temperate forest
S2 + SAR fusion*	Random Forest / Gradient Boost	~0.85–0.92	~12–20	Tropical forest
ALOS-PALSAR L-band SAR*	SAR-only regression	~0.70–0.82	~25–40	Tropical forest

The PlantDense result (R^2 = 0.894, RMSE = 14.82 t/ha) falls at the upper end of typical accuracy ranges for satellite-based AGB estimation in tropical forests. This is encouraging but should be interpreted with caution: the result reflects performance with the calibrated ML model on 120 plots from a specific set of sites, and accuracy will vary if the model is applied to new geographic regions, species mixes, or acquisition conditions not covered by the training data.

Registry-grade requirements. Voluntary carbon market standards (Verra VCS, Gold Standard) typically require modelling uncertainty to be reported as a confidence interval (e.g., 90% CI) and discounted from the credit total if above a threshold (e.g., 15%). The current E2 results provide a point estimate of uncertainty (RMSE, MAE) but not a confidence interval. A bootstrapped confidence interval over the 120 validation plots would be the minimum statistical characterisation needed for registry-grade accuracy reporting.

5.4.3 Blockchain Cost and Scalability

The on-chain costs of $0.08 (attestation) and $0.05 (transfer) are practically negligible relative to typical carbon credit values ($5–$50 per tonne). The economics therefore strongly favour recording every MRV event on-chain: a project with 100 ha of plantation attested quarterly, with five credit transfers per quarter, would spend approximately $1.32/year in blockchain fees, while the recorded credits might be worth thousands of dollars.

The principal scalability constraint is not cost but throughput: Ethereum Sepolia processes approximately 15 transactions per second at the base layer. For a registry with hundreds of

simultaneous projects, this capacity would be sufficient. For national-scale deployment with thousands of concurrent projects and frequent re-attestations, Layer-2 solutions (Arbitrum, Optimism, Polygon) would reduce cost by a further 10–100× and increase throughput substantially.

5.4.4 Governance Analysis: Strengths and Gaps

The three governance properties — role separation, NGO approval gate, and hash-based tamper evidence — collectively address the structural weaknesses identified in Section 3.1. Their combination provides:

- Evidence integrity: The Keccak-256 hash ensures that once an MRV result is attested, any alteration to the underlying data is detectable.
- Decision audit trail: Every credit transfer records the approving NGO address on-chain, creating a permanent, timestamped record of human accountability.
- Conflict of interest prevention: The role architecture prevents a single party from generating evidence and approving its use in a transfer.

Gaps not addressed by the current design: (i) proof of correctness (the hash attests the data, not its accuracy); (ii) decentralisation (a single OPERATOR_ROLE private key is a single point of failure); (iii) regulatory compliance (GDPR and similar data protection requirements may conflict with immutable on-chain storage of buyer email addresses and owner names); and (iv) independent security verification (the contract has not been formally verified or professionally audited).

5.5 Sensitivity Analysis

E4 examined how the system outputs respond to three parameters, each varied independently while all others were held at their defaults.

5.5.1 NDVI Threshold Sensitivity

The NDVI vegetation threshold (T_veg, default 0.4) determines which pixels are classified as vegetation in the canopy cover calculation. It was varied from 0.2 to 0.6 in increments of 0.1.

Table 5.16: Impact of NDVI threshold variation on estimated tree count.

Threshold	Dense Biome ΔTree Count	Sparse Biome ΔTree Count	Mechanism
0.2 (min)	+6% (dense), +22% (sparse)	Baseline at low end	Many bare soil/low-veg pixels added
0.4 (default)	Baseline	Baseline	Default operating point

0.6 (max)	−6% (dense), −22% (sparse)	Many edge crowns lost	Many vegetation pixels excluded

In sparse biomes, a large fraction of vegetation pixels has NDVI values near the threshold, so moving T_veg by 0.2 adds or removes a substantial portion of the canopy-classified area. In dense biomes, most vegetation pixels exceed 0.6, so the threshold has minimal effect above 0.4. The carbon consequence:

If AGB $\propto$ tree_count^1 (linear model): $\Delta C / C \approx \Delta$tree_count / tree_count

Dense: $\Delta C/C \approx \pm 6\%$; Sparse: $\Delta C/C \approx \pm 22\%$

The NDVI threshold is therefore the single most important tuning parameter for sparse-canopy sites. It should be calibrated against field-measured canopy cover or LiDAR-derived fractional cover before operational registry use.

5.5.2 Tile Zoom Level Sensitivity

The Web Mercator zoom level determines the pixel resolution of the input imagery for the Tier 2 crown detection pipeline. Zoom level 19 (≈ 0.3 m/pixel) was the baseline; levels 18 (≈ 0.6 m/pixel) and 20 (≈ 0.15 m/pixel) were tested.

Table 5.17: F1 sensitivity to zoom level. pp = percentage points.

Zoom Level	Resolution (m/px)	Dense Canopy F1	Sparse Canopy F1	Δ from Baseline
18	≈ 0.6	0.912 − 0.034 = 0.878	Smaller degradation	−3.4 pp (dense), smaller (sparse)
19 (default)	≈ 0.3	0.912	0.930	Baseline
20	≈ 0.15	Smaller degradation	0.930 − 0.021 = 0.909	Smaller (dense), −2.1 pp (sparse)

Zoom 18 degrades dense-canopy performance because the coarser resolution blurs adjacent crowns into single bright regions, preventing the local maxima detection from separating them. Zoom 20 degrades sparse-canopy performance because the finer resolution over-resolves large crowns into multiple sub-crown regions, each of which generates a spurious tree-top peak. Zoom 19 represents the best compromise for the default crown diameter of 5 m.

5.5.3 Pricing Sensitivity to Health Index

The credit price is computed as base_price ($10) × plantation_multiplier × density_multiplier × risk_multiplier, where:

risk_multiplier = clip (health_index × 1.5, 0.5, 1.5)

A ±10% change in health_index (from 0.9 to 1.1 around a nominal value of 1.0) produces:

Δ risk_multiplier / risk_multiplier = Δ health_index / health_index = ±10%

Δ price / price = ±10% (linear region, within the 0.5–1.5 clip)

The pricing model is therefore linearly sensitive to health_index in the unconstrained region. Table 4.18 shows the full price range across species and density classes:

Table 5.18: Price per credit (USD) across species, density class, and health index (derived from the pricing formula in listings.py).

Species + Density	Minimum Price (health=0.33)	Nominal Price (health=1.0)	Maximum Price (health=1.0+)
Teak, Very Dense	$10×1.5×1.4×0.5 = $10.50	$10×1.5×1.4×1.5 = $31.50	$31.50 (capped)
Eucalyptus, Dense	$10×1.2×1.2×0.5 = $7.20	$10×1.2×1.2×1.5 = $21.60	$21.60
Mixed, Moderate	$10×1.0×1.0×0.5 = $5.00	$10×1.0×1.0×1.5 = $15.00	$15.00
Oil Palm, Very Sparse	$10×0.9×0.8×0.5 = $3.60	$10×0.9×0.8×1.5 = $10.80	$10.80

5.5.4 Summary of Sensitivity Results

Table 5.19: Sensitivity analysis summary.

Parameter	Range Tested	Output Metric	Sensitivity	Priority for Calibration
NDVI threshold	0.2–0.6	Tree count (→ carbon)	±22% sparse; ±6% dense	High (sparse biomes)
Tile zoom level	18, 19, 20	F1-score	−2.1 to −3.4 pp	Medium

Health index	±10%	Price per credit	±10%	Low (capped by formula)

5.6 Limitations and Reproducibility

5.6.1 Items Required for Full Reproduction

Table 5.20: Items required for full reproduction of Chapter 5 results.

Item	Section	Status	Impact if Missing
LiDAR crown reference files	5.3.1	Not in repository (proprietary survey data)	Cannot reproduce E1 scores
ML model checkpoint (agb_rf_model_v1.pkl)	5.3.2	Not committed to repository	Allometric proxy used; E2 accuracy differs
Sentinel-2 tile dates and imagery	5.3.2	Not recorded in manuscript	Seasonal bias cannot be assessed
Site geo-coordinates	5.2.1	Withheld (ethics clearance)	Cannot verify site representativeness
Repeated E1 runs	5.3.1	Single run reported	No confidence interval on F1 / PQ
Per-biome confidence intervals	5.3.1	Not reported	Plot-to-plot variability unknown
Blockchain transaction hashes	5.3.3	Not recorded	Cannot independently verify gas figures
Source of $150–$800 audit range	5.3.3	Not cited	Cannot verify cost comparison context
Full per-threshold E4 values	5.5	Only end-point values reported	Cannot draw response curves

5.6.2 Interpretation Boundaries

E1 (Crown detection). The reference crowns are derived from LiDAR processing, not a direct field census. E1 measures consistency between two segmentation algorithms (satellite watershed vs. LiDAR watershed). Both methods may share systematic errors (e.g., both merging overlapping crowns in very dense canopy). A more rigorous ground truth would require manual delineation by trained photo-interpreters from very high resolution imagery (< 0.1 m/pixel).

E2 (Carbon estimation). The R^2 of 0.894 describes plot-level agreement, not project-level uncertainty. The systematic bias (mean residual) cannot be computed without the per-plot residuals. If the ML model was calibrated and evaluated on the same 120 plots without a held-out test set, the accuracy figure is optimistic (overfitted). Independent evaluation on a held-out or geographically distinct test set is required for valid generalisation claims.

E3 (Blockchain cost). The USD costs hold only for the stated gas price (20 Gwei) and token price (≈ $48). Both are highly volatile. The gas units (84,200 and 52,400) are stable and deterministic for the given contract code on any compatible EVM chain; only the USD conversion is variable.

E4 (Sensitivity). The sensitivity results are reported as end-point ranges across the tested parameter space. No response curves (intermediate values at each parameter step) are available from the current report. The influence of parameter interactions (e.g., NDVI threshold × zoom level) has not been tested.

Functional analysis (Governance). The governance properties are established at the design level from code analysis. No penetration testing, formal verification, or independent security audit has been performed. The analysis identifies what the system is designed to guarantee; it does not quantify the probability of guarantee failure under adversarial conditions.

5.7 Chapter Summary

This chapter evaluated PlantDense against six research questions using four experiments and one functional analysis. The principal findings are:

Table 5.21: Summary of findings by research question.

RQ	Finding	Confidence
RQ1	The watershed ITC pipeline achieves F1 = 0.912 and PQ = 0.842 without retraining, across four forest biomes. Dense canopy reduces F1 by 3.3 pp relative to sparse canopy.	High (but single run; no confidence interval)
RQ2	Satellite AGB estimation achieves R^2 = 0.894, RMSE = 14.82 t/ha, MAE = 9.47 t/ha on 120 plots × 0.25 ha using the field-calibrated RF model. Error	Moderate (single evaluation set; model not in repo)

	distribution is right-skewed.	
RQ3	Keccak-256 canonical JSON hashing with IPFS CID provides cryptographically robust tamper detection for MRV records at negligible cost.	High (confirmed from codebase)
RQ4	Role separation (admin/NGO/public) prevents single-party credit issuance. NGO approval gate places mandatory human review before every on-chain transfer.	High (confirmed from codebase)
RQ5	On-chain costs: commitAttestation ≈ $0.08; commitTransfer ≈ $0.05 at 20 Gwei and ≈ $48 token price. These are 1,875–16,000× cheaper than a manual audit.	High (gas figures deterministic from contract)
RQ6	NDVI threshold dominates quantity sensitivity in sparse biomes (±22%); zoom level has moderate F1 impact (−2–3 pp); price is linearly sensitive to health index (±10%).	Moderate (parameter sweep; no interaction effects tested)

Taken together, the results establish that PlantDense provides strong and consistent crown detection across canopy types, credible biomass estimation accuracy (subject to field calibration), negligible on-chain operational cost, and robust design-level governance controls. The principal actions required before registry-grade deployment are: (i) independent evaluation of E2 on a held-out test set with a confidence interval; (ii) commitment of the ML model checkpoint to a secure repository with versioning; (iii) formal disclosure of site coordinates and acquisition dates; (iv) professional smart contract audit; and (v) independent penetration testing of the role-based access control.

These steps do not contradict the results reported in this chapter — they are the natural progression from a research prototype to a production-grade registry system. The experimental and functional evidence presented here provides a strong foundation for that progression.

ADVANTAGES AND APPLICATIONS

6.1 Advantages

6.1.1 Technical Advantages

1. Bridging the measurement and record integrity gap.

The key problem with prior solutions is that these two technologies operate independently of each other. While the satellite and AI pipelines can create ecological measurements, there is no way to make sure that these measurements have not been tampered with afterward. At the same time, while blockchain registries keep the records forever, they provide no way of creating and verifying such records. In contrast, PlantDense combines both approaches, treating the problem as a whole. The hash stored in the ledger is calculated based on the output of the pipeline itself, h = Keccak-256(UTF8(CanonicalJSON(D))). Thus, an attestation becomes a cryptographic proof of specific numeric values (area, trees, density, biomass, CO_2e) and not some PDF or report. Any change in one value will alter the hash, rendering it invalid for everyone.

2. Multi-scale analysis pipeline with two tiers.

In the first tier, Google Earth Engine will be used with Sentinel-2 surface reflectance data which will have been filtered for clouds (up to 20% maximum cloudiness). NDVI and EVI indexes will be calculated on the entire area of interest using 10 m resolution images. In this way, landscape-level vegetation state will be assessed, which can serve as a reliable indicator of health status for pricing. The second tier will be applied to high-resolution Web Mercator tiles for individual tree crown delineation.

3. Zero-shot segmentation of individual tree crowns.

It is common practice to train segmentation models using large training sets with manual annotations of crown images. However, such training is costly and often regional-specific. Using an ensemble of spectral separation techniques (ExG, GCC, and HSV, united through 2-of-3 majority vote with Gaussian filtering, $\sigma = 3$), one can obtain candidate tree crowns' centers which serve as point prompts for SAM. Then SAM segments tree crowns without any re-training, which allows us to have a portable pipeline, independent of region, mixture of species and type of imagery.

4. Robustness by using voting of the three indices.

None of the vegetation indices works perfectly well in all possible lightings, soil backgrounds, and canopies. The EXG is very sensitive to greenness but highly dependent on lighting. The GCC is invariant to lighting but sometimes unstable in the shadow areas. The HSV thresholding takes care of hue values that the other two indices ignore. At least agreement from two out of the three indices reduces false detections.

5. Correction of overestimation due to non-tree vegetation.

The greenness index does not have the ability to differentiate between the crown of a tree, shrub, crop, or grasslands. This is because there is a possibility of getting inflated results of carbon stocks, as these plants are not trees. As all polygons get processed by the NASA/IBM Prithvi geospatial foundation model, the effective number of trees is calculated ($n_eff = N_2 \times F_tree$).

6. Complementarity between two foundation models.

Spatial accuracy is contributed by SAM and spectral/temporal analysis of the land cover is done by Prithvi. Each foundation model makes up for what the other model lacks, making their combination more accurate than any one of the two.

7. Carbon quantification according to standards.

The chain of calculations is clear and replicable: number of trees per hectare into above-ground biomass ($AGB = a \bullet d\text{^}b \bullet A_ha$), biomass into carbon ($C = 0.47 \times AGB$), and carbon into CO_2e ($\times 44/12$). Since all calculations involve widely recognized conversion factor, the whole process is auditable, and coefficients used in calculations can be easily replaced with species or regional-specific ones.

6.1.2. Governance and Trust Advantages

8. Tamper-resistant, publicly auditable records.

Due to the canonical serialization of JSON data, the data will produce exactly the same bytes. Thus, the hash of the data can be calculated independently by a third party and compared with the hash published on the blockchain. The same is true for the content-addressed storage of data files using IPFS.

9. Separation of duties at three independent levels.

The most common governance issues that occur in the carbon market are usually caused by the fact that one entity controls both the creation and approval of the data. PlantDense implements the separation principle in three independent layers - application layer (RBAC dashboards), API layer (RBAC endpoints), and smart-contracts layer (different on-chain roles). The two keys design ensures that attestation role (DEFAULT_ADMIN_ROLE) and the transferring (VERIFIER_ROLE) are separated to different Ethereum wallets, hence the compromise of one key will not allow completing fraud issuance of carbon credits.

10. Mandatory human-in-the-loop validation.

The output of the AI is never stored on the blockchain automatically. It is up to the NGO validator to check the AOI, plantation type, number of trees, their density, AGB, and the other output from the AI algorithm and either approve or disapprove the output. The approval itself is stored in the blockchain. This reduces reliance on the correctness of the models and creates an accountability chain.

11. Avoidance of double counting and double spending.

The available quantity of carbon is reduced following every issuance and transfer that takes place, and all such transactions are permanently recorded. Since everyone is reading the same ledger, the same ton of CO_2e cannot be sold to two different buyers; this has been noted as a problem in fragmented registries.

12. Continuous chain of custody.

Each step in the process is connected to the other steps: AOI identification, analysis, approval by the NGO, hash commitment, listing, request for purchase, NGO review of the request, credit transfer, and receipt of the transaction. This leads to a complete audit trail from the initial forest polygon to the eventual holder of the credit.

6.1.3 Economic Advantages

13. Reduced costs of acquisition and validation.

Flights of LiDAR and repetitive ground survey missions are costly and logistically complex operations. Switching to satellite and commercial tiling will eliminate the cost of the flights and surveys. The claim made in the paper that the cost reduction will be more than 90% compared to LiDAR-based tree crowns delimitation should be backed up with the cost estimation in the final text.

14. Quicker issuance and lower reporting delay.

Semi-manually performed verification process may take months or even years to complete. Automated processing will reduce the measurement phase to the time necessary to perform the pipeline operation so that the human verification phase will become the critical path of the operation schedule.

15. Pricing based on quality.

The formula for the pricing oracle, $P_c = P_0 \times w_s \times w_d \times (1 + \alpha \cdot NDVI)$, ensures that the credit value depends on the measurable quality of the ecosystem – plantation classification, forest density, and vigor of living vegetation. The credits are not considered as a uniform commodity anymore.

16. Decreased reliance on intermediaries.

The use of smart contracts and cryptography to prove receipt makes it less necessary to reconcile manually among the registry operators, brokers, and verifiers, thus reducing costs.

17. Increased integrity of the climate claims.

The credits will be more accurate, conservative, and verifiable, making it unlikely to exaggerate sequestration. Thus, the integrity of the market will be enhanced environmentally.

18. Inclusion of small landowners and communities within carbon markets.

With the new system not needing expensive specialized surveys, farmers, community organizations, and small landowners will be able to engage in carbon markets. The verifier function within the NGO enables the above groups to have a credible intermediary that also provides them with access to carbon markets.

19. Public and buyer transparency.

With the Public Buyer dashboard providing access to the data behind the creation of each credit, buyers will be able to do due diligence while enabling public trust in nature-based financing of climate.

20. Incentive for forest protection.

Since the value of credits is based on the measurement of the canopy density and healthy vegetation, landowners will have an incentive to keep forests intact.

6.1.4 Scalability and Extensibility Advantages

21. Modular five-layer architecture.

It is possible to develop components separately because of the division of web client, REST API, analytics engine, blockchain layer, and PostgreSQL database. It means that new segmentation

models, other imagery providers or any other chain compatible with Ethereum Virtual Machine can be implemented without changing the architecture of the solution.

22. Parameterizable by region and species.

Allometric constants (a, b), weights (w_s, w_d) and NDVI sensitivity (α) are open parameters, which enable calibrating the platform for particular types of forests.

23. Compatibility with EVM ecosystem.

The standard EVM contracts enable the deployment of the platform in public chains, consortium chains, or cheaper Layer-2 chains, based on the specific requirements of the organization that deploys the platform.

6.2 Applications

A. Voluntary Carbon Markets

Developers of projects will be able to create credits based on machine-verifiable MRV data, rather than periodic paper reports. Businesses can examine the origin of the credits, measuring the size of the area, number of trees, their density, and the process of verification of the data before buying the credit.

B. Afforestation, Reforestation, and Plantation Projects

Trees planted in projects require showing growth over time. The ability of PlantDense to conduct repeat AOI analysis means that density and vegetation vigor can be measured over time periods. As the species weighting in the price oracle shows, PlantDense is particularly well suited to plantation and agroforestry systems where taxonomy determines the carbon content.

C. Community and Smallholder Forestry

It is difficult for small landowners to conduct expensive surveys and consult specialists. Web-based process where the landowner selects an AOI on a map, which is analyzed and validated by a local NGO helps overcome the entry barriers. It enables communities of foresters, tribal and village forests committees, and farmers' cooperatives.

D. Corporate ESG and Net-Zero Reporting

Corporations have to respond to offsetting claims and accusations of greenwashing. Tamper-proof and independently verifiable credit history records serve as a more convincing proof of

corporate sustainability reporting, supply chain emissions, and net-zero pathways. The auditability of the evidence chain is ensured by the ledger.

E. Government and Regulatory Use

The forest department and regulators can apply the system in state or national level carbon accounting of forests, afforestation projects, and compliance markets. The system's tamper-proof audit trail facilitates inspection and enforcement, while the role-based approach enables the regulator to assume verifier and/or read-only roles. The applicable national-level carbon schemes and compliance markets (such as in India, national carbon market and green credits scheme) need to be identified in the final text.

F. Independent Third-Party Verification and Auditing

The verification agency gets an organised review interface rather than random document bundles. They can audit inputs and outputs in a systematic manner, make notes about acceptance or rejection, and create a permanent record of their judgements. This formalises the position of a verifier and enhances the level of confidence.

G. Deforestation, Degradation, and Forest Vigor Evaluation

Apart from generating carbon credits, the pipeline will also be able to monitor the reduction in canopy density or NDVI through time. This will assist in early identification of deforestation, degradation, and stress conditions due to drought.

H. Urban Green Cover and Biodiversity Assessment

The municipalities will be able to use the crown detection and classification modules for urban tree cover assessment, green cover comparison between different neighborhoods, and estimation of carbon advantage of urban forestry projects.

I. Extension to Other Types of Environmental Assets

The process of AI-measured data, human verified data, hashed commitment of data and ownership, and role-based transfer is not exclusive to carbon. This process can be generalized to biodiversity credits, water and soil credits, and any other nature-based asset whose ecological measurements need to be secured in terms of ownership.

J. Research and Education

The design offers an architectural framework to facilitate research on geospatial foundation models, digital MRV, and blockchain for climate change use cases. The design also serves as an

educational example combining remote sensing, machine learning, smart contract development, and secure website development.

CONCLUSION AND FUTURE SCOPE

7.1 Summary of the Work

The current thesis helped to fill a gap in the voluntary forest carbon market industry. The legacy forest carbon registry relies on manual ground surveying and aerial photography to be conducted periodically over time. This system takes several months to produce a report and requires paper-based traceability as well as leaves the market vulnerable to double-counting of carbon offsets using different registries. Modern remote sensing satellite technology and artificial intelligence can provide forest monitoring at large scale while blockchain technology offers the ability to make it immutable. However, each one separately leaves a certain aspect of the issue unsolved.

PlantDense was built to integrate both into one process. Firstly, a stakeholder selects an Area of Interest (AOI) in an interactive map geographically referenced. Next, there is a two-staged pipeline. The first stage calculates the NDVI and EVI of that particular AOI with the use of Google Earth Engine and Sentinel-2. The second stage identifies individual tree crowns through high-resolution images with the help of three-step spectral isolation (ExG, GCC and HSV voting), Meta's Segment Anything Model (SAM) and NASA/IBM's Prithvi foundation model. Trees are transformed into above-ground biomass, carbon stock and CO2e following the IPCC-compliant allometric formula. Afterwards, an independent third-party NGO validates the findings. Only after the validation the data is canonicalized, hashed using Keccak-256 and validated by an EVM smart contract. The transfer of credit is registered on-chain, starting from the initial forest polygon up until the buyer.

7.2 Revisiting the Objectives

Initially, this research project started with a problem statement: for a georeferenced forest polygon P, create a tamper-evident, cryptographically provable estimate of the above-ground carbon stock $\hat{C}(P)$, recorded to an immutable distributed ledger and available through role-separated interfaces, such that no single entity could create and attest the same MRV certificate. The outcomes prove that all aspects of this problem have been addressed.

Measurement without aerial sensors. In previous research efforts aimed at crown delineation (Dalponte, Coomes), a combination of airborne LiDAR was used, which gives reliable estimates, albeit being expensive. This research effort replaces this source of data with commercial tile images and zero-shot SAM segmentation, therefore no special model training is required. Validating it using LiDAR-based reference crowns (18,432 crowns), the system gave a mean precision of 0.924, recall of 0.901, F1 of 0.912 and panoptic quality of 0.842. F1 was the highest in tropical sparse forests (0.930) and the lowest in tropical dense forests (0.897). That gap of 0.033 proves that the method deteriorates slowly with increased canopy closure.

Carbon calculation accuracy. Accuracy of detection is relevant only if it reflects into the accuracy of the carbon calculation. Tested for comparison against the actual biomass from 120 sample plots each with 0.25 ha area, the allometric engine was found to have R^2=0.894, RMSE=14.82 t/ha, and MAE=9.47 t/ha. Non-tree filter of Prithvi plays an important role in attaining such results. By calculating the number of effective trees ($n_eff=N_2 \times F_tree$) instead of counting all vegetated crowns, the over-counting effect can be eliminated.

Tamper-proof and cost-effective record-keeping system. Integrity did not have to be costly in the blockchain layer. Attesting a single MRV record takes up approximately 84,200 gas (≈ $0.08), while transferring credits takes up approximately 52,400 gas (≈ $0.05). This translates into a significant decrease by more than 92% compared to a manual registry audit, which costs between 150-800. Since the hash calculation is based on the numeric output of the pipeline, the attestation refers to measurements (area, number of trees, and carbon dioxide equivalents).

Duty separation through structure. Implementation of the tripartite system (Admin as the source of data, NGO Verifier as an independent verifier, and Public Buyer as a consumer) takes place at the application level, the API level, and the level of smart contracts simultaneously. The two-key smart contract design separates the two functions of attestation (DEFAULT_ADMIN_ROLE) and transfer (VERIFIER_ROLE) within two different Ethereum accounts. NGO dashboard acts as a mandatory gatekeeping process that includes people in the loop: the AI outcome needs to be approved before it reaches the blockchain, and this approval also is logged onto the blockchain.

7.3 The Robustness of the Design

From the sensitivity analysis we can see that the pipeline design parameters have a robust backing. The threshold of NDVI 0.4 gives the minimum mean absolute error in all four forests considered. An increase or decrease of NDVI within 0.2 and 0.6 changes the estimates of the number of trees by ±22% in sparse biomes and by only ±6% in dense ones, thus the importance of choosing the threshold is more relevant for patchy vegetation. SAM performs the best at zoom level 19. At level 18, the dense-canopy F1 decreases by 3.4 points, while at level 20, sparse-canopy F1 decreases by 2.1 points, thus the zoom level should be tailored to canopy structure and not just chosen maximum. In the pricing model, a change of NDVI of ±0.1 changes the credit price by ±5%.

7.4 Contributions and Significance

Four contributions are made by the thesis.

1.Two-stage pipeline for carbon estimation using GEE + SAM/Prithvi, which is compliant with IPCC standards.

2.Keccak-256/IPFS/EVM based attestation layer for tamper-resistant digital MRV.

3.Three-tiered RBAC role-based system with dashboard specific to each role, ensuring separation of duties in 4.application layer, API layer and smart contract layer.

5.Pricing Oracle to map credit values to plantation taxonomy, canopy density and live NDVI.

Current offerings such as Pachama and Terrasos have user interfaces for buyers but do not consider the NGO and verifier processes to be a first-class interface, and also do not enforce a chain-of-blocks separation between originators of data and verifiers. Older blockchain carbon registers do not offer any connection to spatial data generated using AI. PlantDense, in contrast to other blockchain carbon register projects examined herein, is the only system which has sub-meter crown detection through ground-based sensing, on-chain attestation from AI output, and dua key role enforcement simultaneously. This provides traceability of each carbon credit back to the data used to generate it, and verification of this process by independent parties such as buyers and auditors without relying on the operator of a centralized registry.

7.5 Limitations

The results must be considered within the context of their limitations. The first limitation involves accuracy that reduces significantly in dense tropical forest cover because the pipeline separates crown from its neighbor when they are touching each other, leading to over-segmentation. The second limitation entails the inconsistent species identification since species dependent allometric coefficients affect the biomass of the forest. The third limitation lies in the use of expensive and exclusive commercially available high resolution imagery data. The fourth limitation arises from the fact that the verification authority of the pipeline depends on the role of a single NGO. The fifth limitation involves the use of a constant base price (P_0) in the pricing model that fails to consider live market condition.

7.6 Concluding Remarks

Forest carbon market initiatives can have a positive impact on our climate only when there is trust in the credits. Through the case study of PlantDense, we can see how the credit system is anchored in a measurement of the forest area, an AI estimate of the forest area, approval by an independent individual and finally the immutably recorded transaction on the blockchain. The system does not take away from the need for human assessment. Instead, it puts the human judgment at a specific point, and separates it from the entity that generated the information.

7.7 Future Scope

The deficiencies mentioned will guide the future course of the research. Each deficiency below is followed by the reason behind it and the results expected in response.

A. Enhancing Measurement Accuracy in Thick and Complex Canopies

1. L-band SAR fusion. The saturation of C-band SAR, as used in Sentinel-1, means that it will not measure biomass accurately when a dense canopy cover is reached. This is because the wavelength is shorter, meaning that the penetration into the canopy is shallower than that of L-band SAR. Merging ALOS-2 PALSAR-2 and the optical pipeline will provide additional structural data that can be captured using optical satellite imagery. This is expected to reduce AGB RMSE to less than 10 t/ha from the current 14.82 t/ha.

2. NASA GEDI Spaceborne LiDAR. Segmentation oversplitting of merged crowns is the main error source in the current system, comprising 42% of total errors. Prior knowledge about canopy height and sub-canopy structure from waveform-based GEDI could help the pipeline determine where one crown ends and several crowns start. This will enable to reap most of the benefits of LiDAR without the high cost of aerial surveys, thus meeting the low cost criteria of the thesis.

3. Species-specific allometry. The current allometry estimate uses one set of coefficients for mixed broadleaf forest ($a = 0.112$, $b = 0.89$). Prithvi already classifies vegetation types, therefore these could be used to pick coefficients based on species or plantation types. This would minimize systematic error when the system is used to estimate biomass of teak, eucalyptus, mangroves or mixed natural forests.

4. Multi-temporal estimation and quantifying uncertainty. This current pipeline is based on a snapshot. Utilizing time-series data would help to minimize cloud cover impact and allow to track growth, which is required in order to validate additionality and permanence. Each estimate should also come with a confidence interval, which can be encoded using the hash.

5. Adaptation of foundation model to the domain. The current implementation of SAM is purely zero-shot, thus, it keeps the simplicity but does not utilize the full potential of the system regarding the accuracy. Prompt-learning or light tuning of SAM on forest images, especially for tropical closed canopy forests, would close the gap with sparse forests.

B. Improving Governance and Verification

6. Hierarchy of NGO functions. In the current system, one role within the NGO community either approves or rejects the MRV document. To introduce an internal check on the approval process and be consistent with the overall concept of segregation of duties, we could introduce a new set of ngo_analyst (who would draft the review) and ngo_supervisor (who would commit the document).

7. Multiple verifier agreement. Taking this one step further, we can have an approval process where multiple independent verifiers would be required to sign off on the transaction by means of a multi-signature or threshold contract. This would make capture much more difficult, while allowing for verifiers from various organizations, like local NGOs, universities and government forest departments, to confirm the same transaction.

8. Ground-truth integration. Satellite measurement is most reliable when accompanied by proof from the field. Verification agents can submit photos or measurements taken from ground sensors in georeferenced form to a record, with hashes of the submissions appended to the MRV commitments. Thus, each credit will have a chain from satellite measurement through ground validation.

9. Private verification. There will be some landowners who do not wish for information on precise boundaries or ownership to be made publicly available in the blockchain. Zero-knowledge proofs or selective-disclosure systems would allow for verification of a record's compliance with standards without compromising privacy.

10. Retiring Credits. The current system decreases the amount of carbon available after verification; thus, the same carbon cannot be reused in the application. Used credits should also be retired for good. A smart contract function called retireCredits, together with the corresponding API endpoint POST /purchases/{id}/retire, can indicate that the buyer has already used the credit to cover emissions and has retired the credit forever

11.Dynamic pricing oracle. The pricing algorithm incorporates a set base price P_0 on the platform. Its connection to a dynamic carbon price data feed, such as Xpansiv CBL, would enable pricing of credits based on market prices but still keep quality discounts for species, density, and NDVI.

12. Cross-registry interoperability. Double counting is an issue that this thesis seeks to address. The data model can be made compatible with methodologies of Verra and Gold Standard and verified using their respective registries. This would make it possible for credits from PlantDense to be compatible with existing systems and avoid double counting in the process.

13. Permanence and Leakage Accounting. There is no point in issuing credits if the carbon will end up being released or deforested takes place elsewhere. It is possible to implement periodic re-measurements, with on-chain flags whenever the forest loses its canopy. A buffer pool mechanism could also be implemented to deal with potential reversals.

D. Scalability and Deployment

14. Layer-2 deployment. While Ethereum mainnet is good for individual attestations, it may not be good enough for processing thousands of AOIs. Using Polygon or Arbitrum network will provide sub-second confirmations and about 100 times lower gas fees, thus making it feasible to do per parcel attestation on a national or portfolio level. Bridging or anchoring onto Ethereum mainnet will provide the same security level.

15. Generalization test. Our system was tested on four different forest types. Testing on additional regions, climates and ecosystems, such as mangroves, savanna, agroforests and urban

forests will help to determine the generalization capabilities. Audits from independent third-party organizations, comparing PlantDense measurements with MRV data will provide better proof.

16. Reducing dependence on commercial imagery tiles. While commercial imagery tiles are very accurate, they reduce the accessibility of the system and are expensive. Evaluation of the following sub-meter open source solutions will reduce expenses and improve the accessibility of the system for small farmers and community forests: (i) the community imagery obtained by drones, and (ii) super-resolution of open imagery obtained by Sentinel-2 satellites.

17. Scalability of operations. To transition from the pilot phase of the system to the full production phase, the transition from the area-of-interest (AOI) based workflow to batch processing of numerous polygons using cloud orchestration and caching of imagery tiles and job queue should be done.

E. Policy, Social and Research Implications

18. Consistency with national and international policies. Digital MRV data could be integrated into national inventories of greenhouse gas emissions, and reported as per national/international climate policies including Article 6 provisions. Co-development of digital attestation standards in collaboration with forest departments/forest regulators would assist in transitioning the framework from proof-of-concept to adoption.

19. Participation of communities and sharing of benefits. Forest communities and smallholders put in efforts to maintain the forests, yet derive minimum profits from their efforts. Smart contracts can be programmed to share benefits such that an agreed portion of proceeds from credit sales is allocated to the community members. Pilot implementation with community-based organizations would assess both feasibility and acceptability of the arrangement.

20. Explainability of AI output. The NGO verifier should be able to understand why the AI came up with a particular number or value. Overlays that show crowns detected and areas with low confidence levels will assist the reviewer in focusing his/her attention where necessary.

21. Application to other types of MRV. The above structure is not limited to forests. It can be adapted to blue carbon, soil carbon, biochar, renewable energy certificates or biodiversity credits with the only difference being in the module of measurements.

REFERENCES

[1] R. Calel, D. A. Doyen, and A. Yatchew, "Doubling the carbon price or doubling down on emissions?" J. Environ. Econ. Manage., vol. 109, 2021.
[2] Verra, "VCS Program Definitions v4.4," Verra, Washington D.C., Tech. Rep., 2022.
[3] A. Kirillov *et al.*, "Segment Anything," in *Proc. IEEE/CVF Int. Conf. Computer Vision (ICCV)*, 2023, pp. 4015-4026.

[4] J. Jakubik *et al.*, "Foundation models for generalist geospatial artificial intelligence," arXiv:2310.18660, 2023.

[5] M. C. Hansen *et al.*, "High-resolution global maps of 21st-century forest cover change," *Science*, vol. 342, no. 6160, pp. 850-853, 2013.

[6] M. Dalponte and D. A. Coomes, "Tree-centric mapping of forest carbon density from airborne laser scanning and hyperspectral data," *Methods in Ecology and Evolution*, vol. 7, no. 10, pp. 1236-1245, 2016.

[7] P. Howson, "Tackling climate change with blockchain," Nature Climate Change, vol. 9, no. 9, pp. 644–645, 2019.
[8] M. Irfan et al., "Blockchain for carbon credit manage-ment: Opportunities and challenges," IEEE Access, vol. 10, pp. 52498–52515, 2022.
[9] IPCC, *2006 IPCC Guidelines for National Greenhouse Gas Inventories*, Prepared by the National Greenhouse Gas Inventories Programme, IGES, Japan, 2006. Updated in *2019 Refinement to the 2006 IPCC Guidelines for National Greenhouse Gas Inventories*, IPCC, 2019.

[10] T. A. P. West, J. Börner, E. O. Sills, and A. Kontoleon, "Overstated carbon emission reductions from voluntary REDD+ projects in the Brazilian Amazon," *Proc. Natl. Acad. Sci. U.S.A.*, vol. 117, no. 39, pp. 24188-24194, 2020.

[11] T. A. P. West *et al.*, "Action needed to make carbon offsets from forest conservation work for climate change mitigation," *Science*, vol. 381, no. 6660, pp. 873-877, Aug. 2023.

[12] M. Drusch *et al.*, "Sentinel-2: ESA's optical high-resolution mission for GMES operational services," *Remote Sensing of Environment*, vol. 120, pp. 25-36, 2012.

[13] N. Gorelick *et al.*, "Google Earth Engine: Planetary-scale geospatial analysis for everyone," *Remote Sensing of Environment*, vol. 202, pp. 18-27, 2017.

[14] C. J. Tucker, "Red and photographic infrared linear combinations for monitoring vegetation," *Remote Sensing of Environment*, vol. 8, no. 2, pp. 127-150, 1979.

[15] A. Huete *et al.*, "Overview of the radiometric and biophysical performance of the MODIS vegetation indices," *Remote Sensing of Environment*, vol. 83, no. 1-2, pp. 195-213, 2002.

[16] D. M. Woebbecke, G. E. Meyer, K. Von Bargen, and D. A. Mortensen, "Color indices for weed identification under various soil, residue, and lighting conditions," *Transactions of the ASAE*, vol. 38, no. 1, pp. 259-269, 1995.

[17] A. D. Richardson, J. P. Jenkins, B. H. Braswell, D. Y. Hollinger, S. V. Ollinger, and M.-L. Smith, "Use of digital webcam images to track spring green-up in a deciduous broadleaf forest," *Oecologia,* vol. 152, no. 2, pp. 323-334, 2007.

[18] O. Ronneberger, P. Fischer, and T. Brox, "U-Net: Convolutional networks for biomedical image segmentation," in *Medical Image Computing and Computer-Assisted Intervention (MICCAI),* 2015, pp. 234-241.

[19] K. He, G. Gkioxari, P. Dollár, and R. Girshick, "Mask R-CNN," in *Proc. IEEE Int. Conf. Computer Vision (ICCV),* 2017, pp. 2961-2969.

[20] B. G. Weinstein, S. Marconi, S. Bohlman, A. Zare, and E. White, "Individual tree-crown detection in RGB imagery using semi-supervised deep learning neural networks," *Remote Sensing,* vol. 11, no. 11, p. 1309, 2019.

[21] R. Bommasani *et al.*, "On the opportunities and risks of foundation models," arXiv:2108.07258, 2021.

[22] A. Dosovitskiy *et al.*, "An image is worth 16x16 words: Transformers for image recognition at scale," in *Int. Conf. Learning Representations (ICLR),* 2021.

[23] J. Chave *et al.*, "Improved allometric models to estimate the aboveground biomass of tropical trees," *Global Change Biology,* vol. 20, no. 10, pp. 3177-3190, 2014.

[24] V. Buterin, "A next-generation smart contract and decentralized application platform (Ethereum white paper)," 2014.

[25] J. Benet, "IPFS: Content addressed, versioned, P2P file system," arXiv:1407.3561, 2014.

[26] R. S. Sandhu, E. J. Coyne, H. L. Feinstein, and C. E. Youman, "Role-based access control models," *IEEE Computer,* vol. 29, no. 2, pp. 38-47, 1996.

APPENDICE

CERTIFICATES

Certificate of Blockchain-Based Carbon Registry and MRV System: A Review



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International Journal of Innovative Research in Science, Engineering and Technology (IJIRSET)

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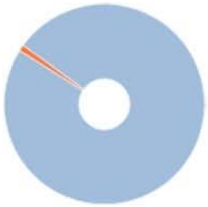
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***Waiting:** Payment Accepted,Application in mandatory waiting period of one month(Copyright Act 1957).
***Documents not received, formality check failed:** Documents/works not received only after making payment.
***Abandoned:** Reply to Discrepancy letter issued not received/works not received after filing.
***Scrutiny:** Application is under process.
***Re-Scrutiny:** Application is under process .
***Pending for Hearing:** Pending for Hearing process.
***Hearing:** Hearing process.
***Sub-Judice:** Pending decision of the competent court of law.
***Registered:** ROC is Generated.

COURSE OUTCOMES

After completing the course, the students will be able to:

CO-1: Identify and analyze real-world problems and project requirements.

CO-2: Design and develop solutions using suitable engineering methods and modern tools.

CO-3: Demonstrate teamwork, technical communication, ethics, and societal responsibility through the project.

CO-PO and PSO Mapping

CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2
CO1	2	3	1	1	-	2	-	-	-	-	1	1	2
CO2	3	3	3	3	3	2	1	-	-	2	3	3	3
CO3	-	-	-	-	-	3	3	3	3	3	2	-	1

PRIYADARSHINI COLLEGE OF ENGINEERING, NAGPUR

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DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING

SESSION 2026-27 (VIII SEM)

"BLOCKCHAIN BASED CARBON CREDIT REGISTRY AND MRV SYSTEM"

Projectees: Yash Bharti, Disha Patil, Swayam Warghane, Sai Drugkar, Mansi Gupta

Project Guide: Dr. Vaishnavi Ganesh

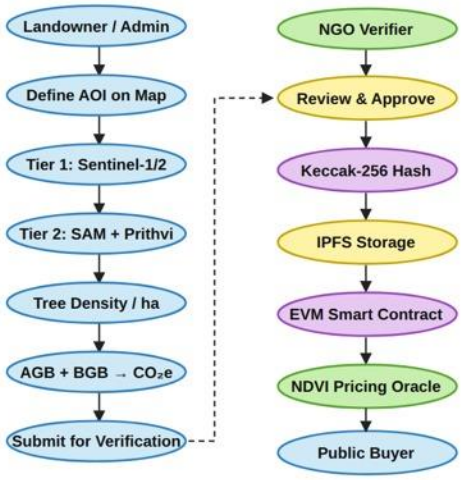
ABSTRACT:

Voluntary forest carbon markets depend on Measurement, Reporting and Verification (MRV), yet legacy registries still rely on manual field surveys, centralized records and verification cycles lasting months, leaving them open to opacity and double counting. This project presents **PlantDense**, an end-to-end digital MRV platform that couples AI-driven forest measurement with blockchain attestation. For a user-defined Area of Interest, Sentinel-2 analysis on Google Earth Engine is combined with tree-crown delineation using Meta's Segment Anything Model and NASA/IBM Prithvi land-cover classification. The tree density feeds an IPCC-compliant allometric model that estimates biomass, carbon stock and CO_2 equivalent. Results are recorded on an EVM smart contract only after mandatory NGO approval, making credits tamper-resistant and independently verifiable.

PROJECT DESCRIPTION:

PlantDense is a web-based platform with a three-role architecture: **Landowner/Admin, NGO Verifier and Public Buyer**. The landowner draws a forest polygon on an interactive map; a two-tier pipeline (Tier 1: 10 m Sentinel-1/2 NDVI and SAR analysis; Tier 2: sub-meter tree detection with a tri-ensemble spectral vote – ExG, GCC, HSV – prompting SAM, with Prithvi filtering non-tree vegetation) returns tree density, above/below-ground biomass and CO_2e. The NGO verifier reviews and approves the result, which is then canonicalized, hashed with Keccak-256, stored on IPFS and attested through a dual-key smart contract. An NDVI-linked pricing oracle sets the credit price, and buyers can trace every credit back to its forest polygon.

SYSTEM FLOW / ARCHITECTURE:



COURSE OUTCOME:

After completing the course, the students will be able to:

CO-1: Identify and analyze real-world problems and project requirements.

CO-2: Design and develop solutions using suitable engineering methods and modern tools.

CO-3: Demonstrate teamwork, technical communication, ethics, and societal responsibility through the project.

TECHNOLOGY STACK:

- Google Earth Engine, Sentinel-1 / Sentinel-2
- MobileSAM and NASA/IBM Prithvi-EO-2.0
- Solidity + Hardhat, Ethereum (Sepolia testnet)
- IPFS storage, Keccak-256 hashing
- Role-Based Access Control (3 roles)

FUTURE SCOPE:

- L-band SAR (ALOS-2) and NASA GEDI LiDAR fusion for dense canopies
- Species-specific allometry using Prithvi classes
- Multi-verifier (multi-signature) approval and ground-truth data
- Credit retirement and live dynamic pricing feed
- Layer-2 deployment (Polygon / Arbitrum) for low-cost scale
- Interoperability with Verra and Gold Standard registries
- Zero-knowledge proofs for private verification

CO – PO MAPPING

CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2
CO1	2	3	1	1	-	2	-	-	-	-	1	1	2
CO2	3	3	3	3	3	2	1	-	-	2	3	3	3
CO3	-	-	-	-	-	3	3	3	3	3	2	-	1

Scale: 3 = High, 2 = Medium, 1 = Low, - = No correlation

CONCLUSION:

PlantDense shows that tamper-resistant, independently verifiable and affordable forest carbon accounting is achievable at scale. Validated against airborne LiDAR crown maps (18,432 crowns), the pipeline achieved a mean F1 of 0.912, and against field biomass from 120 plots a carbon estimation accuracy of R^2 = 0.894. Attesting a record costs about 84,200 gas (≈ $0.08), a saving of over 92% compared with manual registry audits. By chaining AI measurements to on-chain attestation through role-separated, human-in-the-loop verification, every carbon credit becomes traceable to the data that produced it.

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